Market Access and the Production of Information

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ABSTRACT

Investors often face restrictions on the assets they may trade. Beyond risk sharing and capital flows, these restrictions can change what investors learn and what prices reveal. In a noisy rational expectations equilibrium model, I study how market access affects information production by reshaping the allocation of scarce data-processing capacity and the aggregation of information through prices. Investors with partial access may acquire information about assets they cannot hold because it helps them value assets they can trade. This cross-asset learning is strongest at intermediate levels of informativeness. Making an asset investable can reduce price informativeness, or increase it while lowering average posterior precision. Calibrating the model to the Chinese stock market, I find large informational effects despite small foreign ownership. Expanding foreign investability has little impact, whereas relaxing the short-sale constraint substantially improves price informativeness.

Keywords: market access; information acquisition; price informativeness; cross-asset learning; financial liberalization

JEL classification: D83, G11, G14, G15

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I. Introduction

Market-access restrictions—from international market segmentation to investor-eligibility rules—determine who may trade particular assets. A large literature studies their implications for risk sharing, asset prices, and the cost of capital. Yet in an increasingly data-rich financial environment, market access may also operate through an information channel. When investors differ in what they already know and in their ability to process data, how does market access affect the allocation of information across investors and assets, and its aggregation into prices?

I study this question in a noisy rational-expectations equilibrium model with endogenous information acquisition. The model features investor groups that differ in their informational endowments and data-processing technologies. Two groups of investors have offsetting informational advantages: one possesses soft information, while the other has superior data-processing technology. Both groups can acquire hard information by processing data. Before trading, investors allocate scarce data-processing capacity across assets, and equilibrium prices aggregate the resulting information.

The central implication is that market access and information relevance need not coincide. Investors with access to only a subset of assets may nevertheless acquire information about an inaccessible asset when correlated fundamentals make that information useful for valuing an accessible one. Market access thus determines not only feasible portfolios but also the return to learning about different assets. Information production can extend beyond the boundaries of investors' portfolios. Cross-asset learning is strongest at intermediate levels of informativeness: at low precision, the signal is not worth acquiring; at high precision, prices already reveal most of its value, leaving little private gain from producing it.

Market access can drive a wedge between the information investors acquire and the information equilibrium prices reveal. Granting market access to a larger investor base can incur counterintuitive market outcomes: price informativeness can increase while investors' average posterior precision decreases, or price informativeness itself can fall. These effects arise because access changes both who produces information and how scarce processing capacity is reallocated across assets. Foreign information also changes informed demand and hence the price signal observed by domestic investors, inducing them to change their own information choices. Market access therefore operates through a feedback loop between information allocation, portfolio choice, and price aggregation.

Portfolio constraints introduce a further margin of endogenous participation. With a short-sale constraint, information acquisition affects not only the size of an investor's position but also whether the investor participates in the market at all. Information that would affect unconstrained demand may therefore be only imperfectly reflected in prices when the desired position is negative. Expanding the set of assets investors may hold and relaxing constraints on how they trade are consequently distinct policies: the former changes investability, while the latter changes the extent to which private information can be translated into demand.

I quantify these mechanisms using the Shanghai–Hong Kong Stock Connect program. The setting provides a natural application of partial market access: foreign investors can trade an eligible subset of domestic shares, while other shares remain inaccessible. I find that a small foreign ownership share can coexist with large informational effects. The reason is that ownership is not a sufficient statistic for informational influence. Foreign investors affect marginal informed demand and hence the public price signal observed by the much larger domestic sector; domestic investors then update their beliefs, reallocate data-processing capacity, and adjust portfolios across both eligible and ineligible shares.

The counterfactuals distinguish broader investability from greater trading flexibility. At the estimated Chinese benchmark, expanding foreign access from the eligible segment to the full market has little effect when the short-sale constraint remains in place. Relaxing the short-sale constraint instead substantially improves price informativeness. Thus the quantitatively important policy margin is not incremental foreign investability by itself, but whether informed foreign demand can respond sufficiently to private information.

The remainder of the paper proceeds as follows. Section II reviews the related literature. Section III presents the general framework, while Section IV studies information allocation and aggregation across market-access regimes, and Section V introduces portfolio constraints and endogenous participation. Section VI briefly discusses welfare. Section VII calibrates the model to Stock Connect, evaluates counterfactual policies, and presents reduced-form evidence. The final section concludes.

II. Literature

Market access determines not only who can trade an asset but also what information heterogeneous investors choose to produce. This paper brings this information-allocation implication

to the literature on market segmentation and financial liberalization. A large literature studies risk sharing, valuation, and the cost of capital around liberalization episodes (Henry, 2000, 2003; Bekaert and Harvey, 2000; Kim and Singal, 2000; Chari and Henry, 2004), while related research documents changes in domestic information environments, including analyst activity, disclosure, and the incorporation of firm-specific information into prices (Bae et al., 2006; Chan and Kwok, 2017; Griffin et al., 2010). By making data acquisition endogenous, the paper shows that access restrictions govern not only who may hold a security but also where investors direct scarce data-processing capacity. Expanding market access can therefore affect prices through information allocation and aggregation, in addition to the conventional risk-sharing and capital-flow mechanisms.

In light of this information channel, empirical work has documented both local information advantages and foreign advantages associated with sophistication and information processing (Choe et al., 2005; Dvořák, 2005; Hau, 2001; Grinblatt and Keloharju, 2000). This paper fills the corresponding gap in the theoretical literature and formalizes this multidimensional heterogeneity by separating locally endowed soft information from universally available hard information that can be acquired through processing data (Liberti and Petersen, 2019; Gao and Zeng, 2023; Gao and Huang, 2020). It studies how asset-specific market access governs the interaction of these two advantages. The distinction is important because local information and processing technology need not rank investor types in the same way. Domestic institutions may be better placed to interpret local conditions, while foreign institutions may extract higher precision from a common data source.

The theoretical framework builds on noisy rational-expectations models of price aggregation (Grossman and Stiglitz, 1980; Hellwig, 1980; Diamond and Verrecchia, 1981; Verrecchia, 1982), which study how prices aggregate dispersed private information, and on endogenous-information-acquisition models (Van Nieuwerburgh and Veldkamp, 2009), which determine investors' information choices jointly with their portfolio decisions. In these frameworks, the assets investors may trade are generally taken as given. This paper instead makes asset-specific market access an input into that joint problem, so that information about an inaccessible asset can improve beliefs about an accessible one. The resulting cross-asset learning differs from conventional diversification: investors acquire information about an inaccessible asset not to hold it directly, but because it improves the valuation and trading of an accessible asset.

III. General Framework

This section develops a general framework for asset-specific market access across heterogeneous investor groups. The central ingredients are differences in investors ' initial information and data-processing technologies, access restrictions that determine which assets each group may trade, and endogenous data acquisition. Before trading, institutional investors allocate scarce data-processing capacity across assets, and equilibrium prices aggregate the resulting information.

The domestic–foreign structure that follows is a convenient implementation of this broader environment. Domestic investors possess locally generated soft information, while foreign institutions have superior data-processing technology. The framework provides a common basis for the access regimes studied in later sections.

A. Assets

There is a domestic safe asset with gross return r. Domestic firms issue two aggregate risky securities: a type-I security that is eligible for international trade under partial access, and a type-II risky asset that remains closed to international investors under partial access and becomes investable under full access. Let $\boldsymbol{d}$ denote the dividend state known at the start of the trading date. The end-of-date dividend $\widetilde{\boldsymbol{d}}$ follows $\widetilde{\boldsymbol{d}} = \boldsymbol{\mu} + \boldsymbol{G}\boldsymbol{d} + \widetilde{\boldsymbol{y}}$, where $\boldsymbol{G} \equiv \text{diag}(g_1, g_2)$ is the dividend-growth matrix and $\widetilde{\boldsymbol{y}} \overset{i.i.d.}{\sim} N(0, \boldsymbol{\Sigma}_y)$ is the dividend shock. We maintain $r > \max\{g_1, g_2\}$, so discounted dividend values are finite and the asset-specific mappings from effective information into absolute signal-to-noise ratios are positive. Write $\boldsymbol{\Omega}_y \equiv \boldsymbol{\Sigma}_y^{-1}$ and let $\mathcal{C}_y \equiv (\boldsymbol{\Omega}_y)_{12}$ denote its off-diagonal cross-precision. In the two-asset Gaussian case, positive fundamental covariance corresponds to $\mathcal{C}_y < 0$. The dividend is paid out at the end of the trading date. Denote the ex-dividend continuation value at the end of the trading date by $\widetilde{\boldsymbol{v}}$. Then $\widetilde{\boldsymbol{v}} = \boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\widetilde{\boldsymbol{d}}$, where $\boldsymbol{\mu}_v = \frac{r}{r-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu}$.

B. Investors

There is a continuum of investors with four potential types: domestic institutional investors, domestic individual investors, foreign institutional investors, and foreign individual investors, with masses λ_I, λ_J, λ_K, and λ_H, respectively. Which types are present is regime-specific: let $\mathcal{L}^R \subseteq \mathcal{L} \equiv \{I, J, K, H\}$ denote the types active at the trading date under regime R; the regime-

specific active sets used below are stated at the end of this section. For simplicity, the following sections use $l \in \{i, j, k, h\}$ to refer to an investor of any type, and $L \in \{I, J, K, H\}$ to refer to an investor type. Investors of type L are endowed with initial wealth w_L. Which types are active is determined by the access regime alone: the model is a single trading-date economy, and there is no entry or exit across dates. Individual investors do not acquire or process data, while institutional investors acquire and process data, obtaining private signals about the dividend innovation. All domestic investors are additionally endowed with soft information about the dividend innovation.

All investors have CARA utility over end-of-date wealth, with risk-aversion coefficient ρ, so $U_l = -\exp(-\rho \widetilde{w}'_l)$, $l \in \{i, j, k, h\}$. Investors make portfolio choices $\boldsymbol{q}_l$ to maximize expected utility conditional on their respective information sets and subject to a one-period budget constraint

$$w'_l = r w_l + \boldsymbol{q}'_l(\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r) \tag{1}$$

The initial wealth w_K of foreign investors is entirely invested in domestic securities markets; their wealth invested in foreign economies is not modeled.

C. *Market access and portfolio constraints*

The model has a single trading date. We compare equilibria across access regimes $R \in \{\text{cl}, \text{pa}, \text{fu}\}$ (closed, partial, and full access) and across portfolio regimes, with or without a short-sale constraint, at the same trading date rather than across a sequence of periods. Partial access without a short-sale constraint is unmarked.

Domestic investors I and J can trade both types of risky assets in every access regime, with feasible set $\mathbb{R}^2_+$ with a short-sale constraint and $\mathbb{R}^2$ without one. Foreign investors are absent under closed access. Under partial access, foreign institutions K can trade only the Type-I asset, so their feasible set is $\mathbb{R} \times \{0\}$ or $\mathbb{R}_+ \times \{0\}$, while foreign individuals are still absent. Under full access, foreign institutions and individuals can trade both assets, with feasible sets $\mathbb{R}^2$ or $\mathbb{R}^2_+$.

D. *Information structure*

Before making portfolio choices at the trading date, institutional investors can acquire data to generate private signals about the end-of-date dividend innovation $\widetilde{\boldsymbol{y}}$. In addition, domestic in-

vestors are endowed with soft information[1] about $\widetilde{\boldsymbol{y}}$. The equilibrium price aggregates all private information.

Investors have different information sets at different stages of the decision process. When making data-acquisition decisions, domestic institutional investor i has only the prior and soft information $\boldsymbol{\xi}_i$, denoted $\mathcal{I}_{i^-} = \{\boldsymbol{\xi}_i\}$, while foreign institutional investor k, when active, has no relevant information beyond the prior. Let $\boldsymbol{s}_l$, $l \in \{i, k\}$, denote the private signals generated from data, and let $\boldsymbol{s}_p$ denote the price signal. When investor i selects its portfolio, the choice is based on the equilibrium price, soft information, and the private signal generated from acquired data, so $\mathcal{I}_i = \{\boldsymbol{s}_p, \boldsymbol{s}_i, \boldsymbol{\xi}_i\}$. Domestic individual investor j does not acquire or process data and bases its portfolio choice only on endowed soft information and the equilibrium price, so $\mathcal{I}_j = \{\boldsymbol{s}_p, \boldsymbol{\xi}_j\}$. Foreign individual investors H, when active, have no soft information and base their portfolio choice only on the equilibrium price, so $\mathcal{I}_h = \{\boldsymbol{s}_p\}$.

The general equilibrium definition below does not restrict the price mapping. In the unrestricted economies of Section IV, the equilibrium price is a linear function of the signals, and since signals are normally distributed, the price can be expressed as a linear function of shocks:

$$\boldsymbol{p} = \boldsymbol{A} + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{C}\widetilde{\boldsymbol{y}} + \boldsymbol{D}\widetilde{\boldsymbol{x}}. \tag{2}$$

When $\boldsymbol{D}$ is nonsingular, define the signal-to-noise ratio (SNR) as $\boldsymbol{\zeta} \equiv \boldsymbol{D}^{-1}\boldsymbol{C}$. An affine transformation of the equilibrium price $\boldsymbol{p}$ generates an unbiased signal about the end-of-date dividend shock. At the trading date, investors extract the signal $\boldsymbol{s}_p = \widetilde{\boldsymbol{y}} + \boldsymbol{\zeta}^{-1}\widetilde{\boldsymbol{x}}$, whose precision is $\boldsymbol{\Omega}_p = \boldsymbol{\zeta}'\boldsymbol{\Omega}_x\boldsymbol{\zeta}$. Under the short-sale regimes of Section V, exact constrained aggregation makes the global price mapping nonlinear, and that section characterizes its local linear response around a reference state using the same price-signal representation.

Throughout the paper, we distinguish data from information. Data are purchasable inputs, such as accounting information, prices and trading volume, or textual disclosures. Information is the private signal about the unknown dividend innovation $\widetilde{\boldsymbol{y}}$ that results from processing those data. Investors may differ in their data sources, acquired quantities, and data-processing technologies.

Specifically in the model, institutional investors can acquire a total of $\bar{M}$ data points from

[1]Domestic institutional investors may plausibly possess more soft information than domestic individual investors. Because this distinction is not central to the paper and does not alter the model's structure or results, the model assumes that all domestic investors have the same soft-information precision.

two data sources. The data acquisition costs are normalized to zero. Thus investors acquire data at full capacity $\bar{M}$. Denote the numbers of data points acquired by investor l at the trading date for type-I and type-II risky assets as $M_{l,(\mathrm{I})}$ and $M_{l,(\mathrm{II})}$, respectively. Each data source corresponds to one type of firm. The model represents processing technology directly through the quality of the resulting unbiased hard-information signal. A foreign institution that processes a Type-N observation obtains

$$s^{(m)}_{K,(N)} = \widetilde{y}_{(N)} + u^{(m)}_{K,(N)}, \qquad u^{(m)}_{K,(N)} \sim N(0, \Pi^{-1}_{e,(N)}),$$

whereas a domestic institution obtains

$$s^{(m)}_{I,(N)} = \widetilde{y}_{(N)} + u^{(m)}_{I,(N)}, \qquad u^{(m)}_{I,(N)} \sim N\big(0, (\varphi^2 \Pi_{e,(N)})^{-1}\big)\,.$$

Signal noises are independent across observations, investors, and data sources, and $\mathbf{\Pi}_e$ is diagonal. Thus one foreign-processed observation contributes $\Pi_{e,(N)}$ to posterior precision, while one domestically processed observation contributes $\varphi^2 \Pi_{e,(N)}$. For continuity with the technology notation used below, write $\mathbf{\Pi}_e \equiv \beta^2 \mathbf{\Omega}_e$ and $\varphi \equiv \beta_I/\beta \in (0,1)$, where β/β_I indexes the foreign–domestic processing advantage. This is a reduced-form signal-quality wedge: it does not represent domestic institutions as applying an incorrect loading to the same raw observation. The economically operative objects are therefore $(\mathbf{\Pi}_e, \varphi)$.

The resulting signal is *hard information*. By contrast, *soft information* cannot be stored or purchased as data and is held only by domestic investors. For each domestic investor $l \in \{i, j\}$, let

$$\widetilde{\boldsymbol{\xi}}_l = \widetilde{\boldsymbol{y}} + \widetilde{\boldsymbol{\epsilon}}_l, \qquad \widetilde{\boldsymbol{\epsilon}}_l \overset{i.i.d.}{\sim} N(0, \mathbf{\Sigma}_\epsilon).$$

Although domestic institutional investors have an informational advantage in soft information, they have a technological disadvantage in processing data. Thus, a priori, it is unclear which group has an overall information advantage.

D.1. Equilibrium

The regime-specific structure can be summarized by two objects: the set of active types $\mathcal{L}^R$ and the feasible portfolio correspondence $\mathcal{Q}^R_L \subseteq \mathbb{R}^2$. At the trading date, $\mathcal{L}^R = \{I, J\}$ if market access is denied to foreign investors, $\mathcal{L}^R = \{I, J, K\}$ if partial access is granted to foreign institutional

investors, and $\mathcal{L}^R = \{I, J, K, H\}$ if full access is granted to all foreign investors. Partial access imposes $\mathcal{Q}_K^{\text{pa}} = \mathbb{R} \times \{0\}$, whereas full access has $\mathcal{Q}_K^{\text{fu}} = \mathbb{R}^2$; with a short-sale constraint these become $\mathbb{R}_+ \times \{0\}$ and $\mathbb{R}_+^2$, respectively, and every unrestricted feasible set $\mathbb{R}^2$ is replaced by its nonnegative counterpart $\mathbb{R}_+^2$. For an institutional investor, the feasible allocation of data-processing capacity is $\mathcal{M} \equiv \left\{(M_{(\text{I})}, M_{(\text{II})}) \in \mathbb{R}_+^2 : M_{(\text{I})} + M_{(\text{II})} = \bar{M}\right\}$.

Definition 1 (Market equilibrium). *Fix a market regime R. A rational-expectations equilibrium consists of a price function $\boldsymbol{P}^R(\boldsymbol{d}, \widetilde{\boldsymbol{y}}, \widetilde{\boldsymbol{x}})$, data-acquisition strategies for all institutional investors active in regime R, and portfolio strategies $\{\boldsymbol{q}_l^R\}_{l \in \mathcal{L}^R}$, such that:*

(i) Each institutional investor chooses its allocation of data-processing capacity from $\mathcal{M}$ to maximize expected utility at the trading stage conditional on its information at the data-choice stage, taking the equilibrium price-information environment and the subsequent optimal portfolio choice as given;

(ii) For every investor l of type $L \in \mathcal{L}^R$, $\boldsymbol{q}_l^R \in \arg\max_{\boldsymbol{q} \in \mathcal{Q}_L^R} \mathbb{E}\left[-\exp(-\rho \widetilde{w}_l') \mid \mathcal{I}_l\right]$, subject to the budget constraint (1)*;*

(iii) Investors' conditional beliefs are formed from their private information, soft information when available, and the information contained in equilibrium prices;

(iv) The price function clears the market: the aggregate investor holdings of the risky assets equal the exogenous net position $\bar{\boldsymbol{x}} + \widetilde{\boldsymbol{x}}$.

IV. Market Access and Endogenous Information Allocation

This section studies how asset-specific market access shapes endogenous information acquisition and price aggregation in the absence of portfolio constraints. When an investor group can trade only a subset of assets, they allocate scarce data-processing capacity according to the value of information for the assets it can trade. If fundamentals are correlated, information about an inaccessible asset can nevertheless improve the valuation and trading of an accessible asset.

In the domestic–foreign language, domestic institutions can trade both asset types and possess locally generated soft information. Foreign institutions have superior data-processing technology but, under partial access, can trade only Type-I shares; Type-II shares remain inaccessible to them. The domestic and foreign labels therefore discipline the later quantitative application, while the

underlying mechanism applies more broadly to any groups that differ in information, processing technology, and asset-specific access.

A. Partial access

The modeling framework is specified to the partial-access regime where $\mathcal{L} = \{I, J, K\}$, $\mathcal{Q}_I = \mathcal{Q}_J = \mathbb{R}^2$ and $\mathcal{Q}_K = \mathbb{R} \times \{0\}$.

A.1. Portfolio choices

At the trading stage, investor l faces a simple one-shot portfolio choice problem, which has an optimal solution[2]

$$\boldsymbol{q}_l = \frac{1}{\rho}\hat{\boldsymbol{\Omega}}_L \hat{\boldsymbol{\mu}}_l, \tag{3}$$

where $\hat{\boldsymbol{\mu}}_l \equiv \mathbb{E}[\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r \mid \mathcal{I}_l]$ and $\hat{\boldsymbol{\Omega}}_L \equiv \mathbb{V}ar^{-1}(\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r \mid \mathcal{I}_l)$ are the conditional mean and precision of excess payoffs.

A.2. Equilibrium price relation

Definition 2 (Cross-precision). *Let $z \equiv \hat{\mathcal{C}}_{Ky}/\hat{\Omega}_{Ky,(\mathrm{II})}$ denote the cross-precision of the foreign posterior, the off-diagonal entry of $\hat{\boldsymbol{\Omega}}_{Ky}$ normalized by its $\{2,2\}$ element.*

Economically, z measures the intensity of foreign *cross-asset learning* in Type-I demand.

Signal-to-noise ratio The SNR enters the market clearing condition and is related to other variables through

$$\boldsymbol{\zeta} = -\frac{1}{\rho r}(r\boldsymbol{I} - \boldsymbol{G})\Big[\Big(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \Big(\boldsymbol{E}_{11} - z\boldsymbol{E}_{12}\Big)\boldsymbol{M}_K\Big)\boldsymbol{\Pi}_e + \lambda_1 \boldsymbol{\Omega}_\epsilon\Big], \tag{4}$$

where $\lambda_1 = \lambda_I + \lambda_J$, $\mathbb{E}_{11} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $\mathbb{E}_{12} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. An interesting observation is that the SNR matrix is not symmetric. If foreign institutional investors choose to acquire data on type-II risky asset, the information they obtain after data processing is not directly aggregated into the equilibrium price of type-II risky asset since it is not eligible for international trade. However,

[2]For notational simplicity, this paper adopts the notational style that uses a hat to distinguish conditional moments at the portfolio choice stage (*interim*) with those at the data choice stage (ex ante).

this information will be partially reflected in the equilibrium price of type-I risky asset, if the two types of assets are correlated, in which case information on type-II asset also affects their posterior belief on type-I. This creates informational inefficiency.

Average posterior precision Market clearing aggregates investors' posterior precision into an eligibility-weighted object $\bar{\hat{\boldsymbol{\Omega}}}_y$. Under partial access it need not be symmetric, because it is an equilibrium aggregate rather than an individual Bayesian precision matrix. Foreign entry dilutes type-I soft information because the mass of investors eligible to hold that asset rises from λ_1 to λ_2; it does not mechanically dilute type-II soft information, because type-II remains closed to foreign investors. Foreign institutions add private information directly when they process type-I data. When the assets are correlated, their type-II data can also enter the type-I price through the cross-asset SNR, thereby changing the common price-information component $\boldsymbol{\Omega}_p$ and, in particular, potentially raising $\hat{\Omega}_{p,(\mathrm{II})}$ through the cross-price signal. Limited investability creates two countervailing forces. Foreign Type-II information can improve the information embedded in prices through cross-asset learning, but the trading restriction limits the precision that foreign institutions can use directly in Type-I demand.

Price relation Let $\boldsymbol{\Delta}_{\text{pa}}$ denote the eligibility correction induced by foreign Type-II ineligibility; it vanishes under full access. In equilibrium, the price coefficients are characterized by

$$\boldsymbol{A} = \frac{1}{r}\boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - \rho r(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Lambda}^{-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\bar{\boldsymbol{x}}, \tag{5}$$

$$\boldsymbol{B} = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}, \tag{6}$$

$$\boldsymbol{C} = (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\left[\boldsymbol{I} - \bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Omega}_y + \bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Delta}_{\text{pa}}\right], \tag{7}$$

$$\boldsymbol{D} = \boldsymbol{C}\boldsymbol{\zeta}^{-1}, \tag{8}$$

where $\bar{\hat{\boldsymbol{\Sigma}}}_y \equiv \bar{\hat{\boldsymbol{\Omega}}}_y^{-1}$. $\boldsymbol{B}$ is the constant discounted dividend-growth rate; together, $\boldsymbol{A}$ and $\boldsymbol{B}$ govern the level of the equilibrium price. $\boldsymbol{C}$ measures the price sensitivity to dividend innovations, and $\boldsymbol{D}$ governs the amount of demand noise embedded in prices. The amount of purchased data $\boldsymbol{M}_L$, data-processing capacity $\bar{M}$, effective data precision $\boldsymbol{\Pi}_e$, relative data-processing efficiency φ, and the precision of soft information $\boldsymbol{\Omega}_\epsilon$ all enter the price coefficients through the SNR and the average posterior precision of dividend innovations.

A.3. Endogenous information allocation

The central equilibrium choice is how each institutional investor allocates the fixed capacity $\bar{M}$ across the two data sources. More generally, investor groups face different allocation problems when they differ in endowed information and in the assets to which they have access. In the implementation, domestic investors possess soft information while foreign institutions are granted access only to the Type-I asset. The two investor problems are characterized in turn.

Domestic information allocation The relevant sufficient statistic for the allocation problem is summarized in the following lemma.

LEMMA 1. *The data choice of domestic institutional investor i at the trading date enters the ex ante utility function only via* $\det(\hat{\boldsymbol{\Sigma}}_i)$*, where $\hat{\boldsymbol{\Sigma}}_i$ is the posterior variance of excess payoffs.*

Because the data choice $(M_{i,(\mathrm{I})}, M_{i,(\mathrm{II})})$ enters the ex ante utility function only through the determinant of the posterior payoff-variance matrix, $\det(\hat{\boldsymbol{\Sigma}}_i)$, the problem reduces to allocating data points to maximize the determinant of posterior precision, $\det(\hat{\boldsymbol{\Omega}}_i)$. The first-order condition can be written in effective-precision units as $\frac{\hat{\Omega}_{iy,(\mathrm{I})}}{\varphi^2 \Pi_{e,(\mathrm{I})}} = \frac{\hat{\Omega}_{iy,(\mathrm{II})}}{\varphi^2 \Pi_{e,(\mathrm{II})}}$. Thus domestic institutional investor i equalizes posterior precision measured in units of the effective precision generated by one processed domestic data point. This normalization motivates the following definition.

DEFINITION 3 (Equivalent data amount). *For asset type $N \in \{\mathrm{I}, \mathrm{II}\}$ and access regime a, the equivalent data amount is* $m_{e,(N);a}(M) \equiv (\Omega_{y,(N)} + \hat{\Omega}_{p,(N);a} + \Omega_{\epsilon,(N)})/(\varphi^2 \Pi_{e,(N)}) + M$.

It is the posterior precision for Type-N dividends, expressed in units of the effective precision from one processed domestic data point. Under partial access we suppress the regime tag and write $m_{e,(N)}$. Conditional on the equilibrium price signal, $m_{e,(N);a}(M) = m_{e,(N);a}(0) + M$, so that one additional processed data point raises the corresponding equivalent-data amount by one. This is an individual-choice identity: an atomistic investor takes $\hat{\Omega}_{p,(N);a}$ as given when reallocating processing capacity. Across equilibrium regimes, price precision is endogenous, so the level of the equivalent-data amount can change even holding M fixed. The interior first-order condition is therefore $m_{e,(\mathrm{I})}(M_{i,(\mathrm{I})}) = m_{e,(\mathrm{II})}(M_{i,(\mathrm{II})})$.

Denote the optimal Type-I data allocation of domestic institutional investor i by M_i^*; the

corresponding Type-II allocation is $\bar{M} - M_i^*$. The optimal Type-I allocation is

$$M_i^* = \begin{cases} 0, & \text{if } m_{e,(\mathrm{I})}(0) \geq m_{e,(\mathrm{II})}(\bar{M}), \\ \bar{M}, & \text{if } m_{e,(\mathrm{II})}(0) \geq m_{e,(\mathrm{I})}(\bar{M}), \\ \frac{1}{2}\big[m_{e,(\mathrm{II})}(\bar{M}) - m_{e,(\mathrm{I})}(0)\big], & \text{otherwise.} \end{cases} \tag{9}$$

The two corner conditions compare the equivalent-data amounts at the opposite endpoints of the capacity constraint; otherwise, the unique interior allocation equalizes the two equivalent-data amounts. Thus the domestic allocation follows a water-filling rule: more processing capacity is directed to the asset with the lower initial equivalent-data stock.

Foreign cross-asset learning Under partial access, foreign institutions can trade Type I but cannot trade Type II. Correlated fundamentals nevertheless make Type-II data relevant for their Type-I valuation. The following lemma identifies the sufficient statistic for this information choice in the domestic–foreign implementation.

Lemma 2 (Foreign data choice under partial access). *Conditional on the equilibrium price signal, a foreign institution allocates data to maximize the Type-I Schur-complement precision*

$$\hat{\Omega}^S_{Ky,(\mathrm{I})} = \hat{\Omega}_{Ky,(\mathrm{I})} - \frac{\hat{\mathcal{C}}^2_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}}.$$

Define the marginal value of this cross-asset learning as follows.

Definition 4 (Information spillover rate). *Conditional on the equilibrium price signal, let* $M \equiv M_{k,(\mathrm{II})}$ *and write*

$$z(M) \equiv \frac{\mathcal{C}_y + \hat{\mathcal{C}}_p}{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + M\Pi_{e,(\mathrm{II})}}.$$

The information spillover rate is

$$\kappa(M) \equiv z(M)^2 \frac{\Pi_{e,(\mathrm{II})}}{\Pi_{e,(\mathrm{I})}}.$$

It is the marginal Type-I precision generated indirectly by one processed Type-II observation, relative to that generated directly by one processed Type-I observation. The foreign allocation is

$(\bar{M} - M^*_{k,(\mathrm{II})}, M^*_{k,(\mathrm{II})})$, where

$$M^*_{k,(\mathrm{II})} = \begin{cases} 0, & \kappa(0) \leq 1, \\ \bar{M}, & \kappa(\bar{M}) \geq 1, \\ M^{\text{int}}, & \kappa(0) > 1 > \kappa(\bar{M}). \end{cases} \tag{10}$$

Here, $M^{\text{int}} \in (0, \bar{M})$ denotes the interior allocation.

Proposition 1 (Cross-asset learning under partial access). *Under partial access, if*

$$\kappa(0) > 1 > \kappa(\bar{M}),$$

the foreign institution allocates a strictly positive amount of processing capacity to Type-II data. The allocation is unique and satisfies

$$\kappa(M^*_{k,(\mathrm{II})}) = 1,$$

Proposition 1 establishes that information about an inaccessible asset can have private trading value when fundamentals are correlated. At an interior allocation, the foreign institution divides its fixed processing capacity until the marginal improvement in its Type-I valuation is equal from direct Type-I data and indirect Type-II data. Thus, portfolio access determines where the institution may trade, but not which information is relevant for its trading decision.

Cross-asset learning requires fundamental dependence. When Type-I and Type-II dividend innovations are uncorrelated, Type-II data does not improve the foreign institution's valuation of tradable Type-I shares, and the foreign institution allocates no capacity to Type-II processing. The following comparative static characterizes how this learning varies with effective Type-II data precision when both information choices and the price signal adjust in equilibrium.

Proposition 2 (Intermediate-precision cross-asset learning). *Maintain $\mathcal{C}_y < 0$. Let $E \equiv \Pi_{e,(\mathrm{II})}$ vary with primitive Type-II raw-data precision, holding β and all other primitives fixed. Then there exist $0 < \underline{E} < \overline{E} < \infty$ such that*

$$M^*_{k,(\mathrm{II})}(E) = 0$$

for all $E \in (0, \underline{E}) \cup (\overline{E}, \infty)$.

Under the sufficient condition for interior cross-asset learning stated in the Internet Appendix, there exist $0 < E_- < E_+ < \infty$ such that

$$M^*_{k,(\mathrm{II})}(E) > 0$$

for every $E \in (E_-, E_+)$.

At low Type-II precision, indirect learning is unproductive. At high precision, the relevant information is increasingly incorporated into prices and therefore has less private value. These forces generate the intermediate-precision region. Formal bounds, sufficient conditions, and numerical illustrations are provided in the Internet Appendix.

B. Effects of Partial Access

Market access affects both price informativeness and expected returns through its effects on information allocation, price aggregation, and risk bearing. These effects arise from the interaction between heterogeneity in information and processing technology and restrictions on the assets for which investors can deploy their processing capacity. In the domestic–foreign implementation, foreign entry changes the composition of informed demand in the Type-I market and, therefore, the equilibrium price signal observed by domestic investors, who then reallocate data-processing capacity across the two asset types.

The resulting changes in the price signal and in private data acquisition determine the asset-specific average posterior precision, summarized by the diagonal entries $\bar{\hat{\Omega}}_{y,(N)}$ of the eligibility-weighted precision aggregate $\bar{\hat{\mathbf{\Omega}}}_y$.

Market access also changes risk bearing. By enlarging the set of investors able to hold eligible shares, foreign entry changes the risk premium embedded in prices, consistent with the risk-sharing component of liberalization emphasized in the literature (Chari and Henry, 2004; Chan and Kwok, 2017).

B.1. Domestic information reallocation

The closed-access counterfactual is readily available from the partial-access model by setting $\lambda_K = 0$. Using Definition 3, $m_{e,(N)}$ without a regime tag denotes the equivalent data amount evaluated in the partially liberalized equilibrium, while $m_{e,(N);\mathrm{cl}}$ denotes the same normalization evaluated in the closed-access equilibrium ($\lambda_K = 0$). Accordingly, $\hat{\Omega}_{p,(N);\mathrm{cl}}$ is the closed-equilibrium price-signal precision with all foreign-investor terms set to zero. These are regime evaluations of a single concept rather than separate definitions; their levels can differ because equilibrium price precision changes with market access.

B.2. Price informativeness and the signal-to-noise ratio

Partial market access affects the SNRs of type-I and type-II risky assets. Beyond the common channel through changes in domestic institutional investors' data choices, which affects both asset types, the effects differ across assets.

For type-I risky assets, foreign entry can affect the SNR directly or indirectly. If foreign institutions learn about type-I assets, the information extracted from the acquired and processed data is incorporated into the Type-I equilibrium price, directly affecting the Type-I SNR. The effect increases with the foreign investor mass, processing capacity, and data-signal precision. If foreign institutions instead learn about type-II assets, the resulting information is not incorporated directly into the Type-II equilibrium price; it affects the Type-I price only through information spillovers and therefore affects the Type-I SNR indirectly.

For type-II risky assets, domestic institutional investors are the sole contributors to the SNR. Since market access does not affect their endowed soft information, the direction of the SNR change depends only on domestic institutions' data choices: if they learn less about type-II assets, i.e. $M_{(\mathrm{II})} < M_{(\mathrm{II});\mathrm{cl}}$, the Type-II SNR falls; if they learn more, it rises.

Let $\zeta_{(\mathrm{I})}$ and $\zeta_{(\mathrm{II})}$ denote the own-asset SNRs, which measure the loading of each asset's own dividend-innovation signal relative to noise in its equilibrium price. Let $\zeta_{(\mathrm{I}\leftarrow\mathrm{II})}$ denote the cross-asset SNR, which measures the loading of type-II dividend information into the type-I price relative to type-I noise. These ratios depend on the risk-free rate, dividend growth, risk aversion, investor masses, effective data precision $\mathbf{\Pi}_e$, relative processing efficiency φ, and institutional data choices.

The cross-asset SNR $\zeta_{(\mathrm{I}\leftarrow\mathrm{II})}$ is present only if foreign institutional investors choose to learn about type-II assets. Under the type-II corner $\kappa(\bar{M}) \geq 1$, foreign institutions allocate all processing capacity to type-II data. Their information therefore contributes to the cross-asset SNR rather than directly to either own-asset SNR. Any change in the two own-asset SNRs is then driven by domestic institutions' reallocation of their fixed processing capacity across the two asset types.

The Type-I response reflects two distinct frictions. The positive channel is driven primarily by investability: foreign hard information is aggregated into the Type-I price, with its strength governed by foreign presence and processing technology relative to domestic institutions. A negative response requires the reduction in domestic Type-I learning to be sufficiently strong; the dilution of domestic soft information enters this threshold.

At this all-Type-II foreign-processing corner, the changes normalized by the asset-specific mapping from effective data precision into price informativeness are exactly equal and opposite:

$$\frac{\Delta\left|\zeta_{(\mathrm{I})}\right|}{(r-g_1)\Pi_{e,(\mathrm{I})}} = -\frac{\Delta\left|\zeta_{(\mathrm{II})}\right|}{(r-g_2)\Pi_{e,(\mathrm{II})}}. \tag{11}$$

Thus, at this corner, more domestic Type-I processing raises the Type-I own-asset SNR and lowers the Type-II own-asset SNR; the signs reverse under a reallocation toward Type II, and both changes are zero without reallocation. This opposite-sign movement follows from fixed domestic processing capacity, while the investability restriction explains why foreign Type-II information enters through the cross-asset SNR rather than either own-asset SNR. Equation (11) removes the asset-specific scaling and makes the one-for-one reallocation transparent.

The comparison emphasizes that making type-II data investable does not mechanically increase its price informativeness. Foreign institutional investors bring additional effective precision, but this gain can be offset when entry crowds out domestic type-II data acquisition. The exact condition is therefore a comparison between lost domestic effective information and newly created foreign effective information rather than a threshold on foreign investor mass alone.

B.3. Total information and posterior precision

Foreign entry changes total information through three channels: it adds foreign hard information, changes domestic data acquisition, and changes the density of domestic soft information among eligible investors. The last channel mechanically affects Type I but not Type II, because foreign investors become eligible only for Type-I shares under partial access.

The net effect on posterior precision is therefore not determined by price informativeness alone. At the all-Type-I foreign-processing corner, a weak decline in Type-I price informativeness implies a strict decline in average Type-I posterior precision because foreign entry can both reduce domestic information density and impose the portfolio-constraint discount. For Type II, the eligible-investor mass is unchanged, and the cross-price signal can offset a decline in domestic Type-II data acquisition.

C. Closed access

The closed-access regime provides the benchmark in which the second investor group is excluded from the two risky-asset markets. In the domestic–foreign implementation, the domestic financial market remains closed to foreign investors at the trading date, so that only domestic investors trade the two risky securities. The environment is the specialization of Section III in which foreign investor masses are set to zero while the domestic primitives are held fixed. This makes closed access the natural theoretical benchmark for isolating the effect of foreign entry and asset eligibility.

Only domestic institutions produce hard information in this regime, and the two asset prices aggregate domestic information alone. In particular, the cross-asset price channel generated by foreign Type-II learning is absent: $\zeta_{(\mathrm{I}\leftarrow\mathrm{II});\mathrm{cl}} = 0$. The domestic allocation rule in Section III, evaluated at $\lambda_K = 0$, determines the closed-access data allocation, price signal, and posterior precisions used in the partial- and full-access comparisons.

D. Full access

Under full access, all domestic risky securities are available to foreign investors. More generally, this regime removes the asset-specific restriction on the second investor group. It is useful both as a theoretical comparison with partial access and as a counterfactual for a further expansion of investability in partially open markets.

In the full-access economy, all securities are available for international trade. In addition, λ_H foreign individual investors enter the domestic market. Foreign institutional investors may allocate data-processing capacity across both types of risky assets. The entry of foreign individual investors also lowers the average density of private information.

The full-access equilibrium specializes Definition 1 to $\mathcal{L} = \{I, J, K, H\}$, with $\mathcal{Q}_L = \mathbb{R}^2$ for every investor type.

D.1. Full-access equilibrium

The optimal portfolio choices have the same form as in the partial-access model. Domestic institutions face the same data-choice problem, whereas foreign institutions now allocate $\bar{M}$ across both asset types. The price coefficients change with the set of eligible investors.

Under full access, both assets have the same eligible-investor mass $\lambda_3 \equiv \lambda_I + \lambda_J + \lambda_K + \lambda_H$. For $N \in \{\mathrm{I}, \mathrm{II}\}$, the own-asset SNR and the asset-specific average posterior precision are

$$\zeta_{(N);\mathrm{fu}} = -\frac{r - g_N}{\rho r}\left[(\lambda_I \varphi^2 M_{I,(N);\mathrm{fu}} + \lambda_K M_{K,(N);\mathrm{fu}})\Pi_{e,(N)} + \lambda_1 \Omega_{\epsilon,(N)}\right], \tag{12}$$

$$\bar{\hat{\Omega}}_{y,(N);\mathrm{fu}} = \Omega_{y,(N)} + \hat{\Omega}_{p,(N);\mathrm{fu}} + \frac{\lambda_I \varphi^2 M_{I,(N);\mathrm{fu}} + \lambda_K M_{K,(N);\mathrm{fu}}}{\lambda_3}\Pi_{e,(N)} + \frac{\lambda_1}{\lambda_3}\Omega_{\epsilon,(N)}, \tag{13}$$

where $\hat{\Omega}_{p,(N);\mathrm{fu}} = \zeta^2_{(N);\mathrm{fu}}\Omega_{x,(N)}$. Unlike partial access, no portfolio-constraint Schur discount appears because both risky assets are tradable by foreign institutions.

D.2. Comparative statics of full access

Full access affects the SNRs differently. Once the previously inaccessible asset becomes tradable, information produced by the access-restricted group is aggregated directly in that asset's market. In the domestic–foreign implementation, this is information that foreign institutional investors have acquired about Type-II risky assets. Domestic institutional investors may adjust their data acquisition accordingly. The change in average posterior precision from partial to full access contains both a price-information component and a change in the density of privately held information.

Proposition 3 (Market access, price informativeness, and posterior precision). *(i) Type-II price informativeness is lower under full than under partial access if and only if*

$$\left(\zeta^2_{(\mathrm{II});\mathrm{fu}} - \zeta^2_{(\mathrm{II})}\right)\Omega_{x,(\mathrm{II})} < \zeta^2_{(\mathrm{I}\leftarrow\mathrm{II})}\Omega_{x,(\mathrm{I})}.$$

(ii) Consider the all-Type-I foreign-processing corner, and let $\mathcal{H}_{\mathrm{I}}$ denote the resulting change in aggregate Type-I hard-information precision. Relative to closed access, Type-I price informativeness increases while average posterior precision among Type-I- eligible investors decreases if

$$0 < \mathcal{H}_{\mathrm{I}} < \frac{(\lambda_K/\lambda_2)C_{\mathrm{I}} + \mathcal{Q}_{\mathrm{I}}}{A_{\mathrm{I}}},$$

where $A_{\mathrm{I}}, C_{\mathrm{I}} > 0$ and $\mathcal{Q}_{\mathrm{I}} \geq 0$ are defined in Internet Appendix Section IA.K.1.

Proposition 3 shows that price informativeness is an equilibrium aggregation outcome rather than a monotone consequence of investability. In part (i), the Type-I price transmits informa-

tion about Type-II fundamentals under partial access. Full access can therefore reduce Type-II price informativeness when the loss of this cross-price channel exceeds the information newly incorporated directly into the Type-II price.

In part (ii), $\mathcal{H}_{\mathrm{I}}$ measures the net increase in Type-I hard-information production. The stated condition implies that this increase improves the Type-I price signal but is insufficient to offset the dilution of information across eligible investors and the portfolio-constraint discount. Market access can thus make prices more informative even as the average posterior precision of investors eligible to trade the asset declines.

V. Portfolio Constraints and Endogenous Participation

Market access determines which assets an investor may trade, but it does not determine whether private information can be translated into trade. That second margin is governed by portfolio constraints. In the unconstrained model, private information changes desired positions continuously. With a short-sale constraint, unfavorable information can instead lead an investor to take no position at all. Information therefore affects both the size of demand conditional on participation and the decision to participate in an asset market.

This distinction makes expanding market access and relaxing trading constraints economically different interventions. Expanding access changes the set of assets an investor may hold. Relaxing a short-sale constraint changes the extent to which private information can be expressed in price-setting demand for assets that are already investable. Because participation is endogenous, the constraint also changes aggregate risk-bearing capacity and the information content of equilibrium prices.

I incorporate this participation margin by imposing nonnegative risky-asset positions. The analysis retains exact constrained portfolio choice and continuum aggregation. Because these constraints generally make the global equilibrium price function nonlinear, I characterize the price-information map locally around the equilibrium reference state. The following analysis formalizes these points and derives the resulting implications for information aggregation.

For an investor of type L, the portfolio problem under a short-sale constraint is

$$\max_{\boldsymbol{q}\in\mathcal{Q}_L}\left\{\boldsymbol{q}'\hat{\boldsymbol{\mu}}_l-\frac{\rho}{2}\boldsymbol{q}'\hat{\boldsymbol{\Sigma}}_L\boldsymbol{q}\right\},$$

where $\mathcal{Q}_L$ incorporates both asset-specific access and nonnegative-position constraints. For an investor eligible to trade both risky assets, the Kuhn–Tucker conditions are

$$\boldsymbol{q}_l \geq 0, \qquad \rho\hat{\boldsymbol{\Sigma}}_L\boldsymbol{q}_l - \hat{\boldsymbol{\mu}}_l \geq 0, \qquad \boldsymbol{q}_l \odot \left(\rho\hat{\boldsymbol{\Sigma}}_L\boldsymbol{q}_l - \hat{\boldsymbol{\mu}}_l\right) = 0.$$

Thus, when one position is constrained to zero, the remaining position is reoptimized; constrained demand is not obtained by simply truncating negative components of unconstrained demand.

The constraint introduces endogenous market participation. Let

$$P_{L,n} \equiv \Pr\big(q_{l,(n)} > 0 \mid L\big)$$

denote the probability that an eligible investor of type L holds asset n. Aggregate demand is therefore

$$\boldsymbol{Q}(\boldsymbol{p}) = \sum_{L\in\mathcal{L}^R} \lambda_L \mathbb{E}[\boldsymbol{q}_l(\boldsymbol{p}) \mid L]\,,$$

and its response to prices depends both on the positions of participating investors and on changes in the set of investors who participate. Private information can consequently affect prices through two margins: the size of demand conditional on participation and the probability of participation itself.

Exact constrained market clearing generally produces a nonlinear price function. I therefore retain exact constrained portfolio choice and aggregation, while characterizing the price-information map locally around a reference state:

$$\boldsymbol{p} \approx \bar{\boldsymbol{p}} + \boldsymbol{C}_c\left(\widetilde{\boldsymbol{y}} - \bar{\boldsymbol{y}}\right) + \boldsymbol{D}_c\left(\widetilde{\boldsymbol{x}} - \bar{\boldsymbol{x}}\right), \qquad \boldsymbol{\zeta}_c \equiv \boldsymbol{D}_c^{-1}\boldsymbol{C}_c.$$

The associated local price signal is $\boldsymbol{s}_p = \widetilde{\boldsymbol{y}} + \boldsymbol{\zeta}_c^{-1}\widetilde{\boldsymbol{x}}$.

The local price map incorporates both the intensive margin of portfolio demand and the extensive margin of participation induced by the short-sale constraint. Accordingly, expanding market access and relaxing a short-sale constraint are distinct interventions: the former changes the assets an investor may hold, whereas the latter changes the positions an eligible investor may take.

Proposition 4 (Participation-weighted foreign demand). *Under partial access and a nonnegative-position constraint, let m_{K1} and s_{K1}^2 denote the mean and variance of the foreign institution's poste-*

rior expected excess payoff on Type-I shares, and let $\alpha_K = m_{K1}/s_{K1}$*. Its aggregate Type-I demand and local demand response are*

$$\overline{q}_{K,\mathrm{I}} = \frac{s_{K1}\phi(\alpha_K) + m_{K1}\Phi(\alpha_K)}{\rho\widehat{\Sigma}_{K,11}}, \qquad \frac{\partial \overline{q}_{K,\mathrm{I}}}{\partial m_{K1}} = \frac{\Phi(\alpha_K)}{\rho\widehat{\Sigma}_{K,11}}.$$

The probability that the short-sale constraint binds is

$$B_{K,\mathrm{I}} = \Phi(-\alpha_K).$$

Relative to the unconstrained model, the foreign institution's response to a change in expected Type-I payoffs is scaled by $\Phi(\alpha_K)$, its endogenous probability of holding the asset. A short-sale constraint therefore attenuates the price-setting response of foreign demand even though Type-I shares remain investable.

VI. Welfare Analysis

Having characterized decentralized data acquisition, this section compares it with the allocation chosen by a planner and identifies the wedge between private and social marginal values. The section evaluates the constrained efficiency of the information allocation within a given regime, not the welfare effect of changing market access itself. Within that fixed regime the welfare question is general: access restrictions can make privately valuable information about one asset affect price formation and allocation in another.

The cross-asset learning mechanism does not by itself establish a welfare loss. To evaluate efficiency, consider a constrained second-best planner that chooses only the common data allocation of each institutional-investor type, taking market access, short-sale constraints, private signals, decentralized portfolio choice, and rational-expectations market clearing as given.

Throughout this section, a indexes the access regime: cl denotes closed access, the partially open benchmark carries no access tag, and fu denotes full access. Because the comparison holds the short-sale constraint fixed, decentralized welfare objects carry the tag c. Thus $\mathrm{CE}_{L;a,\mathrm{c}}(\boldsymbol{m})$ denotes type L's ex ante certainty equivalent in the constrained decentralized equilibrium under regime a, for a vector $\boldsymbol{m}$ of institutional data allocations. Planner allocations append the tag pl; for example, $\boldsymbol{m}_{G;a,\mathrm{c},\mathrm{pl}}$ is the global planner's allocation. Finally, $\mathcal{L}_a$ denotes the investor types

active under regime a.

We consider one planner objective at a time. The global planner weights every active investor type and solves

$$\boldsymbol{m}_{G;a,\mathrm{c,pl}} \in \arg\max_{\boldsymbol{m}} \sum_{L\in\mathcal{L}_a} \lambda_L \,\mathrm{CE}_{L;a,\mathrm{c}}(\boldsymbol{m}), \tag{14}$$

whereas the domestic regulator values only domestic investors and solves

$$\boldsymbol{m}_{D;a,\mathrm{c,pl}} \in \arg\max_{\boldsymbol{m}} \left[\lambda_I \,\mathrm{CE}_{I;a,\mathrm{c}}(\boldsymbol{m}) + \lambda_J \,\mathrm{CE}_{J;a,\mathrm{c}}(\boldsymbol{m})\right]. \tag{15}$$

Both are subject to the same capacity, portfolio, price-consistency, and market-clearing conditions as the decentralized economy. In the main benchmark, information-processing capacity is an exogenously committed budget: for every data-producing type L, $M_{L,(\mathrm{I})} + M_{L,(\mathrm{II})} = \bar{M}$. Thus the planner can reallocate capacity across the two assets but cannot leave it idle. This restriction preserves the paper's allocation problem and isolates whether decentralized investors direct a given processing budget to the socially efficient source.

An atomistic investor takes the price law as given when choosing data. For either planner objective, the planner additionally internalizes how a common change in type L's data allocation changes prices, posterior moments, and the probability that trading constraints bind. Denote this omitted marginal equilibrium effect by $\mathcal{X}_{L;a,\mathrm{c}}$. At an interior decentralized allocation, the private marginal value is zero, so

$$\left.\frac{dW_{a,\mathrm{c}}}{dm_L}\right|_{\boldsymbol{m}_{a,\mathrm{c}}} = \mathcal{X}_{L;a,\mathrm{c}}(\boldsymbol{m}_{a,\mathrm{c}}). \tag{16}$$

Consequently, a nonzero $\mathcal{X}_{L;a,\mathrm{c}}$ establishes that the decentralized information allocation is not constrained efficient.

For the partial-opening mechanism, let $n_K = M_{K,(\mathrm{II})}$ be the foreign information-capacity choice assigned to the inaccessible asset. Its decentralized value under partial access is $n_{K;\mathrm{c}}$. By the wedge condition of (16), since $\mathcal{X}_{K;\mathrm{c}}(\boldsymbol{m}_{\mathrm{c}}) < 0$—equivalently $-\mathcal{X}_{K;\mathrm{c}}(\boldsymbol{m}_{\mathrm{c}}) > 0$—a marginal reduction in n_K raises welfare; under strict concavity, the global planner chooses less foreign processing of the inaccessible asset. The following proposition states this partial-access cross-asset result in a specialized fixed-active-set benchmark and reports the wedge in closed form. In that benchmark, c denotes the posterior cross-precision of the foreign investor's signal about the two assets, $\Pi_{e,(n)} = \beta^2\Omega_{e,(n)}$ the effective data precisions, and $0 \le \chi \le 1$ the endogenous social

pass-through of the private cross-learning benefit, i.e. the fraction of that benefit that remains in weighted social surplus.

Proposition 5 (Global planner and the non-investable asset). *In the fixed-active-set benchmark, when both the decentralized and the global-planner data choices are interior and $c \neq 0$, the global planner allocates weakly less capacity to information about the non-investable asset, and*

$$n_{K;\mathrm{c}} - n_{K;G,\mathrm{c,pl}} = \frac{|c|\left(1-\sqrt{\chi}\right)}{\sqrt{\Pi_{e,(\mathrm{I})}\Pi_{e,(\mathrm{II})}}}. \tag{17}$$

The wedge is positive when fundamentals are correlated and $\chi < 1$, vanishes when correlation is zero, and shrinks when broader access raises the social pass-through χ. The Internet Appendix derives this result. Equations (16)–(17) therefore provide conditional analytical predictions rather than a calibrated sign restriction. The quantitative section evaluates the decentralized and planner allocations under the estimated primitives and reports the associated certainty-equivalent welfare gains separately.

VII. Quantitative Analysis

This section takes the model to the Shanghai–Hong Kong Stock Connect setting. It first maps the institutional environment into the calibration, then evaluates the benchmark, policy counterfactuals, and reduced-form evidence. More broadly, the framework applies to settings in which investors differ in market access, data acquisition, and trading constraints.

A. Institutional setting

The domestic Chinese stock market (the "A-share" market) was established in 1990 with two independently operated domestic exchanges: the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE). Historically, the Chinese stock market has been only weakly integrated with international markets. The Shanghai–Hong Kong Stock Connect Program was an important step in China's capital-market liberalization. Launched on November 17, 2014, it allows Hong Kong and international investors to trade SSE shares (northbound trading) through the Hong Kong Stock Exchange (HKEX). Shanghai–Hong Kong Stock Connect primarily cov-

ers large-capitalization index constituents listed on the SSE and HKEX. The Internet Appendix provides the trading arrangement, eligibility criteria, and international comparisons.

China's gradual liberalization provides a useful setting because prior evidence suggests that Chinese stock prices have become more informative about future profits over time (Carpenter et al., 2021). This paper studies how changes in market access can reshape information allocation and price aggregation within that broader institutional evolution.

B. Parameterization and Estimation

The calibration proceeds in three steps. First, the investor-type shares and the risk-free rate are calculated directly from the data. Second, the price coefficients are estimated by OLS from cross-sectional regressions. The estimated matrix $\widehat{\boldsymbol{C}}$ provides the price-loading moment residuals, while $\widehat{\boldsymbol{B}}$ is used entirely in a preliminary step to recover the dividend-growth parameters $\boldsymbol{G}$ through $\boldsymbol{G} = r\widehat{\boldsymbol{B}}(\boldsymbol{I} + \widehat{\boldsymbol{B}})^{-1}$; with $\boldsymbol{G}$ fixed, the model-implied $\boldsymbol{B}$ coincides with $\widehat{\boldsymbol{B}}$ by construction, so $\widehat{\boldsymbol{B}}$ residuals carry no identifying information and are excluded from the third-stage objective. Third, the remaining structural parameters are estimated by weighted minimum distance, minimizing the price-loading ($\widehat{\boldsymbol{C}}$), price-noise-covariance, ownership-share, and holdings-ratio residuals, all evaluated at the fully constrained local equilibrium with the data choices solved endogenously as an inner fixed point and the short-sale constraints imposed asset by asset.

The different sets of moments identify distinct components of the model. Ownership-share moments identify the relative investor masses $\boldsymbol{\lambda}$ after accounting for equilibrium portfolio demand. Price-loading moments identify the covariance structure of information and noise, because endogenous data acquisition affects equilibrium price sensitivities. Price-noise moments discipline the supply-noise parameters conditional on the equilibrium price signal. The foreign cross-asset precision object z is an endogenous equilibrium outcome rather than an additional data moment; the spillover channel is disciplined through the price coefficients and the equilibrium structure.

B.1. Measured inputs and calibration targets

The observed investor-type shares and the risk-free rate are measured inputs rather than estimated parameters: the shares are calibration targets for the investor masses $\boldsymbol{\lambda}$, and r is a fixed input. The scale of the information technology is set by three further choices: $\beta_I = 1$ and $\bar{M} = 2$

are normalizations, whereas $\beta = 3$ (equivalently $\varphi = \beta_I/\beta = 1/3$) is a substantive fixed technology assumption. The supplementary analyst-forecast exercise is informative against equal processing efficiency but does not sharply identify the magnitude of the gap: joint or conditional re-estimation improves fit only by moving risk aversion to an economically extreme region. We therefore retain $\beta = 3$ as the maintained benchmark and report the forecast construction, profiles, and joint estimates in the Internet Appendix.

Market share Aggregate data on the composition of shareholders are obtained from the China Securities Regulatory Commission (CSRC). The largest holder of Chinese A-shares is industrial capital, including non-financial companies and general legal persons. Because industrial capital and original shareholders' holdings cannot be freely traded[3], they are not included in the analysis. In the same vein, market shares are calculated with *negotiable market capitalization*, i.e. market capitalization for tradable common listed shares only.

In 2013, individual investors held more than 33% of total A-share market capitalization, peaking above 39% in 2015 and declining below 30% by 2023; domestic institutional investors held around 15%, and foreign capital about 3%. These ratios are computed against total market capitalization, which includes non-tradable shares predominantly held by industrial capital and original shareholders. Since only negotiable market capitalization is used in this model, the ownership shares are restated on a negotiable basis. For the window surrounding the launch of the Shanghai-Hong Kong Stock Connect, domestic individual investors, domestic institutional investors, and foreign institutional investors hold 66%, 28%, and 6% of negotiable market capitalization, respectively, which sum to one.

Risk-free rate The risk-free rate is set to the gross return $r = 1.042$, corresponding to a mean yield to maturity of approximately 4.2% per annum on the 10-year government bond in 2014. The government bond yield curve is published by China Central Depository & Clearing Co., Ltd. (CCDC).

[3]These are common shares held by state legal persons or original promoters. They cannot be traded on the exchange but can be traded by auction or transfer by agreement under the approval of the CSRC.

B.2. OLS estimation for price coefficients

Data on daily stock prices and the eligibility details of the Stock Connect Programs are obtained from the China Stock Market & Accounting Research (CSMAR) database. Following the measurement approach in the literature, unexpected earnings serve as a proxy for the dividend innovation $\widetilde{y}$. Earnings surprises are measured by residuals from the corresponding time-series forecasting models. To estimate the price coefficients, we run the following cross-sectional regressions:

$$ma_{i,t,(\mathrm{I})} = \hat{A}_{t,(\mathrm{I})} + \hat{B}_{t,(\mathrm{I})} ea_{i,t-1,(\mathrm{I})} + \hat{C}_{t,(\mathrm{I})} ue_{i,t,(\mathrm{I})} + \hat{C}^{(c)}_{t} ue_{i,t,(\mathrm{II})} + \hat{K}_{i,t,(\mathrm{I})} Z_{i,t,(\mathrm{I})} + \hat{\varepsilon}_{i,t,(\mathrm{I})},$$
$$ma_{i,t,(\mathrm{II})} = \hat{A}_{t,(\mathrm{II})} + \hat{B}_{t,(\mathrm{II})} ea_{i,t-1,(\mathrm{II})} + \hat{C}_{t,(\mathrm{II})} ue_{i,t,(\mathrm{II})} + \hat{K}_{i,t,(\mathrm{II})} Z_{i,t,(\mathrm{II})} + \hat{\varepsilon}_{i,t,(\mathrm{II})},$$

where $ma_{i,t}$ is the logarithm of market capitalization divided by total assets; $ea_{i,t-1}$ is the previous year's EPS times shares outstanding, divided by total assets; $ue_{i,t}$ is the earnings surprise normalized by total assets; and $Z_{i,t}$ is a set of 2012 CSRC sub-industry dummies. The Internet Appendix provides detailed variable definitions. Because the outcome is the log market-to-assets ratio, the estimated coefficients are semi-elasticities of market capitalization with respect to earnings and earnings surprises—the units in which the equilibrium price-loading moments are matched. These regression coefficients provide empirical moments corresponding to the model's equilibrium price sensitivities $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$, rather than direct estimates of structural coefficients. The residuals provide an empirical proxy for the price-noise component $\boldsymbol{D}\boldsymbol{\Sigma}_x\boldsymbol{D}'$: omitted fundamentals, measurement error, and misspecification orthogonal to the controls are also absorbed into the residual covariance.

The supply-noise precisions $\Omega_{x,(\mathrm{I})}$ and $\Omega_{x,(\mathrm{II})}$ are therefore jointly identified with the other structural parameters under this measurement assumption rather than observed directly. The Internet Appendix reports a robustness exercise that re-weights the price-noise block and leaves the estimates and the allocation corners unchanged; the exercise demonstrates robustness to the block's weight rather than validation of this maintained measurement mapping, because the regression residual covariance also absorbs omitted fundamentals, measurement error, and misspecification.

The growth matrix $\boldsymbol{G}$ is identified directly from the estimated equilibrium price-loading matrix $\widehat{\boldsymbol{B}}$ through the restriction $\boldsymbol{B} = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}$. This identification follows from the present-value restriction linking equilibrium prices to expected dividend growth rather than from an independent estimate of future growth rates. The estimates yield $g_1 > g_2$. This ordering is eco-

nomically plausible given the selection of foreign-investable securities toward large, liquid, and relatively strong-performing firms, although it is not imposed by the model and should not be interpreted as a mechanical consequence of market access itself.

B.3. Weighted minimum-distance estimation

Let $m_L(\boldsymbol{\lambda})$ denote the model-implied market share of type-L investors,

$$m_L(\boldsymbol{\lambda}) \equiv \frac{\lambda_L \, \boldsymbol{q}_L' \boldsymbol{p}}{\sum_{L'} \lambda_{L'} \boldsymbol{q}_{L'}' \boldsymbol{p}}, \qquad L \in \{I, J, K\}. \tag{18}$$

Both price weights and market shares are homogeneous of degree zero in $\boldsymbol{\lambda}$, so it is convenient to normalize $\lambda_I + \lambda_J + \lambda_K = 1$.

Treating the observed market shares of each investor type as the exogenous weights λ_L is problematic. This mapping is exact only when equilibrium demands are identical across types; with information-driven dispersion in demands, the observed market shares are equilibrium objects that also embed the cross-type dispersion of posterior expectations. Therefore $\boldsymbol{\lambda} = (\lambda_I, \lambda_J, 1 - \lambda_I - \lambda_J)$ is to be estimated by moment matching.

Because price informativeness and posterior variances depend on endogenous data choices, each candidate parameter vector is evaluated at the fixed point of the constrained partial-access equilibrium. The calibration matches price loadings, price-noise covariance, three investor ownership shares, and the domestic institutional holdings ratio. Effective data precision $\boldsymbol{\Pi}_e$ is estimated directly, while $\beta = 3$, $\beta_I = 1$, and $\bar{M} = 2$ are maintained benchmark choices.

The Type-II effective-precision parameter is set-identified at the estimated corner because no investor processes Type-II data there; we report a corner-supporting evaluation point and do not interpret its level as a point-identified economic fact. The holdings-ratio moment disciplines the domestic allocation through equilibrium restrictions. The full moment vector, weighting, local-rank diagnostics, and joint-estimation robustness exercises for relative processing efficiency are in the Internet Appendix. Table I reports the estimated primitives and fit. Ownership shares are equilibrium outcomes rather than investor-mass parameters, and institutional information choices are solved endogenously within estimation.

Calibration results The parameter estimates, targeted moments, endogenous information allocation, and participation outcomes are summarized in Tables I and II. As an untargeted diag-

nostic, the estimated model reproduces the direction of the data-based price variance reduction: 9.32% versus 12.37% in the data for type-I shares and 4.15% versus 4.40% for type-II shares. Because this moment is not separately targeted in estimation, the approximate agreement is reassuring; it is not, however, an independent out-of-sample validation, since the diagnostic is built from the same price-information objects that underlie the targeted $\boldsymbol{C}$ and price-noise moments. The residual tension in Table I now concentrates in the price-noise moments: the model's $\Sigma_{11}^{p,\text{noise}}$ is 0.3008 against data of 0.23479 (marginal residual 3.00) and $\Sigma_{22}^{p,\text{noise}}$ is 0.2257 against 0.32837 (marginal residual -3.51), while the own-price loadings fit well, with C_{22} at 2.2874 against 2.2683 (marginal residual 0.07). The reparameterized estimation moves the residual tension from the non-investable own-price loading toward the price-noise covariance; a fuller account of this trade-off appears in the Internet Appendix sensitivity discussion. Re-weighting the price-noise block at half its sampling weight and re-estimating leaves every parameter estimate and the allocation corners unchanged (Internet Appendix), so the imperfect fit of these two moments does not drive the estimated information allocation. A related point concerns the cross-loading: the reduced-form cross-price coefficient is positive ($C_{12}^{\text{data}} = 1.1766$), yet the estimated structural equilibrium generates a much smaller cross-loading ($C_{12}^{\text{model}} = 0.0632$). This is not a statistical failure (marginal residual -1.05), and it is economically coherent: investors internalize correlated fundamentals through private data acquisition, so the incremental informational content of prices for the cross fundamental is small even though the reduced-form comovement—which also reflects discounting and equilibrium trading—is sizable.

C. Benchmark Equilibrium and Information Mechanism

In the calibrated model, foreign institutional investors hold a 6.02% share of negotiable market value (Table I), close to the 6% observed in the data. The first quantitative question is why granting foreign investors access only to Type-I shares can nevertheless affect market outcomes for both Type-I and Type-II shares, and why the two responses need not be alike. Table II collects the equilibrium objects behind this transmission. Foreign entry directly changes informed demand and risk bearing in Type I. Because the resulting Type-I price is a public signal about correlated fundamentals, domestic investors update beliefs, reallocate data-processing capacity, and adjust portfolios across both asset types. Partial access can therefore generate market-wide and heterogeneous effects even though foreigners directly trade only one segment. The small foreign

Table I
Structural parameter estimates and benchmark model fit.

Panel A. Structural parameters

Panel A reports the economically central primitives of the estimated partial-access economy with short-sale constraints. $\Pi_{e,(\mathrm{II})}$ is a corner-supporting, set-identified value; the Internet Appendix characterizes its identified region and reports the maintained technology choices, remaining parameters, and full identification diagnostics.

Parameter	Economic interpretation	Value
$\Pi_{e,(\mathrm{I})}$	Effective data precision, Type I	7,438.0460
$\Pi_{e,(\mathrm{II})}$	Effective data precision, Type II (corner-supporting)	73.8648
$\Omega_{\epsilon,(\mathrm{I})}$	Domestic soft-information precision, Type I	1,596.7535
$\Omega_{\epsilon,(\mathrm{II})}$	Domestic soft-information precision, Type II	1,108.4729
ρ	Risk aversion	1.4613
λ_K	Foreign institutional investor mass	0.0195

Panel B. Targeted moments and benchmark fit

Panel B reports the headline targeted moments. The Internet Appendix reports the full moment vector, weighting, sampling standard errors, marginal residuals, objective, and robustness diagnostics.

Moment	Data	Model
C_{11}	3.7123	3.6565
C_{12}	1.1766	0.0632
C_{22}	2.2683	2.2874
$\Sigma_{11}^{p,\text{noise}}$	0.23479	0.3008
$\Sigma_{12}^{p,\text{noise}}$	0.00007	0.0040
$\Sigma_{22}^{p,\text{noise}}$	0.32837	0.2257
Foreign institutional ownership share	0.0600	0.0602
Domestic institutional Type-I/II holdings ratio	1.5953	1.6232

Table II
Benchmark equilibrium and information mechanism.

Values refer to the estimated partial-access equilibrium with short-sale constraints. Panels report ownership and participation, information technology and capacity allocation, zero-holding probabilities, and variance reduction relative to the prior. Capacity is measured relative to $\bar{M} = 2$; the active foreign mass equals λ_K times the Type-I participation probability.

	Benchmark value
Panel A. Ownership, mass, and participation	
Foreign institutional ownership share	6.02%
Estimated foreign institutional model mass, λ_K	1.95%
Probability of a zero type-I holding	92.8%
Type-I participation probability	7.2%
Participating foreign mass, $\lambda_K \times 7.2\%$	0.14%
Panel B. Information technology and allocation	
Foreign processing technology, β	3.000
Domestic institutional processing technology, β_I	1.000
Relative domestic processing efficiency, $\varphi \equiv \beta_I/\beta$	1/3
Effective precision per processed observation, foreign relative to domestic	9×
Domestic capacity allocated to Type I / Type II	2.000 / 0.000
Foreign capacity allocated to Type I / Type II	2.000 / 0.000
Panel C. Probability of a zero holding	
Domestic institutions, I: Type I / Type II	77.7% / 73.7%
Domestic individuals, J: Type I / Type II	72.8% / 73.7%
Foreign institutions, K: Type I	92.8%
Panel D. Market outcomes	
Price-based variance reduction, Type I / Type II	9.32% / 4.15%
Total variance reduction, Type I / Type II	83.97% / 67.23%

sector reinforces this mechanism: its estimated mass is $\lambda_K = 0.0195$ (1.95% of the investor population), and its endogenous Type-I participation probability is just 7.2%, so the actively participating foreign mass is approximately 0.14% of the population. Ownership, investor mass, active participation, and informational influence are distinct equilibrium objects.

The first reason is that observed ownership is an equilibrium outcome, not the foreign investor mass. Market shares are determined by equilibrium holdings (18): a low observed share can reflect a high probability of a zero holding induced by portfolio constraints, smaller positive positions conditional on participation, differences in beliefs and posterior precision, or asset-access restrictions. The calibration estimates a foreign institutional model mass of $\lambda_K = 0.0195$, or 1.95% of the investor population (Table II), even though foreign institutions hold only 6.02% of market value in the benchmark equilibrium. The gap reflects the short-sale constraint, which binds frequently: the type-I zero-holding probability is 92.8% (Table II), so the participation probability is only 7.2%, and the participating foreign mass is approximately $0.0195 \times 0.072 = 0.14\%$. The estimated foreign mass is itself small, and infrequent participation makes the actively trading foreign sector smaller still. Ownership share, investor mass, and informational influence are three distinct objects: even a quantitatively tiny participating foreign mass can exert disproportionate informational influence if its information is sufficiently productive and is transmitted through equilibrium prices. This conclusion is conditional on the benchmark identification region: an almost equally fitting specification with equal type-I and type-II effective precision (objective 1.2919 versus 1.2587) flips the domestic information allocation to type-II and eliminates the type-I foreign-entry effect (Internet Appendix), so the data do not yet globally select this mechanism.

The second reason is that foreign institutions have a processing advantage under the maintained benchmark technology: $\beta = 3$, whereas domestic institutional technology is normalized to $\beta_I = 1$ (Table II), so $\varphi = 1/3$ and $\varphi^2 = 1/9$. The supplementary forecast-error exercise does not sharply identify this gap: appending the moment can improve fit only by moving risk aversion to an economically extreme region, and the profile is nearly flat over a wide range of processing advantages. We therefore retain $\beta = 3$ as a maintained benchmark and report the forecast construction, profile, and joint-estimation results in the Internet Appendix. With effective data precision $\mathbf{\Pi}_e \equiv \beta^2 \mathbf{\Omega}_e$, one correctly processed observation contributes $\mathbf{\Pi}_e$ to posterior precision when processed by a foreign institution and $\varphi^2 \mathbf{\Pi}_e$ when processed by a domestic institution: holding the data source fixed, a foreign-processed observation delivers nine times the

effective precision of the same observation processed with the domestic institutional technology. A marginal foreign investor is therefore informationally different from a marginal domestic holder. In the benchmark, both foreign and domestic institutions allocate their entire processing capacity to type-I data (Table II); this concentration is consistent with the targeted ratio of domestic institutional holdings in type-I to type-II shares of about 1.60 in the data and 1.62 in the model. A small ownership stake can accordingly be associated with a concentrated, high-precision contribution to the information environment of the investable market.

Third, foreign information does not remain private to the foreign sector. Foreign informed demand transmits part of that information into equilibrium prices, and the price is a public signal observed by all investors: foreign entry and foreign data processing shift informed demand, which changes the equilibrium price signal, which updates domestic posterior beliefs, which alter domestic information allocation and portfolio demand, and thereby feed back into the price. A foreign sector can therefore have market-wide effects even when its direct holdings are small, because it affects the marginal price signal rather than the average ownership share. This is fundamentally different from a partial-equilibrium ownership-share calculation.

Fourth, domestic institutions do not hold their data acquisition fixed when foreign investors enter. Their processing capacity is allocated endogenously across the two asset types, and a change in price informativeness changes the marginal value of private data. Foreign entry therefore has both a direct information effect, through foreign processing and informed demand, and an indirect crowding-in or crowding-out effect through the reoptimization of domestic data acquisition. The equilibrium effect of foreign entry need not be proportional to the foreign ownership share.

The high zero-holding probabilities reported in Table II (Panel C) warrant emphasis, because they are central to the large short-sale effects documented below. The short-sale constraint binds frequently in the estimated economy: Table II (Panel C) reports zero-holding probabilities of 77.7% and 73.7% for domestic institutions, 72.8% and 73.7% for domestic individuals, and 92.8% for foreign institutions in type-I; equivalently, the positive-holding probabilities are 22.3%, 26.3%, 27.2%, 26.3%, and 7.2%, respectively. The combination that looks most extreme—foreign institutions allocate their entire processing capacity to type-I data ($M_{K,(\mathrm{I})} = 2$) and yet hold type-I shares only about 7% of the time—is economically coherent rather than contradictory. Information has option value: it raises posterior precision in the states in which the constraint is slack and the investor participates, so a low participation probability need not imply a low ex ante value of

data. More generally, the information choice is evaluated ex ante over all states, whereas portfolio participation is determined ex post after observing signals and prices; the two decisions therefore need not move together. The high binding probabilities are a property of the estimated primitives (large position-size exposures relative to the aggregate supply noise), not an input. Because the estimated binding probabilities are high, the Internet Appendix reports sensitivity analyses assessing the quantitative importance of the short-sale constraint.

Fifth, correlated fundamentals and domestic general-equilibrium responses transmit the effect across asset types. Although foreign institutions process only Type-I data in the calibrated benchmark, changes in foreign Type-I informed demand alter the price-information environment faced by domestic investors, who learn about and trade both assets. The model therefore explains why the effects of foreign access need not be confined to the shares that foreigners are permitted to hold.

Finally, foreign entry also changes the composition of the information-bearing investor population in the opposite direction. Foreign institutions have superior hard-information processing but no domestic soft information, so their entry simultaneously adds precise processed hard information and dilutes the average density of domestic soft information. The net effect on market outcomes is therefore not determined by the foreign ownership share alone; it depends on the interaction of the foreign mass, processing technology, soft-information precision, data allocation, asset access, and portfolio constraints.

These mechanisms clarify why the effect of partial access need not be confined to the securities that foreigners can hold and why the two asset types can respond differently. In the constrained closed-to-partial comparison, price-based variance reduction changes from 8.85% to 9.32% for Type I (about a 5.3% increase relative to the closed-market level) and from 4.18% to 4.15% for Type II (Table III). The price-information effect is therefore positive but modest for Type I and essentially flat for Type II in this particular statistic. The broader mechanism is nevertheless market-wide: foreign entry changes the Type-I price signal, and that public signal feeds into domestic beliefs, data allocation, and portfolios in both markets. The 0.14% actively participating foreign mass illustrates why direct participation is not a sufficient statistic for that transmission. The much larger gains documented in the unrestricted counterfactual below arise primarily from relaxing the short-sale constraint rather than from incremental foreign investability.

D. Counterfactual Full Market Access

The benchmark shows how partial access can transmit foreign entry beyond the directly investable segment through equilibrium prices and domestic responses. We next use the estimated model to ask what would happen if foreign institutions could trade both share types. Holding the estimated Chinese primitives fixed, we compare two scenarios: full access with a short-sale constraint maintained, in which foreign institutions can trade both Type-I and Type-II shares but all risky-asset positions remain nonnegative; and full access without a short-sale constraint. Table III reports the outcomes, together with the partial-access benchmark for comparison.

The comparison identifies the short-sale constraint, rather than incremental investability, as the dominant quantitative margin. With the constraint maintained, full access leaves foreign processing concentrated on Type I and price-based variance reduction essentially unchanged relative to partial access (Table III). Once short sale is allowed, prices aggregate substantially more private information: Panel D reports 65.92% and 37.99% price-based variance reduction for Type I and Type II, respectively. Portfolio constraints therefore impede the translation of private information into price-setting demand without materially altering the concentrated Type-I information allocation at the Chinese benchmark. We next evaluate the efficiency of the resulting decentralized information allocation.

E. Information Allocation Efficiency

For compactness, the tables report variance reduction relative to the prior. Price-based variance reduction uses the posterior based on the price signal alone, while total variance reduction uses a population-wide posterior-precision statistic with a common population denominator. This quantitative statistic is distinct from the asset-eligible average posterior precision $\bar{\hat{\Omega}}_{y,(N)}$ used in the analytical comparative statics above. For the population-wide statistic, full symmetric individual posterior precision matrices are averaged before inversion; both variance-reduction statistics invert a full precision matrix before taking its diagonal because dividend innovations are correlated.

Table III
Market-access and short-sale counterfactuals.

All counterfactuals hold estimated Chinese primitives fixed. Closed removes foreign investors; constrained full access permits foreign Type-II trading but retains a short-sale constraint; Panel D also permits short sale. Panel E reports constrained-minus-unrestricted differences in percentage points. Both full-access equilibria converge; entries below 0.0005 in absolute value are reported as zero.

	Closed Counterfactual	**Partial (Estimated)**	Full access† (Constrained)
Panel A. Information production			
Domestic $M_{I,(\mathrm{I})}$	2.000	2.000	2.000
Domestic $M_{I,(\mathrm{II})}$	0.000	0.000	0.000
Foreign $M_{K,(\mathrm{I})}$	n.a.	2.000	2.000
Foreign $M_{K,(\mathrm{II})}$	n.a.	0.000	0.000
Panel B. Price-based variance reduction			
Type I, VR^{p}_{I}	8.85%	9.32%	9.32%
Type II, VR^{p}_{II}	4.18%	4.15%	4.15%
Panel C. Total variance reduction			
Type I, $VR^{\mathrm{tot}}_{\mathrm{I}}$	82.32%	83.97%	83.97%
Type II, $VR^{\mathrm{tot}}_{\mathrm{II}}$	67.23%	67.23%	66.81%
Panel D. Full access, short sale allowed			
Foreign capacity allocated to Type I			100.0%
Foreign capacity allocated to Type II			0.0%
Type I, VR^{p}_{I}			65.92%
Type II, VR^{p}_{II}			37.99%
Type I, $VR^{\mathrm{tot}}_{\mathrm{I}}$			87.60%
Type II, $VR^{\mathrm{tot}}_{\mathrm{II}}$			72.04%
Panel E. Effect of a short-sale constraint (constrained minus unrestricted, pp)			
$\Delta VR^{p}_{\mathrm{I}}$	−51.56	−56.60	−56.60
$\Delta VR^{p}_{\mathrm{II}}$	−34.71	−33.84	−33.84
$\Delta VR^{\mathrm{tot}}_{\mathrm{I}}$	−3.56	−3.63	−3.63
$\Delta VR^{\mathrm{tot}}_{\mathrm{II}}$	−5.29	−5.11	−5.23

†Full access with a short-sale constraint maintained.

Table IV
Decentralized and efficient information allocation.

The planner chooses a common institutional allocation of capacity $\bar{M}$; prices and private portfolios are re-solved under each row's trading rules. Welfare is the investor-mass-weighted sum of ex ante certainty equivalents. Solutions are checked on a global grid, allocation boundaries, and local refinement. Full $+c$ is the converged constrained full-access equilibrium.

Panel A. Information-allocation wedges

Regime	$\Delta M_{I,\mathrm{I}}$	$\Delta M_{I,\mathrm{II}}$	$\Delta M_{K,\mathrm{I}}$	$\Delta M_{K,\mathrm{II}}$
Closed	-2.000	$+2.000$	n.a.	n.a.
Closed $+c$	-2.000	$+2.000$	n.a.	n.a.
Partial	-2.000	$+2.000$	-1.944	$+1.944$
Partial $+c$	-2.000	$+2.000$	-1.872	$+1.872$
Full	-2.000	$+2.000$	-1.950	$+1.950$
Full $+c$	-2.000	$+2.000$	-1.872	$+1.872$

Panel B. Welfare

Regime	Decentralized W_r	Planner $W_{r,\mathrm{pl}}$	Wedge ΔW_r
Closed	0.8382	0.8898	0.0516
Closed $+c$	0.6502	0.6562	0.0060
Partial	0.8011	0.8923	0.0912
Partial $+c$	0.6440	0.6514	0.0074
Full	0.8018	0.8931	0.0913
Full $+c$	0.6490	0.6561	0.0071

F. Reduced-Form Evidence

The structural analysis characterizes how market access affects equilibrium information allocation and prices. This subsection documents whether the Shanghai–Hong Kong Stock Connect announcement and launch were accompanied by differential changes in information-related market outcomes between Type-I shares, which became foreign-investable, and Type-II shares, which remained inaccessible. These patterns provide an empirical counterpart to the model's focus on how a targeted change in investability can reshape the information environment.

Since market participants could anticipate the policy change before trading started, the empirical design distinguishes the announcement-to-launch interval from the post-launch period and estimates

$$Y_{it} = \alpha_i + \gamma_{j(i),t} + \beta_A(\text{Eligible}_i \times \text{Announcement}_t) + \beta_L(\text{Eligible}_i \times \text{Post}_t) + \varepsilon_{it}, \tag{19}$$

where α_i denotes a firm fixed effect and $\gamma_{j(i),t}$ an industry-by-day fixed effect. The eligible group is fixed using the April 10, 2014 reference universe and retained throughout the sample. The preferred comparison restricts both groups to Shanghai-listed stocks: 562 eligible stocks and 443 never-eligible stocks. This same-exchange comparison avoids confounding the treatment contrast with Shanghai–Shenzhen movements. The preferred specification omits contemporaneous firm controls, which can respond to liberalization and are mechanically related to return-based outcomes.

The outcome variable is idiosyncratic volatility, the annualized 30-trading-day standard deviation of market-model residuals. β_A is the announcement-to-launch differential relative to the pre-announcement baseline, β_L is the post-launch differential, and $\beta_L - \beta_A$ is the incremental post-launch contrast. These return-based measures are not direct empirical counterparts of the model's precision objects; the exercise documents differential market outcomes around the opening rather than tests the structural mechanism.

Table V reports the preferred same-exchange specification and a size-adjusted variant. In the preferred specification, eligible stocks have lower idiosyncratic volatility both between announcement and launch (-0.0388, $s.e. = 0.0066$) and after launch (-0.0385, $s.e. = 0.0065$), while the incremental post-launch contrast is indistinguishable from zero. Size adjustment attenuates the announcement-period estimate but leaves the post-launch differential negative. Thus the broad differential is robust, whereas its timing is not cleanly attributable to anticipation versus

implementation.

Table V
Stock Connect and idiosyncratic volatility.

Estimates compare announcement-eligible with never-eligible Shanghai stocks using equation (19). Column (1) includes firm and industry-by-day fixed effects; column (2) additionally includes pre-treatment size-decile-by-day effects on the common 2013-size sample. Standard errors are two-way clustered by firm and date. The final row is $\beta_L - \beta_A$.

	(1)	(2)
Eligible × Announcement-to-launch period	-0.0388^{***}	-0.0148^{*}
	(0.0066)	(0.0080)
Eligible × Post-launch period	-0.0385^{***}	-0.0277^{***}
	(0.0065)	(0.0078)
Post-launch − Announcement	0.0002	-0.0129^{*}
	(0.0061)	(0.0073)
Sample	Shanghai	Shanghai
Firm fixed effects	Yes	Yes
Industry × day fixed effects	Yes	Yes
Pre-treatment size decile × day fixed effects	No	Yes
Controls	No	No
Clustering	Firm & date	Firm & date
Observations	761,168	723,108
Within R^2	0.0044	0.0015

VIII. Conclusion

Market access changes not only who can hold assets but also how investors allocate data-processing capacity and how private information is aggregated into equilibrium prices. When investors differ in soft information and data-processing technology, access restrictions also change the return to acquiring data and the information environment faced by other investors through equilibrium prices.

The theory delivers three main implications. First, market access and information relevance need not coincide: under partial access, foreign institutions may acquire data about an asset they cannot hold because correlated fundamentals make that information useful for trading an accessible asset. Second, this cross-asset learning is an intermediate-precision phenomenon: suf-

ficiently noisy data are not worth acquiring, while at high precision prices already reveal most of their value. Third, market access, private information, and price informativeness need not move together. Making an asset newly investable can reduce its price informativeness when the loss of cross-price information exceeds the gain in own-price information, and more informative prices can coexist with lower average posterior precision.

Calibration to the Chinese stock market illustrates how these mechanisms operate under partial access. Stock Connect gives foreign investors direct access to Type-I shares only, yet foreign entry changes the public price signal observed by domestic investors, who update beliefs, reallocate data-processing capacity, and adjust portfolios across both asset types. Partial access can therefore have market-wide and heterogeneous informational effects even on securities that remain inaccessible to foreign investors. Large informational effects can arise even when foreign ownership and participation are limited; neither is a sufficient statistic for informational influence. The transmission operates through marginal informed demand and equilibrium prices.

These counterfactuals point to a broader lesson. The informational consequences of market access depend not only on which assets investors may hold, but also on whether they can translate private information into positions. In the China calibration, expanding foreign investability while retaining the short-sale constraint has little effect on price informativeness, whereas relaxing that constraint produces substantial improvements. Policies that broaden formal access without allowing informed investors to translate private information into demand may therefore yield little improvement in informational efficiency.

Appendix: Proofs of Main Theoretical Results

This appendix supplies proofs of the paper's main theoretical results. It covers the information-allocation proposition under heterogeneous market access (Proposition 1), the intermediate-precision cross-asset-learning proposition (Proposition 2), the price-informativeness proposition (Proposition 3), and the portfolio-constraints proposition (Proposition 4). Supporting lemmas, the complete constrained-equilibrium characterization, welfare derivations, and quantitative details appear in the Internet Appendix.

A.1. Proof of Proposition 4

Proof. Under partial access, the foreign institution's constrained Type-I position is

$$q^*_{k,\mathrm{I}} = \max\left\{0, \frac{\widehat{\mu}_{k,\mathrm{I}}}{\rho\widehat{\Sigma}_{K,11}}\right\}.$$

Since $\widehat{\mu}_{k,\mathrm{I}} \sim N(m_{K1}, s^2_{K1})$, the truncated-normal identity $\mathbb{E}[X_+] = s\phi(m/s) + m\Phi(m/s)$ gives the expression for $\overline{q}_{K,\mathrm{I}}$. Differentiating with respect to m_{K1} gives the stated local response, and $\Pr(q^*_{k,\mathrm{I}} = 0) = \Pr(\widehat{\mu}_{k,\mathrm{I}} \leq 0) = \Phi(-\alpha_K)$. The complete derivation appears in Internet Appendix Section IA.F. □

A.2. Proof of Proposition 1

Proof of Proposition 1. For domestic institutions, Lemma 1 reduces the data-choice problem to maximizing posterior precision subject to the binding capacity constraint. The interior first-order condition equalizes posterior precision in effective-data units, and the endpoint comparisons give (9).

For foreign institutions under partial access, Lemma 2 reduces the objective to Type-I Schur-complement precision. With $M \equiv M_{k,(\mathrm{II})}$ and $M_{k,(\mathrm{I})} = \bar{M} - M$, the marginal objective is $\frac{d\hat{\Omega}^S_{kyt,(\mathrm{I})}}{dM} = \Pi_{e,(\mathrm{I})}[\kappa(M) - 1]$. Holding the equilibrium price signal fixed, $\kappa(M)$ is continuous and, whenever the cross-precision is nonzero, strictly decreasing. The two endpoint conditions therefore give the Type-I and Type-II corners, while $\kappa(0) > 1 > \kappa(\bar{M})$ yields the unique interior solution $\kappa(M^{\text{int}}) = 1$. These are exactly the cases in (10). □

A.3. Proof of Proposition 2

Proof of Proposition 2. At the candidate all-Type-I foreign-processing corner, the foreign allocation rule is governed by the marginal spillover rate $\kappa_E(0)$. As $E \to 0$, the direct precision contribution of a Type-II observation vanishes, so $\kappa_E(0) \to 0$ and the Type-II allocation is zero. At the other extreme, equilibrium price precision rises sufficiently quickly with E that the cross-precision ratio z_E vanishes faster than the direct precision gain increases: $z_E^2 E \to 0$. Thus $\kappa_E(0) \to 0$ also as $E \to \infty$, and the Type-II allocation is again zero.

Internet Appendix Section IA.A.4.2 constructs a primitive lower bound for $\kappa_E(0)$ that is single-peaked in E. Under the stated sufficient condition, that lower bound exceeds one on a nonempty bounded interval. Hence $\kappa_E(0) > 1$ throughout that interval, the all-Type-I corner violates the foreign allocation rule, and strictly positive Type-II processing is optimal. The full bounds, high-precision case split, and lower-bound derivation are provided in that Internet Appendix section. □

A.4. Proof of Proposition 3

Proof. **Part (i).** The Type-II price-precision decomposition in Internet Appendix Corollary 11 separates the change in the own-price component from the cross-price component available under partial access. Thus Type-II price informativeness falls if and only if

$$\left(\zeta^2_{(\mathrm{II});\mathrm{fu}} - \zeta^2_{(\mathrm{II})}\right)\Omega_{x,(\mathrm{II})} < \zeta^2_{(\mathrm{I}\leftarrow\mathrm{II})}\Omega_{x,(\mathrm{I})}.$$

Part (ii). At the all-Type-I foreign-processing corner, the Type-I SNR change has the same sign as the net change $\mathcal{H}_{\mathrm{I}}$ in aggregate Type-I hard-information precision. The posterior-precision accounting derived in the Internet Appendix is

$$\Delta\bar{\hat{\Omega}}_{y,(\mathrm{I})} = A_{\mathrm{I}}\mathcal{H}_{\mathrm{I}} - \frac{\lambda_K}{\lambda_2}C_{\mathrm{I}} - \mathcal{Q}_{\mathrm{I}},$$

where $A_{\mathrm{I}}, C_{\mathrm{I}} > 0$ and $\mathcal{Q}_{\mathrm{I}} \geq 0$ is the foreign portfolio-constraint discount. If Type-I price informativeness weakly falls, then $\mathcal{H}_{\mathrm{I}} \leq 0$, making the first and third terms nonpositive and the dilution term strictly negative. Average Type-I posterior precision therefore falls strictly. Conversely, when price informativeness rises, the joint outcome in the proposition occurs whenever

$$0 < \mathcal{H}_{\mathrm{I}} < \frac{(\lambda_K/\lambda_2)C_{\mathrm{I}} + \mathcal{Q}_{\mathrm{I}}}{A_{\mathrm{I}}}.$$

The left inequality makes Type-I price informativeness rise, while the right inequality makes average Type-I posterior precision fall. The full rearrangement is in the Internet Appendix. □

References

Bae, K.H., Bailey, W., Mao, C.X., 2006. Stock market liberalization and the information environment. *Journal of International Money and Finance* 25, 404–428.

Bekaert, G., Harvey, C.R., 2000. Foreign speculators and emerging equity markets. *The Journal of Finance* 55, 565–613.

Carpenter, J.N., Lu, F., Whitelaw, R.F., 2021. The real value of china' s stock market. *Journal of Financial Economics* 139, 679–696.

Chan, M.K., Kwok, S., 2017. Risk-sharing, market imperfections, asset prices: Evidence from china' s stock market liberalization. *Journal of Banking & Finance* 84, 166–187.

Chari, A., Henry, P.B., 2004. Risk sharing and asset prices: evidence from a natural experiment. *The Journal of Finance* 59, 1295–1324.

Choe, H., Kho, B.C., Stulz, R.M., 2005. Do domestic investors have an edge? the trading experience of foreign investors in korea. *The Review of Financial Studies* 18, 795–829.

Diamond, D.W., Verrecchia, R.E., 1981. Information aggregation in a noisy rational expectations economy. *Journal of Financial Economics* 9, 221–235.

Dvořák, T., 2005. Do domestic investors have an information advantage? evidence from indonesia. *The Journal of Finance* 60, 817–839.

Gao, C., Zeng, W., 2023. Near is dear: Remoteness, soft information, and stock returns. *SSRN Working Paper No. 4509746* URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4509746, doi:10.2139/ssrn.4509746. first version June 2022.

Gao, M., Huang, J., 2020. Informing the market: The effect of modern information technologies on information production. *The Review of Financial Studies* 33, 1367–1411.

Griffin, J.M., Kelly, P.J., Nardari, F., 2010. Do market efficiency measures yield correct inferences? a comparison of developed and emerging markets. *The Review of Financial Studies* 23, 3225–3277.

Grinblatt, M., Keloharju, M., 2000. The investment behavior and performance of various investor types: a study of finland's unique data set. *Journal of Financial Economics* 55, 43–67.

Grossman, S.J., Stiglitz, J.E., 1980. On the impossibility of informationally efficient markets. *The American Economic Review* 70, 393–408.

Hau, H., 2001. Location matters: An examination of trading profits. *The Journal of Finance* 56, 1959–1983.

Hellwig, M.F., 1980. On the aggregation of information in competitive markets. *Journal of Economic Theory* 22, 477–498.

Henry, P.B., 2000. Stock market liberalization, economic reform, and emerging market equity prices. *The Journal of Finance* 55, 529–564.

Henry, P.B., 2003. Capital-account liberalization, the cost of capital, and economic growth. *American Economic Review* 93, 91–96.

Kim, E.H., Singal, V., 2000. The fear of globalizing capital markets. *Emerging Markets Review* 1, 183–198.

Liberti, J.M., Petersen, M.A., 2019. Information: Hard and soft. *Review of Corporate Finance Studies* 8, 1–41.

Van Nieuwerburgh, S., Veldkamp, L., 2009. Information immobility and the home bias puzzle. *The Journal of Finance* 64, 1187–1215.

Verrecchia, R.E., 1982. Information acquisition in a noisy rational expectations economy. *Econometrica: Journal of the Econometric Society* 50, 1415–1430.

Internet Appendix to "Market Access and the Production of Information"

Complete Derivations, Proofs, and Quantitative Analysis

This version: October 31, 2026

This Internet Appendix follows the convention in the manuscript. Its sections, figures, tables, and equations use the prefix "IA"; cross-references to the main text resolve through the companion manuscript source.

Throughout the Internet Appendix, following the main text, $\mathbf{\Pi}_e \equiv \beta^2 \mathbf{\Omega}_e$ denotes *effective data precision* and $\varphi \equiv \beta_I/\beta \in (0,1)$ denotes the *relative data-processing efficiency* of domestic institutions. We use $(\mathbf{\Pi}_e, \varphi)$ throughout the baseline and access-comparison derivations. The primitive notation $(\beta, \beta_I, \mathbf{\Omega}_e)$ labels the reduced-form processing-technology wedge and its variations; the baseline model uses unbiased processed signals with precisions $\mathbf{\Pi}_e$ and $\varphi^2 \mathbf{\Pi}_e$, respectively. The symbol χ^P used in the welfare analysis below is a distinct social pass-through object.

The first sections provide the unrestricted partial-, closed-, and full-access derivations and their supporting proofs. They are followed by institutional background and the complete short-sale-constrained derivations for partial, closed, and full access. In particular, Internet Appendix Section IA.F supplies the state-by-state portfolio characterization, exact continuum aggregation, and local price representation underlying Proposition 4 in the main text. The remaining sections document welfare, calibration, and reduced-form implementation and robustness. Supplementary technical corollaries are collected at the end so that the empirical and computational materials remain together.

IA.A Partial Access: Solutions and Proofs

IA.A.1 Conditional moments

IA.A.1.1. Computing conditional moments

Recall that the dividend $\widetilde{\boldsymbol{d}}$ follows an AR(1) process, $\widetilde{\boldsymbol{d}} = \boldsymbol{\mu} + \boldsymbol{G}\boldsymbol{d} + \widetilde{\boldsymbol{y}}$. If we view the asset as a claim to an infinite series of dividend payments, then $\widetilde{\boldsymbol{v}}$, the ex-dividend value of the firm at the

end of Period t can be interpreted as

$$\widetilde{\boldsymbol{v}} = \mathbb{E}\left[\sum_{s=1}^{\infty} \frac{\widetilde{\boldsymbol{d}}_s}{r^s} \mid \mathcal{I}_{l+}\right] = \boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\widetilde{\boldsymbol{d}},$$

where $\boldsymbol{\mu}_v = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu}\sum_{s=1}^{\infty}[(1/r)^s I - (\boldsymbol{G}/r)^s] = (r/(r-1))(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu}$.

The excess payoff that investors obtain at the trading date is $\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r$. It is normally distributed, and its conditional mean and variance are central to the model. This section calculates them under different information sets. Throughout this Appendix, hats distinguish conditional moments at the portfolio-choice stage (*interim*) from those at the data-choice stage (ex ante).

IA.A.1.2. Conditional moments of dividend innovations

At the trading date, the price signal is

$$\boldsymbol{s}_p = \boldsymbol{C}^{-1}(\boldsymbol{p}_t - \boldsymbol{A} - \boldsymbol{B}\boldsymbol{d}) = \widetilde{\boldsymbol{y}} + \boldsymbol{C}^{-1}\boldsymbol{D}\widetilde{\boldsymbol{x}}, \tag{IA.1}$$

with price noise $\widetilde{\boldsymbol{\nu}}_p = \boldsymbol{C}^{-1}\boldsymbol{D}\widetilde{\boldsymbol{x}} \sim N(0, \boldsymbol{\Sigma}_p)$, where $\boldsymbol{\Sigma}_p = (\boldsymbol{C}^{-1}\boldsymbol{D})\boldsymbol{\Sigma}_x(\boldsymbol{C}^{-1}\boldsymbol{D})' = \boldsymbol{\Omega}_p^{-1}$. Thus $\boldsymbol{\Omega}_p = \boldsymbol{\zeta}'\boldsymbol{\Omega}_x\boldsymbol{\zeta}$.

Domestic individual investor

At the portfolio choice stage, domestic individual investor j is endowed with inalienable information $\widetilde{\boldsymbol{\xi}}_j$, obtains the price signal $\boldsymbol{s}_p$, but does not process data. Thus she updates the conditional expectation and variance about $\widetilde{\boldsymbol{y}}$ following

$$\begin{aligned}
\hat{\boldsymbol{\Sigma}}_{Jy} &\equiv \mathbb{V}ar(\widetilde{\boldsymbol{y}} \mid \mathcal{I}_j) = \mathbb{V}ar(\widetilde{\boldsymbol{y}} \mid \boldsymbol{p}_t) = (\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \boldsymbol{\Omega}_\epsilon)^{-1}, \\
\hat{\boldsymbol{\mu}}_{Jy} &\equiv \mathbb{E}[\widetilde{\boldsymbol{y}} \mid \mathcal{I}_j] = \mathbb{E}[\widetilde{\boldsymbol{y}} \mid \boldsymbol{p}_t] = \hat{\boldsymbol{\Sigma}}_{Jy}\boldsymbol{\Omega}_p\boldsymbol{s}_p.
\end{aligned}$$

Domestic institutional investor

Apart from information contained in the price, domestic institutional investor i is endowed with inalienable information and also extracts information from data. The conditional moments are

$$\begin{aligned}
\hat{\boldsymbol{\Sigma}}_{Iy} &\equiv \mathbb{V}ar(\widetilde{\boldsymbol{y}} \mid \mathcal{I}_i) = \mathbb{V}ar(\widetilde{\boldsymbol{y}} \mid \boldsymbol{p}_t, \boldsymbol{s}_i, \boldsymbol{\xi}_i) = (\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \varphi^2\boldsymbol{M}_i\boldsymbol{\Pi}_e + \boldsymbol{\Omega}_\epsilon)^{-1}, \\
\hat{\boldsymbol{\mu}}_{iy} &\equiv \mathbb{E}[\widetilde{\boldsymbol{y}} \mid \mathcal{I}_i] = \mathbb{E}[\widetilde{\boldsymbol{y}} \mid \boldsymbol{p}_t, \boldsymbol{s}_i, \boldsymbol{\xi}_i] = \hat{\boldsymbol{\Sigma}}_{Iy}(\boldsymbol{\Omega}_p\boldsymbol{s}_p + \varphi^2\boldsymbol{\Pi}_e\boldsymbol{\iota}_i\boldsymbol{s}_i + \boldsymbol{\Omega}_\epsilon\boldsymbol{\xi}_i).
\end{aligned}$$

Here, $\boldsymbol{\iota}_i\boldsymbol{s}_i = \begin{pmatrix} \sum_{m\in\{\text{(I)}\}} s_i^{(m)} \\ \sum_{m\in\{\text{(II)}\}} s_i^{(m)} \end{pmatrix}$ denotes the vector of data signals acquired by investor i in period t,

summed separately for each risky-asset type. Domestic processing produces an unbiased signal with lower precision, $\varphi^2\boldsymbol{\Pi}_e$, rather than a signal generated by an incorrect loading; this reduced-form processing wedge affects both posterior beliefs and portfolio choices.

International institutional investor

International institutional investors are active under partial and full access. Apart from the observed price $\boldsymbol{p}_t$, international institutional investor k also extracts information from data. For an arbitrary feasible allocation $\boldsymbol{M}_K$, one has $\boldsymbol{M}_K = \begin{pmatrix} M_{K,(\mathrm{I})} & 0 \\ 0 & M_{K,(\mathrm{II})} \end{pmatrix}$, $M_{K,(\mathrm{I})} + M_{K,(\mathrm{II})} = \bar{M}$ and $\boldsymbol{\iota}_K \boldsymbol{s}_K = \begin{pmatrix} \sum_{m\in\mathcal{M}_{K,\mathrm{I}}} s_K^{(m)} \\ \sum_{m\in\mathcal{M}_{K,\mathrm{II}}} s_K^{(m)} \end{pmatrix}$. Then the conditional moments become

$$\begin{aligned}
\hat{\boldsymbol{\Sigma}}_{Ky} &\equiv \mathbb{V}ar(\widetilde{\boldsymbol{y}} \mid \mathcal{I}_k) = \mathbb{V}ar(\widetilde{\boldsymbol{y}} \mid \boldsymbol{p}_t, \boldsymbol{s}_k) = (\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \boldsymbol{M}_K\boldsymbol{\Pi}_e)^{-1}, \\
\hat{\boldsymbol{\mu}}_{ky} &\equiv \mathbb{E}[\widetilde{\boldsymbol{y}} \mid \mathcal{I}_k] = \mathbb{E}[\widetilde{\boldsymbol{y}} \mid \boldsymbol{p}_t, \boldsymbol{s}_k] = \hat{\boldsymbol{\Sigma}}_{Ky}(\boldsymbol{\Omega}_p\boldsymbol{s}_p + \boldsymbol{\Pi}_e\boldsymbol{\iota}_K\boldsymbol{s}_k).
\end{aligned}$$

At the portfolio choice stage, the current payoff $\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r$ can be analyzed as

$$\begin{aligned}
\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}_t r \mid \mathcal{I}_l &= \boldsymbol{\mu}_v + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\widetilde{\boldsymbol{d}} - \boldsymbol{p}_t r \\
&= \boldsymbol{\mu}_v + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\widetilde{\boldsymbol{y}} - \boldsymbol{p}_t r.
\end{aligned}$$

Its conditional variance and expectation are

$$\hat{\boldsymbol{\Sigma}}_L = r^2(r\boldsymbol{I} - \boldsymbol{G})^{-1}\hat{\boldsymbol{\Sigma}}_{Lyt}(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}, \tag{IA.2}$$

$$\begin{aligned}
\hat{\boldsymbol{\mu}}_l = \boldsymbol{\mu}_v &+ r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} \\
&+ r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\hat{\boldsymbol{\mu}}_{lyt} - \boldsymbol{p}_t r,
\end{aligned} \tag{IA.3}$$

respectively. At the data choice stage, the current excess payoff can be written as

$$\begin{aligned}
&\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}_t r \mid \mathcal{I}_{lt^-} \\
&= \boldsymbol{\mu}_v + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - r\boldsymbol{A} + [r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G} - r\boldsymbol{B}]\boldsymbol{d} \\
&\quad + [r(r\boldsymbol{I} - \boldsymbol{G})^{-1} - r\boldsymbol{C}]\widetilde{\boldsymbol{y}} - r\boldsymbol{D}\widetilde{\boldsymbol{x}} \mid \mathcal{I}_{l1}.
\end{aligned}$$

Its conditional variance and expectation are

$$\boldsymbol{\Sigma}_L = [r(r\boldsymbol{I}-\boldsymbol{G})^{-1} - r\boldsymbol{C}]\boldsymbol{\Sigma}_{Lyt}[r(r\boldsymbol{I}-\boldsymbol{G})^{-1} - r\boldsymbol{C}]' + r^2\boldsymbol{D}\boldsymbol{\Sigma}_x\boldsymbol{D}', \tag{IA.4}$$

$$\begin{aligned}\boldsymbol{\mu}_l = \boldsymbol{\mu}_v + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{\mu} - r\boldsymbol{A} + \left[r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{G} - r\boldsymbol{B}\right]\boldsymbol{d} \\ + \left[r(r\boldsymbol{I}-\boldsymbol{G})^{-1} - r\boldsymbol{C}\right]\boldsymbol{\mu}_{lyt},\end{aligned} \tag{IA.5}$$

respectively. Before foreign institutional investors make data choice, they have no information on the current dividend innovation $\widetilde{\boldsymbol{y}}$, thus $\boldsymbol{\Sigma}_{Kyt} = \boldsymbol{\Sigma}_y$, $\boldsymbol{\mu}_{kyt} = 0$. Domestic investors have inalienable information $\boldsymbol{\xi}_{lt}$, so $\boldsymbol{\Sigma}_{Lyt} = (\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_\epsilon)^{-1}$ and $\boldsymbol{\mu}_{lyt} = \boldsymbol{\Sigma}_{Lyt}\boldsymbol{\Omega}_\epsilon\boldsymbol{\xi}_{lt}$, $l = i, j$, $L = I, J$.

IA.A.2 Optimal portfolio choice

At the trading date, the portfolio problems of all three investor types have the same form. For any domestic investor l, $l \in \{i, j\}$,

$$\max_{\{\boldsymbol{q}_l\}} \qquad \mathbb{E}[-\exp\left(-\rho w_l^+\right) \mid \mathcal{I}_l] \tag{IA.6}$$

$$\text{s.t.} \qquad w_l^+ = rw_l + \boldsymbol{q}_l'(\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r). \tag{IA.7}$$

This is a one-period problem. Substituting (IA.7) into (IA.6) and using the moment-generating function of a normal random variable[1] yields $\boldsymbol{q}_l^* = \arg\max_{\boldsymbol{q}_l} \boldsymbol{q}_l'\hat{\boldsymbol{\mu}}_l - (\rho/2)\boldsymbol{q}_l\hat{\boldsymbol{\Sigma}}_L\boldsymbol{q}_l'$. The first-order condition gives

$$\boldsymbol{q}_l = \frac{1}{\rho}\hat{\boldsymbol{\Omega}}_L\hat{\boldsymbol{\mu}}_l. \tag{IA.8}$$

Because the payoffs of type-I and type-II risky assets are correlated, $\hat{\boldsymbol{\Omega}}_L$ is not diagonal. Thus the posterior expectation and variance of one asset's payoff, as well as its covariance with the other payoff, affect the position in the other risky asset. Foreign institutional investor k faces a constrained problem with $q_{k,(\mathrm{II})} = 0$. Thus

$$\boldsymbol{q}_k = \begin{pmatrix} \frac{\hat{\mu}_{kt,(\mathrm{I})}}{\rho\hat{\sigma}^2_{K,(\mathrm{I})}} \\ 0 \end{pmatrix}.$$

In this case, the posterior variance and expectation of the type-I risky asset alone determine the position $q_{k,(\mathrm{I})}$. This does not imply that information about the type-II risky asset is irrelevant: it can affect both $\hat{\sigma}^2_{K,(\mathrm{I})}$ and $\hat{\mu}_{kt,(\mathrm{I})}$. To use uniform notation, define $\hat{\boldsymbol{\Omega}}_K = \begin{pmatrix} \hat{\Omega}_{K,(\mathrm{I})} & \hat{C}_K \\ \hat{C}_K & \hat{\Omega}_{K,(\mathrm{II})} \end{pmatrix}$. Then

[1] If $\boldsymbol{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $\mathbb{E}[\exp(\boldsymbol{x})] = \exp(\boldsymbol{\mu} + \boldsymbol{\Sigma}/2)$.

the first diagonal element $\hat{\sigma}^2_{K,(\mathrm{I})}$ of the posterior variance-covariance matrix $\hat{\boldsymbol{\Sigma}}_K$ is $\frac{\hat{\Omega}_{K,(\mathrm{II})}}{\hat{\Omega}_{K,(\mathrm{I})}\hat{\Omega}_{K,(\mathrm{II})}-\hat{\mathcal{C}}^2_K}$. Define $\hat{\boldsymbol{\Omega}}^{\S}_K = \big(\hat{\Omega}_{K,(\mathrm{I})} - \frac{\hat{\mathcal{C}}^2_K}{\hat{\Omega}_{K,(\mathrm{II})}}\big)\boldsymbol{E}_{11}$. With these notations, the portfolio choice of investor k can be written as

$$\boldsymbol{q}_k = \frac{1}{\rho}\hat{\boldsymbol{\Omega}}^{\S}_K\hat{\boldsymbol{\mu}}_k.$$

The first diagonal elements of $\hat{\tilde{\Omega}}_K$ and $\hat{\tilde{\Omega}}^{\S}_K$ differ.

IA.A.3 Equilibrium price coefficients

At the trading date, domestic institutional, domestic individual, and foreign institutional investors all participate in the domestic type-I stock market. For clarity, the following derivation of the price coefficients retains the (I) superscript only on $M_{t,(\mathrm{I})}$, the amount of Type-I data acquired by domestic institutional investors, and suppresses it for all other model parameters and variables. Market clearing requires aggregate investor holdings to equal the exogenous net position:

$$\int_i \boldsymbol{q}_i\,di + \int_j \boldsymbol{q}_j\,dj + \int_k \boldsymbol{q}_k\,dk = \bar{\boldsymbol{x}} + \widetilde{\boldsymbol{x}}. \tag{IA.9}$$

Begin by aggregating investor i's posterior expectations of the dividend innovation. The noise in data signals and in inalienable information integrates to zero. Thus,

$$\int_i \hat{\boldsymbol{\mu}}_{iy}\,di = \hat{\boldsymbol{\Sigma}}_{Iy}\int_i \boldsymbol{\Omega}_p\boldsymbol{s}_p + \varphi^2\boldsymbol{\Pi}_e\boldsymbol{\iota}_i\boldsymbol{s}_i + \boldsymbol{\Omega}_\epsilon\boldsymbol{\xi}_i\,di = \lambda_I\hat{\boldsymbol{\Sigma}}_{Iy}[\boldsymbol{\Omega}_p\boldsymbol{s}_p + (\varphi^2\boldsymbol{M}_I\boldsymbol{\Pi}_e + \boldsymbol{\Omega}_\epsilon)\widetilde{\boldsymbol{y}}].$$

On the other hand, the aggregate demand of all domestic-institutional investors evaluates to

$$\begin{aligned}
&\int_i \boldsymbol{q}_i\,di = \frac{1}{\rho}\hat{\boldsymbol{\Omega}}_I\int_i \hat{\boldsymbol{\mu}}_i\,di \\
&= \frac{1}{\rho r^2}(r\boldsymbol{I}-\boldsymbol{G})\hat{\boldsymbol{\Omega}}_{Iy}(r\boldsymbol{I}-\boldsymbol{G})' \\
&\times \int_i \boldsymbol{\mu}_v + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\hat{\boldsymbol{\mu}}_{iy} - \boldsymbol{p}_t r\,di.
\end{aligned}$$

Substituting $\int_i \hat{\boldsymbol{\mu}}_{iy}\,di$ yields

$$\begin{aligned}
\int_i \boldsymbol{q}_i\,di = {}& \frac{\lambda_I}{\rho r^2}(r\boldsymbol{I}-\boldsymbol{G})\hat{\boldsymbol{\Omega}}_{Iy}(r\boldsymbol{I}-\boldsymbol{G})'[\boldsymbol{\mu}_v + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} - \boldsymbol{p}_t r] \\
&+ \frac{\lambda_I}{\rho r}(r\boldsymbol{I}-\boldsymbol{G})[\boldsymbol{\Omega}_p\boldsymbol{s}_p + (\varphi^2\boldsymbol{M}_I\boldsymbol{\Pi}_e + \boldsymbol{\Omega}_\epsilon)\widetilde{\boldsymbol{y}}].
\end{aligned} \tag{IA.10}$$

The aggregate demand of all domestic individual investors evaluates to

$$\int_j \boldsymbol{q}_j\, dj = \frac{\lambda_J}{\rho r^2}(r\boldsymbol{I}-\boldsymbol{G})\hat{\boldsymbol{\Omega}}_{Jy}(r\boldsymbol{I}-\boldsymbol{G})'[\boldsymbol{\mu}_v + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} - \boldsymbol{p}_t r] + \frac{\lambda_J}{\rho r}(r\boldsymbol{I}-\boldsymbol{G})[\boldsymbol{\Omega}_p \boldsymbol{s}_p + \boldsymbol{\Omega}_\epsilon \widetilde{\boldsymbol{y}}]. \tag{IA.11}$$

Finally, the aggregate demand of all foreign institutional investors evaluates to $\int_k \boldsymbol{q}_k\, dk = \frac{1}{\rho}\hat{\boldsymbol{\Omega}}^{\S}_K \int_k \hat{\boldsymbol{\mu}}_k\, dk$. To aggregate their demands with other investors in a uniform notation, a connection between $\hat{\boldsymbol{\Omega}}^{\S}_K$ and $\hat{\boldsymbol{\Omega}}_{Ky}$ is made. First, let us write out $\hat{\boldsymbol{\Omega}}_{Ky}$. Let $\hat{\boldsymbol{\Omega}}_{Ky} = \left(\begin{smallmatrix} \hat{\Omega}_{Ky,(\mathrm{I})} & \hat{\mathcal{C}}_{Ky} \\ \hat{\mathcal{C}}_{Ky} & \hat{\Omega}_{Ky,(\mathrm{II})} \end{smallmatrix}\right)$. Then

$$\begin{aligned}\begin{pmatrix} \hat{\Omega}_{K,(\mathrm{I})} & \hat{\mathcal{C}}_K \\ \hat{\mathcal{C}}_K & \hat{\Omega}_{K,(\mathrm{II})} \end{pmatrix} \equiv \hat{\boldsymbol{\Omega}}_K &= \frac{1}{r^2}(r\boldsymbol{I}-\boldsymbol{G})\hat{\boldsymbol{\Omega}}_{Ky}(r\boldsymbol{I}-\boldsymbol{G})' \\ &= \begin{pmatrix} \frac{(r-g_1)^2}{r^2}\hat{\Omega}_{Ky,(\mathrm{I})} & \frac{(r-g_1)(r-g_2)}{r^2}\hat{\mathcal{C}}_{Ky} \\ \frac{(r-g_1)(r-g_2)}{r^2}\hat{\mathcal{C}}_{Ky} & \frac{(r-g_2)^2}{r^2}\hat{\Omega}_{Ky,(\mathrm{II})} \end{pmatrix}.\end{aligned}$$

Thus $\hat{\Omega}_{K,(\mathrm{I})} = \frac{(r-g_1)^2}{r^2}\hat{\Omega}_{Ky,(\mathrm{I})}$, $\hat{\Omega}_{K,(\mathrm{II})} = \frac{(r-g_2)^2}{r^2}\hat{\Omega}_{Ky,(\mathrm{II})}$ and $\hat{\mathcal{C}}_K = \frac{(r-g_1)(r-g_2)}{r^2}\hat{\mathcal{C}}_{Ky}$. Define $\hat{\boldsymbol{\Omega}}^{\S}_{Ky} \equiv \big(\hat{\Omega}_{Ky,(\mathrm{I})} - \hat{\mathcal{C}}^2_{Ky}/\hat{\Omega}_{Ky,(\mathrm{II})}\big)\boldsymbol{E}_{11} = \boldsymbol{E}_{11}\hat{\boldsymbol{\Omega}}_{Ky}\boldsymbol{E}_{11} - \frac{\hat{\mathcal{C}}^2_{Ky}}{\hat{\Omega}_{Kyt,(\mathrm{II})}}\boldsymbol{E}_{11}$, where $\boldsymbol{E}_{11} = \left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)$, $\boldsymbol{E}_{12} = \left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$ are matrix units with the $(1,1)$-th and $(1,2)$-th element being 1 and all other elements zero. Then

$$\begin{aligned}\frac{1}{r^2}(r\boldsymbol{I}-\boldsymbol{G})\hat{\boldsymbol{\Omega}}^{\S}_{Ky}(r\boldsymbol{I}-\boldsymbol{G})' &= \frac{(r-g_1)^2}{r^2}\left(\hat{\Omega}_{Ky,(\mathrm{I})} - \frac{\hat{\mathcal{C}}^2_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}}\right)\boldsymbol{E}_{11} \\ &= \left(\hat{\Omega}_{K,(\mathrm{I})} - \frac{\hat{\mathcal{C}}^2_K}{\hat{\Omega}_{K,(\mathrm{II})}}\right)\boldsymbol{E}_{11} = \hat{\boldsymbol{\Omega}}^{\S}_K.\end{aligned}$$

Therefore,

$$\begin{aligned}&\int_k \boldsymbol{q}_k\, dk = \frac{1}{\rho}\hat{\boldsymbol{\Omega}}^{\S}_K \int_k \hat{\boldsymbol{\mu}}_k\, dk \\ &= \frac{1}{\rho r^2}(r\boldsymbol{I}-\boldsymbol{G})\hat{\boldsymbol{\Omega}}^{\S}_{Ky}(r\boldsymbol{I}-\boldsymbol{G})' \\ &\times \int_k \boldsymbol{\mu}_v + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\hat{\boldsymbol{\mu}}_{ky} - \boldsymbol{p}_t r\, dk \\ &= \frac{\lambda_K}{\rho r^2}(r\boldsymbol{I}-\boldsymbol{G})\hat{\boldsymbol{\Omega}}^{\S}_{Ky}(r\boldsymbol{I}-\boldsymbol{G})'[\boldsymbol{\mu}_v + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} - \boldsymbol{p}_t r] \\ &+ \frac{\lambda_K}{\rho r}(r\boldsymbol{I}-\boldsymbol{G})\hat{\boldsymbol{\Omega}}^{\S}_{Ky}\hat{\boldsymbol{\Sigma}}_{Ky}[\boldsymbol{\Omega}_p \boldsymbol{s}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e \widetilde{\boldsymbol{y}}].\end{aligned} \tag{IA.12}$$

The following identity holds:

$$\begin{aligned}\hat{\boldsymbol{\Omega}}^{\S}_{Ky}\hat{\boldsymbol{\Sigma}}_{Ky} &= \begin{pmatrix} \hat{\Omega}_{Ky,(\mathrm{I})} - \frac{\hat{\mathcal{C}}^2_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}} & 0 \\ 0 & 0 \end{pmatrix} \times \frac{1}{\hat{\Omega}_{Ky,(\mathrm{I})}\hat{\Omega}_{Ky,(\mathrm{II})} - \hat{\mathcal{C}}^2_{Ky}} \\ &\quad \times \begin{pmatrix} \hat{\Omega}_{Ky,(\mathrm{II})} & -\hat{\mathcal{C}}_{Ky} \\ -\hat{\mathcal{C}}_{Ky} & \hat{\Omega}_{Ky,(\mathrm{I})} \end{pmatrix} = \begin{pmatrix} 1 & -\frac{\hat{\mathcal{C}}_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}} \\ 0 & 0 \end{pmatrix} \\ &= \boldsymbol{E}_{11} - \frac{\hat{\mathcal{C}}_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12}.\end{aligned}$$

Substituting (IA.1), (IA.10), (IA.11), and (IA.12) into (IA.9) yields

$$\begin{aligned}\rho(\bar{\boldsymbol{x}} + \widetilde{\boldsymbol{x}}) &= \frac{1}{r^2}(r\boldsymbol{I} - \boldsymbol{G})\boldsymbol{\Lambda}\bar{\hat{\boldsymbol{\Omega}}}_y(r\boldsymbol{I} - \boldsymbol{G})'[\boldsymbol{\mu}_v + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} - \boldsymbol{p}_t r] \\ &\quad + \frac{1}{r}(r\boldsymbol{I} - \boldsymbol{G})\left(\boldsymbol{\Lambda} - \lambda_K \frac{\hat{\mathcal{C}}_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12}\right)\boldsymbol{\Omega}_p\boldsymbol{C}^{-1}(\boldsymbol{p}_t - \boldsymbol{A} - \boldsymbol{B}\boldsymbol{d}) \\ &\quad + \frac{1}{r}(r\boldsymbol{I} - \boldsymbol{G})[(\lambda_I\varphi^2\boldsymbol{M}_I + \lambda_K\hat{\boldsymbol{\Omega}}^{\S}_{Ky}\hat{\boldsymbol{\Sigma}}_{Ky}\boldsymbol{M}_K)\boldsymbol{\Pi}_e + \lambda_1\boldsymbol{\Omega}_\epsilon]\widetilde{\boldsymbol{y}},\end{aligned}$$

where $\lambda_1 = \lambda_I + \lambda_J$, $\lambda_2 = \lambda_I + \lambda_J + \lambda_K$, $\boldsymbol{\Lambda} = \left(\begin{smallmatrix} \lambda_2 & 0 \\ 0 & \lambda_1 \end{smallmatrix}\right)$, and $\bar{\hat{\boldsymbol{\Omega}}}_y \equiv \boldsymbol{\Lambda}^{-1}(\lambda_I\hat{\boldsymbol{\Omega}}_{Iy} + \lambda_J\hat{\boldsymbol{\Omega}}_{Jy} + \lambda_K\hat{\boldsymbol{\Omega}}^{\S}_{Ky})$ is the eligibility-weighted precision aggregate entering market clearing. Because eligibility differs by asset, this aggregate need not be symmetric under partial access and should not be interpreted as an individual Bayesian precision matrix. We write $\bar{\hat{\boldsymbol{\Sigma}}}_y \equiv \bar{\hat{\boldsymbol{\Omega}}}_y^{-1}$ for its algebraic inverse. Its diagonal entries are the asset-specific average posterior precisions among investors eligible to hold each asset. Collecting the $\boldsymbol{p}_t$ terms on the left-hand side yields

$$\begin{aligned}&\left[\boldsymbol{I} - (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y\left(\boldsymbol{I} - \frac{\lambda_K\hat{\mathcal{C}}_{Ky}}{\lambda_2\hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12}\right)\boldsymbol{\Omega}_p\boldsymbol{C}^{-1}\right]\boldsymbol{p}_t \\ &= \frac{1}{r}[\boldsymbol{\mu}_v + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d}] \\ &\quad - (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y\left(\boldsymbol{I} - \frac{\lambda_K\hat{\mathcal{C}}_{Ky}}{\lambda_2\hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12}\right)\boldsymbol{\Omega}_p\boldsymbol{C}^{-1}(\boldsymbol{A} + \boldsymbol{B}\boldsymbol{d}) \\ &\quad + (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Lambda}^{-1}[(\lambda_I\varphi^2\boldsymbol{M}_I + \lambda_K\hat{\boldsymbol{\Omega}}^{\S}_{Ky}\hat{\boldsymbol{\Sigma}}_{Ky}\boldsymbol{M}_K)\boldsymbol{\Pi}_e + \lambda_1\boldsymbol{\Omega}_\epsilon]\widetilde{\boldsymbol{y}} \\ &\quad - \rho r(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y\boldsymbol{\Lambda}^{-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}(\bar{\boldsymbol{x}} + \widetilde{\boldsymbol{x}}),\end{aligned}$$

We then compare with $\boldsymbol{p}_t = \boldsymbol{A} + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{C}\widetilde{\boldsymbol{y}} + \boldsymbol{D}\widetilde{\boldsymbol{x}}$ to determine the coefficients.

$\underline{\boldsymbol{A}}$: Collecting the constant terms yields

$$\begin{aligned}\boldsymbol{A} = &\left[\boldsymbol{I} - (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \left(\boldsymbol{I} - \frac{\lambda_K \hat{\mathcal{C}}_{Ky}}{\lambda_2 \hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12}\right)\boldsymbol{\Omega}_p \boldsymbol{C}^{-1}\right]^{-1} \times \\ &\left[\frac{1}{r}\boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \left(\boldsymbol{I} - \frac{\lambda_K \hat{\mathcal{C}}_{Ky}}{\lambda_2 \hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12}\right)\boldsymbol{\Omega}_p \boldsymbol{C}^{-1}\boldsymbol{A}\right. \\ &\left. - \rho r (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \boldsymbol{\Lambda}^{-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\bar{\boldsymbol{x}}\right].\end{aligned}$$

Denote $\boldsymbol{H} = (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \left(\boldsymbol{I} - (\lambda_K \hat{\mathcal{C}}_{Ky}/(\lambda_2 \hat{\Omega}_{Ky,(\mathrm{II})}))\boldsymbol{E}_{12}\right)\boldsymbol{\Omega}_p \boldsymbol{C}^{-1}$ for readability. Moving all the $\boldsymbol{A}$ terms to the left-hand side, we have

$$\begin{aligned}\boldsymbol{A} = &\left[\boldsymbol{I} + (\boldsymbol{I} - \boldsymbol{H})^{-1}\boldsymbol{H}\right]^{-1}[\boldsymbol{I} - \boldsymbol{H}]^{-1}\left[\frac{1}{r}\boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu}\right. \\ &\left. - \frac{\rho r}{\lambda_2}(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y (r\boldsymbol{I} - \boldsymbol{G})^{-1}\bar{\boldsymbol{x}}\right].\end{aligned}$$

Note that

$$\begin{aligned}\left[\boldsymbol{I} + (\boldsymbol{I} - \boldsymbol{H})^{-1}\boldsymbol{H}\right]^{-1}[\boldsymbol{I} - \boldsymbol{H}]^{-1} = \boldsymbol{I} &\iff \left[\boldsymbol{I} + (\boldsymbol{I} - \boldsymbol{H})^{-1}\boldsymbol{H}\right]^{-1} = \boldsymbol{I} - \boldsymbol{H} \\ &\iff \boldsymbol{I} = \boldsymbol{I} - \boldsymbol{H} + \boldsymbol{H} = \boldsymbol{I}.\end{aligned}$$

Therefore,

$$\boldsymbol{A} = \frac{1}{r}\boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - \rho r (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \boldsymbol{\Lambda}^{-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\bar{\boldsymbol{x}}.$$

$\underline{\boldsymbol{B}}$: Collecting the terms in $\boldsymbol{d}$ yields

$$\boldsymbol{B} = [\boldsymbol{I} - \boldsymbol{H}]^{-1}\left[(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G} - \boldsymbol{H}\boldsymbol{B}\right].$$

Moving all $\boldsymbol{B}$ to the left-hand side, we have

$$\begin{aligned}\boldsymbol{B} &= \left[\boldsymbol{I} + (\boldsymbol{I} - \boldsymbol{H})^{-1}\boldsymbol{H}\right]^{-1}[\boldsymbol{I} - \boldsymbol{H}]^{-1}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G} \\ &= (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}.\end{aligned}$$

$\underline{\boldsymbol{C}}$: The coefficient on $\widetilde{\boldsymbol{y}}$ is

$$\boldsymbol{C} = \left[\boldsymbol{I} - (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \left(\boldsymbol{I} - \frac{\lambda_K \hat{\mathcal{C}}_{Ky}}{\lambda_2 \hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{12}\right) \boldsymbol{\Omega}_p \boldsymbol{C}^{-1}\right]^{-1} \times$$
$$(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \boldsymbol{\Lambda}^{-1} \left[\left(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \left(\boldsymbol{E}_{11} - \frac{\hat{\mathcal{C}}_{Ky}}{\hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{12}\right) \boldsymbol{M}_K\right) \boldsymbol{\Pi}_e + \lambda_1 \boldsymbol{\Omega}_\epsilon\right].$$

Left multiplying $\left[\boldsymbol{I} - (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \left(\boldsymbol{I} - (\lambda_K \hat{\mathcal{C}}_{Ky}/(\lambda_2 \hat{\Omega}_{Ky,(\text{II})}))\boldsymbol{E}_{12}\right) \boldsymbol{\Omega}_p \boldsymbol{C}^{-1}\right]$ on both sides leads to

$$\begin{aligned}
\boldsymbol{C} = {} & (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \times \\
& \boldsymbol{\Lambda}^{-1}\Big[\Big(\boldsymbol{\Lambda} - \lambda_K \frac{\hat{\mathcal{C}}_{Ky}}{\hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{12}\Big)\boldsymbol{\Omega}_p \\
& \quad + \Big(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \Big(\boldsymbol{E}_{11} - \frac{\hat{\mathcal{C}}_{Ky}}{\hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{12}\Big) \boldsymbol{M}_K\Big)\boldsymbol{\Pi}_e + \lambda_1 \boldsymbol{\Omega}_\epsilon\Big].
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\bar{\hat{\boldsymbol{\Omega}}}_y &= \boldsymbol{\Lambda}^{-1}\left(\lambda_I \hat{\boldsymbol{\Omega}}_{Iy} + \lambda_J \hat{\boldsymbol{\Omega}}_{Jy} + \lambda_K \hat{\boldsymbol{\Omega}}^{\S}_{Ky}\right) \\
&= \boldsymbol{\Lambda}^{-1}\Bigg[\lambda_I(\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \varphi^2 \boldsymbol{M}_I \boldsymbol{\Pi}_e + \boldsymbol{\Omega}_\epsilon) + \lambda_J(\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \boldsymbol{\Omega}_\epsilon) \\
&\quad + \lambda_K \left(\boldsymbol{E}_{11}(\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e)\boldsymbol{E}_{11} - \frac{\hat{\mathcal{C}}^2_{Ky}}{\hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{11}\right)\Bigg] \\
&= \boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p - \frac{\lambda_K}{\lambda_2}\left(\Omega_{y,12} + \Omega_{p,12}\right)\boldsymbol{E}_{12} + \boldsymbol{\Lambda}^{-1}(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \boldsymbol{E}_{11}\boldsymbol{M}_K)\boldsymbol{\Pi}_e \\
&\quad + \boldsymbol{\Lambda}^{-1}\lambda_1 \boldsymbol{\Omega}_\epsilon - \frac{\lambda_K \hat{\mathcal{C}}^2_{Ky}}{\lambda_2 \hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{11}.
\end{aligned}$$

Accordingly, the two asset-specific eligible-investor average posterior precisions are

$$\begin{aligned}
\bar{\hat{\Omega}}_{y,(\text{I})} &= \Omega_{y,(\text{I})} + \hat{\Omega}_{p,(\text{I})} + \frac{\lambda_I \varphi^2 M_{I,(\text{I})} + \lambda_K M_{K,(\text{I})}}{\lambda_2}\Pi_{e,(\text{I})} + \frac{\lambda_1}{\lambda_2}\Omega_{\epsilon,(\text{I})} - \frac{\lambda_K}{\lambda_2} z^2 \hat{\Omega}_{Kyt,(\text{II})}, \\
\bar{\hat{\Omega}}_{y,(\text{II})} &= \Omega_{y,(\text{II})} + \hat{\Omega}_{p,(\text{II})} + \frac{\lambda_I}{\lambda_1}\varphi^2 M_{I,(\text{II})}\Pi_{e,(\text{II})} + \Omega_{\epsilon,(\text{II})}.
\end{aligned}$$

Because $\boldsymbol{\Omega}_p = \boldsymbol{\zeta}'\boldsymbol{\Omega}_x\boldsymbol{\zeta}$ and the partial-access $\boldsymbol{\zeta}$ is upper triangular, the type-II price-signal preci-

sion is

$$\hat{\Omega}_{p,(\text{II})} = \zeta_{t,(\text{II})}^2 \Omega_{x,(\text{II})} + (\zeta_{t,(\text{I}\leftarrow\text{II})})^2 \Omega_{x,(\text{I})}.$$

Thus foreign entry does not mechanically dilute type-II soft information, because foreign investors are not eligible to hold type-II. Their type-II information can nevertheless improve the type-II posterior through the cross-price signal. Thus $\boldsymbol{\Omega}_p + \boldsymbol{\Lambda}^{-1}(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \boldsymbol{E}_{11} \boldsymbol{M}_K)\boldsymbol{\Pi}_e + \boldsymbol{\Lambda}^{-1}\lambda_1 \boldsymbol{\Omega}_\epsilon = \bar{\hat{\boldsymbol{\Omega}}}_y - \boldsymbol{\Omega}_y + \frac{\lambda_K}{\lambda_2}(\Omega_{y,12} + \Omega_{p,12})\boldsymbol{E}_{12} + \frac{\lambda_K \hat{\mathcal{C}}_{Ky}^2}{\lambda_2 \hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{11}$. Plugging it back, then

$$\begin{aligned}
\boldsymbol{C} &= (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime} \bar{\hat{\boldsymbol{\Sigma}}}_y \Bigg[\bar{\hat{\boldsymbol{\Omega}}}_y - \boldsymbol{\Omega}_y + \frac{\lambda_K}{\lambda_2}\big(\Omega_{y,12} + \Omega_{p,12}\big)\boldsymbol{E}_{12} \\
&\qquad + \frac{\lambda_K \hat{\mathcal{C}}_{Ky}^2}{\lambda_2 \hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{11} - \frac{\lambda_K \hat{\mathcal{C}}_{Ky}}{\lambda_2 \hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{12}\big(\boldsymbol{\Omega}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e\big) \Bigg] \\
&= (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime} \bar{\hat{\boldsymbol{\Sigma}}}_y \Bigg[\boldsymbol{\Omega}_p + \boldsymbol{\Lambda}^{-1}\big(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \boldsymbol{E}_{11} \boldsymbol{M}_K\big)\boldsymbol{\Pi}_e \\
&\qquad + \boldsymbol{\Lambda}^{-1}\lambda_1 \boldsymbol{\Omega}_\epsilon - \frac{\lambda_K}{\lambda_2} z\, \boldsymbol{E}_{12}\big(\boldsymbol{\Omega}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e\big) \Bigg].
\end{aligned}$$

D: The signal-to-noise ratio (SNR) is

$$\boldsymbol{\zeta} \equiv \boldsymbol{D}^{-1}\boldsymbol{C} = -\frac{1}{\rho r}(r\boldsymbol{I} - \boldsymbol{G}) \left[\left(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \left(\boldsymbol{E}_{11} - \frac{\hat{\mathcal{C}}_{Ky}}{\hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{12} \right) \boldsymbol{M}_K \right) \boldsymbol{\Pi}_e + \lambda_1 \boldsymbol{\Omega}_\epsilon \right]. \tag{IA.13}$$

Thus $\boldsymbol{D}$ can be derived from $\boldsymbol{D} = \boldsymbol{C}\boldsymbol{\zeta}^{-1}$.

In sum, the price coefficients are

$$\boldsymbol{A} = \frac{1}{r}\boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - \rho r (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime} \bar{\hat{\boldsymbol{\Sigma}}}_y \boldsymbol{\Lambda}^{-1} (r\boldsymbol{I} - \boldsymbol{G})^{-1} \bar{\boldsymbol{x}}, \tag{IA.14}$$

$$\boldsymbol{B} = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}, \tag{IA.15}$$

$$\begin{aligned}
\boldsymbol{C} = (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime} \bar{\hat{\boldsymbol{\Sigma}}}_y \Big[&\bar{\hat{\boldsymbol{\Omega}}}_y - \boldsymbol{\Omega}_y + \frac{\lambda_K}{\lambda_2}(\Omega_{y,12} + \Omega_{p,12})\boldsymbol{E}_{12} + \frac{\lambda_K \hat{\mathcal{C}}_{Ky}^2}{\lambda_2 \hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{11} \\
&- \frac{\lambda_K \hat{\mathcal{C}}_{Ky}}{\lambda_2 \hat{\Omega}_{Ky,(\text{II})}} \boldsymbol{E}_{12}(\boldsymbol{\Omega}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e) \Big],
\end{aligned} \tag{IA.16}$$

$$\boldsymbol{D} = \boldsymbol{C}\boldsymbol{\zeta}^{-1}. \tag{IA.17}$$

IA.A.4 Optimal data choice

This section derives institutional investors' ex ante expected utility at the trading date and solves their data-choice problems: allocating $\bar{M}$ data points between foreign-investable and non-tradable firms to maximize that utility.

IA.A.4.1. Domestic institutional investors

IA.A.4.2. Foreign institutional investors

Foreign institutional investor k faces a slightly different data-choice problem. She may invest only in Type-I risky assets, and her optimal portfolio depends on the posterior mean and variance of that payoff. Information about type-II remains relevant because the two dividend innovations are correlated. Conditional on the equilibrium price signal, which an atomistic investor takes as given at the data-choice stage, her interim utility is

$$\begin{aligned}\mathbb{E}[-\exp\left(-\rho w_k^{+}\right) \mid \mathcal{I}_k] &= -\exp\left(-\rho r w_k\right) \times \mathbb{E}\left[\exp\left(-\frac{\hat{\mu}_{kt,(\mathrm{I})}}{\hat{\sigma}^2_{kt,(\mathrm{I})}}\left(\widetilde{d}_{t,(\mathrm{I})} + \widetilde{v}_{t,(\mathrm{I})} - p_{t,(\mathrm{I})} r\right)\right) \mid \mathcal{I}_k\right] \\ &= -\exp\left(-\rho r w_k\right) \times \exp\left(-\frac{\hat{\mu}^2_{kt,(\mathrm{I})}}{2\hat{\sigma}^2_{kt,(\mathrm{I})}}\right).\end{aligned}$$

The payoff covariance and precision transformations are, respectively,

$$\begin{aligned}\hat{\boldsymbol{\Sigma}}_K &= r^2(r\boldsymbol{I} - \boldsymbol{G})^{-1}\hat{\boldsymbol{\Sigma}}_{Ky}(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}, \\ \hat{\boldsymbol{\Omega}}_K &= \frac{1}{r^2}(r\boldsymbol{I} - \boldsymbol{G})'\hat{\boldsymbol{\Omega}}_{Ky}(r\boldsymbol{I} - \boldsymbol{G}),\end{aligned}$$

where $\hat{\boldsymbol{\Sigma}}_{Ky} = (\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \boldsymbol{M}_K\boldsymbol{\Pi}_e)^{-1}$ and $\hat{\boldsymbol{\Omega}}_{Ky} = \hat{\boldsymbol{\Sigma}}^{-1}_{Ky}$. Since $\boldsymbol{G}$ is diagonal,

$$\hat{\sigma}^2_{kt,(\mathrm{I})} = \left(\frac{r}{r - g_1}\right)^2 \hat{\sigma}^2_{kyt,(\mathrm{I})},$$

so minimizing the posterior variance of the tradable payoff is equivalent to minimizing the posterior variance of its dividend innovation. Taking expectations over the interim posterior mean

gives

$$\begin{aligned}\mathbb{E}[-\exp\left(-\rho w_k^+\right) \mid \mathcal{I}_{kt^-}] &= -\exp\left(-\rho r w_k\right) \times \mathbb{E}\left[\exp\left(-\frac{\hat{\mu}_{kt,(\mathrm{I})}^2}{2\hat{\sigma}_{kt,(\mathrm{I})}^2}\right) \mid \mathcal{I}_{kt^-}\right] \\ &= -\exp\left(-\rho r w_k\right) \times \left|\sigma_{Kt,(\mathrm{I})}^2/\hat{\sigma}_{kt,(\mathrm{I})}^2\right|^{-\frac{1}{2}} \times \exp\left(-\frac{\mu_{Kt,(\mathrm{I})}^2}{2\sigma_{Kt,(\mathrm{I})}^2}\right).\end{aligned}$$

Hence the foreign data-choice problem reduces to minimizing $\hat{\sigma}_{kyt,(\mathrm{I})}^2$. Writing

$$\hat{\mathbf{\Omega}}_{kyt} = \begin{pmatrix} \hat{\Omega}_{kyt,(\mathrm{I})} & \hat{\mathcal{C}}_{kyt} \\ \hat{\mathcal{C}}_{kyt} & \hat{\Omega}_{kyt,(\mathrm{II})} \end{pmatrix},$$

the inverse of this 2×2 precision matrix gives

$$\hat{\sigma}_{kyt,(\mathrm{I})}^2 = \frac{\hat{\Omega}_{kyt,(\mathrm{II})}}{\hat{\Omega}_{kyt,(\mathrm{I})}\hat{\Omega}_{kyt,(\mathrm{II})} - \hat{\mathcal{C}}_{kyt}^2},$$

and therefore

$$\min \hat{\sigma}_{kyt,(\mathrm{I})}^2 \iff \max \hat{\Omega}_{kyt,(\mathrm{I})}^S, \qquad \hat{\Omega}_{kyt,(\mathrm{I})}^S \equiv \hat{\Omega}_{kyt,(\mathrm{I})} - \frac{\hat{\mathcal{C}}_{kyt}^2}{\hat{\Omega}_{kyt,(\mathrm{II})}}.$$

This is the Type-I Schur-complement precision stated in Lemma 2. Recall that

$$\hat{\mathbf{\Omega}}_{kyt} = \begin{pmatrix} \Omega_{y,(\mathrm{I})} + \hat{\Omega}_{p,(\mathrm{I})} + M_{kt,(\mathrm{I})}\Pi_{e,(\mathrm{I})} & \mathcal{C}_y + \hat{\mathcal{C}}_p \\ \mathcal{C}_y + \hat{\mathcal{C}}_p & \Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + M_{kt,(\mathrm{II})}\Pi_{e,(\mathrm{II})} \end{pmatrix}.$$

Therefore the data choice problem finally reduces to:

$$\max_{\{M_{kt,(\mathrm{I})}, M_{kt,(\mathrm{II})}\}} \quad M_{kt,(\mathrm{I})}\Pi_{e,(\mathrm{I})} - \frac{(\mathcal{C}_y + \hat{\mathcal{C}}_p)^2}{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + M_{kt,(\mathrm{II})}\Pi_{e,(\mathrm{II})}} \tag{IA.18}$$

$$\text{s.t.} \quad M_{kt,(\mathrm{I})} + M_{kt,(\mathrm{II})} \leq \bar{M}. \tag{IA.19}$$

Again the data constraint is satisfied with equality. Substituting $M_{kt,(\mathrm{I})} + M_{kt,(\mathrm{II})} = \bar{M}$ into the objective function, we have $obj = (\bar{M} - M_{kt,(\mathrm{II})})\Pi_{e,(\mathrm{I})} - (\mathcal{C}_y + \hat{\mathcal{C}}_p)^2/(\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + M_{kt,(\mathrm{II})}\Pi_{e,(\mathrm{II})})$.

The interior F.O.C.

$$\frac{\partial obj}{\partial M_{kt,(\mathrm{II})}} = -\Pi_{e,(\mathrm{I})} + (\mathcal{C}_y + \hat{\mathcal{C}}_p)^2 \frac{\Pi_{e,(\mathrm{II})}}{(\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + M_{kt,(\mathrm{II})}\Pi_{e,(\mathrm{II})})^2} = 0.$$

Intuitively, the marginal gain in Type-I posterior precision from directly processing Type-I data equals the marginal gain in Type-I Schur-complement precision obtained indirectly from processing Type-II data. Conditional on the equilibrium price signal, let

$$M \equiv M_{kt,(\mathrm{II})}, \qquad z(M) \equiv \frac{\mathcal{C}_y + \hat{\mathcal{C}}_p}{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + M\Pi_{e,(\mathrm{II})}}.$$

The information spillover rate is

$$\kappa(M) \equiv z(M)^2 \frac{\Pi_{e,(\mathrm{II})}}{\Pi_{e,(\mathrm{I})}}.$$

Holding the cross-precision term fixed,

$$\frac{\partial \hat{\Omega}^S_{kyt,(\mathrm{I})}}{\partial \hat{\Omega}_{kyt,(\mathrm{II})}} = z(M)^2,$$

so $z(M)^2$ is the local cross-asset conversion factor from one additional unit of Type-II posterior precision into Type-I Schur-complement precision. Because $\Pi_{e,(N)} = \beta^2 \Omega_{e,(N)}$ is the effective precision contributed by one correctly processed Type-N observation, one additional Type-II observation therefore contributes $z(M)^2 \Pi_{e,(\mathrm{II})}$ units of Type-I precision indirectly, whereas one additional Type-I observation contributes $\Pi_{e,(\mathrm{I})}$ units directly. Thus $\kappa(M)$ is the ratio of these two marginal information values. Under the binding processing-capacity constraint,

$$\frac{d\, obj}{dM} = \Pi_{e,(\mathrm{I})} \big[\kappa(M) - 1\big],$$

so $\kappa(M) = 1$ is exactly the interior indifference condition.

Interior solution and uniqueness. Holding the equilibrium price signal fixed, define

$$A \equiv \Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} > 0, \qquad b \equiv \Pi_{e,(\mathrm{II})} > 0.$$

The relevant normalized cross-precision is

$$z(M) = \frac{\hat{\mathcal{C}}_{Kyt}}{A + bM},$$

so the spillover rate can be written as

$$\kappa(M) = \frac{\hat{\mathcal{C}}_{Kyt}^2 \Pi_{e,(\mathrm{II})}/\Pi_{e,(\mathrm{I})}}{(A+bM)^2} = \kappa(0)\left(\frac{A}{A+bM}\right)^2.$$

Therefore, whenever $\hat{\mathcal{C}}_{Kyt} \neq 0$,

$$\kappa'(M) = -\frac{2b\,\hat{\mathcal{C}}_{Kyt}^2 \Pi_{e,(\mathrm{II})}/\Pi_{e,(\mathrm{I})}}{(A+bM)^3} < 0, \qquad \kappa''(M) = \frac{6b^2\,\hat{\mathcal{C}}_{Kyt}^2 \Pi_{e,(\mathrm{II})}/\Pi_{e,(\mathrm{I})}}{(A+bM)^4} > 0.$$

Hence $\kappa(M)$ is continuous, strictly decreasing, and convex in M, holding the equilibrium price signal fixed; if $\hat{\mathcal{C}}_{Kyt} = 0$, then $\kappa(M) \equiv 0$. This conditional monotonicity is a property of the individual-choice problem, not a comparative-static statement across rational-expectations equilibria. Across equilibria, $\hat{\mathcal{C}}_p$ and $\hat{\Omega}_{p,(\mathrm{II})}$ vary with aggregate information choices, so z and κ are endogenous summaries of the equilibrium information environment. Under $\kappa(0) > 1 > \kappa(\bar{M})$, continuity implies the existence of an $M^{\text{int}} \in (0, \bar{M})$ satisfying $\kappa(M^{\text{int}}) = 1$, and strict monotonicity makes this solution unique. Solving the indifference condition gives

$$M^{\text{int}} = \frac{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})}}{\Pi_{e,(\mathrm{II})}}\left[\sqrt{\kappa(0)} - 1\right]. \tag{IA.20}$$

The first multiplicative factor, $[\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})}]/[\Pi_{e,(\mathrm{II})}]$, is the pre-existing Type-II information stock measured in equivalent units of processed Type-II data. It scales the additional Type-II processing required to bring the spillover rate down to the interior indifference threshold. The second factor, $\sqrt{\kappa(0)} - 1$, measures the strength of the initial information spillover relative to the threshold $\kappa = 1$: the farther the initial spillover benefit lies above that threshold, the more Type-II data must be processed before the direct and indirect marginal learning benefits are equalized. Because $\hat{\Omega}_{p,(\mathrm{II})}$ is an equilibrium object, equation (IA.20) is a closed-form characterization of the foreign investor's optimal allocation conditional on the equilibrium price-information environment; it is not, by itself, a uniqueness proof for the full rational-expectations equilibrium.

In sum, the foreign Type-II data choice is

$$M^*_{kt,(\mathrm{II})} = \begin{cases} 0, & \kappa(0) \le 1, \\ \bar{M}, & \kappa(\bar{M}) \ge 1, \\ M^{\text{int}}, & \kappa(0) > 1 > \kappa(\bar{M}). \end{cases}$$

Proof of Corollary 6. If $\mathcal{C}_y = 0$, the equilibrium equation for the foreign cross-precision has zero constant term. For $M_{kt,(\mathrm{II})} > 0$, its cubic representation is $z^3 + pz + q = 0$, where $p > 0$ and $q = -\mathcal{C}_y/D$ for $D > 0$. Hence $q = 0$ and $z(z^2 + p) = 0$, whose unique real root is $z = 0$. At the boundary $M_{kt,(\mathrm{II})} = 0$, the same equilibrium equation is linear and gives $z = \mathcal{C}_y/D_0 = 0$ for $D_0 > 0$. Therefore $z = 0$ at every feasible foreign Type-II allocation, so Definition 4 implies $\kappa(M) = 0$ for all $M \in [0, \bar{M}]$. In particular, $\kappa(0) = 0 \le 1$, and equation (10) yields $M^*_{kt,(\mathrm{II})} = 0$. Because the processing-capacity constraint binds, $M^*_{kt,(\mathrm{I})} = \bar{M}$. □

Intermediate-precision cross-asset learning. Let

$$E \equiv \Pi_{e,(\mathrm{II})}.$$

This is an equilibrium comparative static: as E changes, domestic and foreign information choices and the equilibrium price signal change as well. The foreign corner rule is an individual-choice condition evaluated at a given price signal, but the high-E result depends on the equilibrium response of Type-II price precision. If the price signal and a nonzero z were instead held fixed, $\kappa_E(0) = z^2 E/\Pi_{e,(\mathrm{I})}$ would increase linearly in E, so the high-precision zero-allocation result would not follow. If β is held fixed, the same equilibrium experiment can be implemented by varying primitive raw-data precision $\Omega_{e,(\mathrm{II})} = E/\beta^2$. Proposition 2 combines the two zero-learning tails with a primitive sufficient condition for a nonempty positive-learning region between them.

Define

$$\chi_2 \equiv \left(\frac{r - g_2}{\rho r}\right)^2 \Omega_{x,(\mathrm{II})}, \qquad d \equiv \lambda_1 \Omega_{\epsilon,(\mathrm{II})}, \qquad B_{\mathrm{IP}} \equiv \lambda_I \varphi^2 \bar{M},$$

and the primitive lower bound

$$\underline{\kappa}(E) \equiv \frac{\mathcal{C}_y^2 E}{\Pi_{e,(\mathrm{I})} \left[\Omega_{y,(\mathrm{II})} + \chi_2 (d + B_{\mathrm{IP}} E)^2\right]^2}. \tag{IA.21}$$

This lower bound is single-peaked, with unique maximizer

$$E^\dagger = \frac{\sqrt{4d^2 + 3\Omega_{y,(\mathrm{II})}/\chi_2} - d}{3B_{\mathrm{IP}}}. \tag{IA.22}$$

The primitive sufficient condition used in Proposition 2 is

$$\underline{\kappa}(E^{\dagger}) > 1. \tag{IA.23}$$

Proof of Corollary 8. At the all-Type-I foreign-processing corner, $M_{K,(\mathrm{II})} = 0$, the cross-precision ratio is

$$z(0) = \frac{\mathcal{C}_y}{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})}}.$$

Hence Definition 4 gives

$$\kappa(0) = \frac{\mathcal{C}_y^2}{\left(\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})}\right)^2} \frac{\Pi_{e,(\mathrm{II})}}{\Pi_{e,(\mathrm{I})}}.$$

Because $\hat{\Omega}_{p,(\mathrm{II})} \geq 0$,

$$\kappa(0) \leq \frac{\mathcal{C}_y^2}{\Omega_{y,(\mathrm{II})}^2} \frac{\Pi_{e,(\mathrm{II})}}{\Pi_{e,(\mathrm{I})}}.$$

Therefore condition (IA.88) implies $\kappa(0) \leq 1$. By the foreign data-allocation corner rule, this yields

$$M^*_{K,(\mathrm{I})} = \bar{M}, \qquad M^*_{K,(\mathrm{II})} = 0.$$

The primitive upper bound contains no domestic soft-information precision, so this sufficient guarantee of the all-Type-I corner is invariant to that precision. The condition is conservative: when equilibrium Type-II price precision is strictly positive, the actual spillover rate lies strictly below the primitive upper bound. □

In equilibrium, the data choices for domestic and foreign institutional investors are determined simultaneously. They solve the following system of non-linear equations:

$$M_{I,(\mathrm{I})} = \begin{cases} 0, & \text{if } (\Omega_{y,(\mathrm{I})} + \hat{\Omega}_{p,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})})/(\varphi^2 \Pi_{e,(\mathrm{I})}) \geq (\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + \bar{M}\varphi^2 \Pi_{e,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})})/(\varphi^2 \Pi_{e,(\mathrm{II})}), \\ \bar{M}, & \text{if } (\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})})/(\varphi^2 \Pi_{e,(\mathrm{II})}) \geq (\Omega_{y,(\mathrm{I})} + \hat{\Omega}_{p,(\mathrm{I})} + \bar{M}\varphi^2 \Pi_{e,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})})/(\varphi^2 \Pi_{e,(\mathrm{I})}), \\ \frac{1}{2}\Big[\dfrac{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})}}{\varphi^2 \Pi_{e,(\mathrm{II})}} + \bar{M} & \\ \quad - \dfrac{\Omega_{y,(\mathrm{I})} + \hat{\Omega}_{p,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})}}{\varphi^2 \Pi_{e,(\mathrm{I})}}\Big] & \text{otherwise.} \end{cases}$$

$$M_{K,(\mathrm{II})} = \begin{cases} 0, & \text{if } \kappa(0) \leq 1, \\ \bar{M}, & \text{if } \kappa(\bar{M}) \geq 1, \\ M^{\mathrm{int}}, & \text{if } \kappa(0) > 1 > \kappa(\bar{M}). \end{cases}$$

where

$$\hat{\boldsymbol{\Omega}}_{p2} = \boldsymbol{\zeta}_2' \boldsymbol{\Omega}_x \boldsymbol{\zeta}_2,$$
$$\boldsymbol{\zeta}_2 = -\frac{1}{\rho r}(r\boldsymbol{I} - \boldsymbol{G}) \left[\left(\lambda_I \varphi^2 \boldsymbol{M}_{I2} + \lambda_K \left(\boldsymbol{E}_{11} - \frac{\hat{\mathcal{C}}_{Ky2}}{\hat{\Omega}_{Ky,(\mathrm{II})}} \boldsymbol{E}_{12} \right) \boldsymbol{M}_{K2} \right) \boldsymbol{\Pi}_e + \lambda_1 \boldsymbol{\Omega}_\epsilon \right].$$

Recall that

$$\begin{aligned}
\hat{\Omega}_{Ky,(\mathrm{II})} &= \Omega_{y,(\mathrm{II})} + M^{\text{int}} \Pi_{e,(\mathrm{II})} + \left[\frac{r - g_1}{\rho r} \lambda_K z M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} \right]^2 \Omega_{x,(\mathrm{I})} \\
&\quad + \left[\frac{r - g_2}{\rho r} \left[\lambda_I \varphi^2 M_{I,(\mathrm{II})} \Pi_{e,(\mathrm{II})} + \lambda_1 \Omega_{\epsilon,(\mathrm{II})} \right] \right]^2 \Omega_{x,(\mathrm{II})}, \\
\hat{\mathcal{C}}_{Ky2} &= \mathcal{C}_y - \left(\frac{r - g_1}{\rho r} \right)^2 \lambda_K z M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} \\
&\quad \times \left[(\lambda_I \varphi^2 M_{I,(\mathrm{I})} + \lambda_K M_{K,(\mathrm{I})}) \Pi_{e,(\mathrm{I})} + \lambda_1 \Omega_{\epsilon,(\mathrm{I})} \right] \Omega_{x,(\mathrm{I})}.
\end{aligned}$$

First, express $\hat{\Omega}_{Ky,(\mathrm{II})}$ and $\hat{\mathcal{C}}_{Ky2}$ in terms of model parameters and data choices. Let $z = \hat{\mathcal{C}}_{Ky2} / \hat{\Omega}_{Ky,(\mathrm{II})}$. From here we can derive a cubic equation for z:

$$\begin{aligned}
&\mathcal{C}_y - \left(\frac{r - g_1}{\rho r} \right)^2 \lambda_K z M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} \\
&\quad \times \left[(\lambda_I \varphi^2 M_{I,(\mathrm{I})} + \lambda_K M_{K,(\mathrm{I})}) \Pi_{e,(\mathrm{I})} + \lambda_1 \Omega_{\epsilon,(\mathrm{I})} \right] \Omega_{x,(\mathrm{I})} \\
&= z \Bigg\{ \Omega_{y,(\mathrm{II})} + M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} + \left[\frac{r - g_1}{\rho r} \lambda_K z M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} \right]^2 \Omega_{x,(\mathrm{I})} \\
&\quad + \left[\frac{r - g_2}{\rho r} \left[\lambda_I \varphi^2 M_{I,(\mathrm{II})} \Pi_{e,(\mathrm{II})} + \lambda_1 \Omega_{\epsilon,(\mathrm{II})} \right] \right]^2 \Omega_{x,(\mathrm{II})} \Bigg\}.
\end{aligned}$$

If foreign institutional investors do not learn about type-II risky asset, i.e. $M_{K,(\mathrm{II})} = 0$, then the above equation reduces to a linear equation of z, with

$$z = \frac{\mathcal{C}_y}{\Omega_{y,(\mathrm{II})} + \left[\frac{r - g_2}{\rho r} \left[\lambda_I \varphi^2 M_{I,(\mathrm{II})} \Pi_{e,(\mathrm{II})} + \lambda_1 \Omega_{\epsilon,(\mathrm{II})} \right] \right]^2 \Omega_{x,(\mathrm{II})}}.$$

When $M_{K,(\text{II})} > 0$, to write it in the standard form $z^3 + pz + q = 0$, we have

$$
\begin{aligned}
p = &\left\{ \Omega_{y,(\text{II})} + M^{\text{int}} \Pi_{e,(\text{II})} + \left[\frac{r - g_2}{\rho r} \left[\lambda_I \varphi^2 M_{I,(\text{II})} \Pi_{e,(\text{II})} + \lambda_1 \Omega_{\epsilon,(\text{II})} \right] \right]^2 \Omega_{x,(\text{II})} \right. \\
&\left. + \left(\frac{r - g_1}{\rho r} \right)^2 \lambda_K M_{K,(\text{II})} \Pi_{e,(\text{II})} \left[(\lambda_I \varphi^2 M_{I,(\text{I})} + \lambda_K M_{K,(\text{I})}) \Pi_{e,(\text{I})} + \lambda_1 \Omega_{\epsilon,(\text{I})} \right] \Omega_{x,(\text{I})} \right\} \\
&/ \left\{ \left[\frac{r - g_1}{\rho r} \lambda_K M_{K,(\text{II})} \Pi_{e,(\text{II})} \right]^2 \Omega_{x,(\text{I})} \right\} > 0, \\
q = &-\mathcal{C}_y / \left\{ \left[\frac{r - g_1}{\rho r} \lambda_K M_{K,(\text{II})} \Pi_{e,(\text{II})} \right]^2 \Omega_{x,(\text{I})} \right\},
\end{aligned}
$$

which are only functions of model parameters and data choices. Using Cardano's results, we know that $q^2/4 + p^3/27 > 0$, which implies that this cubic equation only has one real root:

$$
\frac{\hat{\mathcal{C}}_{Ky2}}{\hat{\Omega}_{Ky,(\text{II})}} = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}.
$$

For notational simplicity, let us denote the solution by $z(M_{I,(\text{I})}, M_{K,(\text{II})})$. Under the maintained condition $\mathcal{C}_y < 0$, this solution is negative. Indeed, after multiplication by its positive denominator, the cubic has the form $az^3 + bz - \mathcal{C}_y = 0$ with $a, b > 0$; its left-hand side is strictly increasing, is positive at zero, and tends to minus infinity as z tends to minus infinity. Moreover, the preceding cross-precision equation implies $\hat{\mathcal{C}}_{Ky2} = \mathcal{C}_y - Kz$ for $K \geq 0$, whereas $\hat{\mathcal{C}}_{Ky2} = z\hat{\Omega}_{Ky,(\text{II})} < 0$. Hence

$$
\mathcal{C}_y \leq \hat{\mathcal{C}}_{Ky2} < 0, \qquad \left| \hat{\mathcal{C}}_{Ky2} \right| \leq |\mathcal{C}_y|,
$$

with equality when $M_{K,(\text{II})} = 0$.

$$
M_{I,(\text{I})} = \begin{cases}
0, & \text{if } (\Omega_{y,(\text{I})} + \hat{\Omega}_{p,(\text{I})} + \Omega_{\epsilon,(\text{I})})/(\varphi^2 \Pi_{e,(\text{I})}) \geq (\Omega_{y,(\text{II})} + \hat{\Omega}_{p,(\text{II})} + \bar{M}\varphi^2 \Pi_{e,(\text{II})} + \Omega_{\epsilon,(\text{II})})/(\varphi^2 \Pi_{e,(\text{II})}), \\
\bar{M}, & \text{if } (\Omega_{y,(\text{II})} + \hat{\Omega}_{p,(\text{II})} + \Omega_{\epsilon,(\text{II})})/(\varphi^2 \Pi_{e,(\text{II})}) \geq (\Omega_{y,(\text{I})} + \hat{\Omega}_{p,(\text{I})} + \bar{M}\varphi^2 \Pi_{e,(\text{I})} + \Omega_{\epsilon,(\text{I})})/(\varphi^2 \Pi_{e,(\text{I})}), \\
\frac{1}{2}\Big[\dfrac{\Omega_{y,(\text{II})} + \hat{\Omega}_{p,(\text{II})} + \Omega_{\epsilon,(\text{II})}}{\varphi^2 \Pi_{e,(\text{II})}} + \bar{M} & \\
\quad - \dfrac{\Omega_{y,(\text{I})} + \hat{\Omega}_{p,(\text{I})} + \Omega_{\epsilon,(\text{I})}}{\varphi^2 \Pi_{e,(\text{I})}} \Big] & \text{otherwise.}
\end{cases}
$$

$$\hat{\Omega}_{p,(\text{II})} = \left[\frac{r-g_1}{\rho r}\lambda_K \frac{\hat{\mathcal{C}}_{Ky2}}{\hat{\Omega}_{Ky,(\text{II})}} M_{K,(\text{II})}\Pi_{e,(\text{II})}\right]^2 \Omega_{x,(\text{I})} + \left[\frac{r-g_2}{\rho r}\left[\lambda_I \varphi^2 M_{I,(\text{II})}\Pi_{e,(\text{II})} + \lambda_1 \Omega_{\epsilon,(\text{II})}\right]\right]^2 \Omega_{x,(\text{II})},$$

$$\hat{\Omega}_{p,(\text{I})} = \left[\frac{r-g_1}{\rho r}\left[(\lambda_I \varphi^2 M_{I,(\text{I})} + \lambda_K M_{K,(\text{I})})\Pi_{e,(\text{I})} + \lambda_1 \Omega_{\epsilon,(\text{I})}\right]\right]^2 \Omega_{x,(\text{I})}.$$

$$\begin{aligned}
2M_{I,(\text{I})} &= \frac{\Omega_{y,(\text{II})} + \hat{\Omega}_{p,(\text{II})} + \Omega_{\epsilon,(\text{II})}}{\varphi^2 \Pi_{e,(\text{II})}} + \bar{M} - \frac{\Omega_{y,(\text{I})} + \hat{\Omega}_{p,(\text{I})} + \Omega_{\epsilon,(\text{I})}}{\varphi^2 \Pi_{e,(\text{I})}} \\
&= \frac{\Omega_{y,(\text{II})} + (\frac{r-g_2}{\rho r})^2 [\lambda_I \varphi^2 (\bar{M} - M_{I,(\text{I})})\Pi_{e,(\text{II})} + \lambda_1 \Omega_{\epsilon,(\text{II})}]^2 \Omega_{x,(\text{II})}}{\varphi^2 \Pi_{e,(\text{II})}} \\
&\quad + \frac{[\frac{r-g_1}{\rho r}\lambda_K z M_{K,(\text{II})}\Pi_{e,(\text{II})}]^2 \Omega_{x,(\text{I})} + \Omega_{\epsilon,(\text{II})}}{\varphi^2 \Pi_{e,(\text{II})}} + \bar{M} \\
&\quad - \frac{\Omega_{y,(\text{I})} + (\frac{r-g_1}{\rho r})^2 [\lambda_I \varphi^2 M_{I,(\text{I})}\Pi_{e,(\text{I})} + \lambda_K M_{K,(\text{I})}\Pi_{e,(\text{I})} + \lambda_1 \Omega_{\epsilon,(\text{I})}]^2 \Omega_{x,(\text{I})} + \Omega_{\epsilon,(\text{I})}}{\varphi^2 \Pi_{e,(\text{I})}}.
\end{aligned}$$

Moving one $M_{I,(\text{I})}$ to RHS and $(\Omega_{y,(\text{I})} + ((r-g_1)/(\rho r))^2 [\lambda_I \varphi^2 M_{I,(\text{I})}\Pi_{e,(\text{I})} + \lambda_K M_{K,(\text{I})}\Pi_{e,(\text{I})} + \lambda_1 \Omega_{\epsilon,(\text{I})}]^2 \Omega_{x,(\text{I})} + \Omega_{\epsilon,(\text{I})})/(\varphi^2 \Pi_{e,(\text{I})})$ to LHS, we have

$$\begin{aligned}
&\frac{\Omega_{y,(\text{I})} + (\frac{r-g_1}{\rho r})^2 [\lambda_I \varphi^2 M_{I,(\text{I})}\Pi_{e,(\text{I})} + \lambda_K M_{K,(\text{I})}\Pi_{e,(\text{I})} + \lambda_1 \Omega_{\epsilon,(\text{I})}]^2 \Omega_{x,(\text{I})} + \Omega_{\epsilon,(\text{I})} + M_{I,(\text{I})}\varphi^2 \Pi_{e,(\text{I})}}{\varphi^2 \Pi_{e,(\text{I})}} \\
&= [\Omega_{y,(\text{II})} + (\frac{r-g_2}{\rho r})^2 [\lambda_I \varphi^2 (\bar{M} - M_{I,(\text{I})})\Pi_{e,(\text{II})} + \lambda_1 \Omega_{\epsilon,(\text{II})}]^2 \Omega_{x,(\text{II})} \\
&+ \left[\frac{r-g_1}{\rho r}\lambda_K z M_{K,(\text{II})}\Pi_{e,(\text{II})}]^2 \Omega_{x,(\text{I})} + \Omega_{\epsilon,(\text{II})} + (\bar{M} - M_{I,(\text{I})})\varphi^2 \Pi_{e,(\text{II})}\right] / (\varphi^2 \Pi_{e,(\text{II})})
\end{aligned}$$

This is a quadratic equation (or, in a knife-edge case, a linear equation) in $M_{I,(\text{I})}$. Multiplying the equation by the positive constant β^2 does not change its roots. We therefore write the rescaled equation as

$$aM_{I,(\text{I})}^2 + bM_{I,(\text{I})} + c = 0,$$

where

$$a = \lambda_I^2 \varphi^4 \Pi_{e,(\mathrm{I})} \Pi_{e,(\mathrm{II})} \left[\left(\frac{r - g_1}{\rho r} \right)^2 \Pi_{e,(\mathrm{I})} \Omega_{x,(\mathrm{I})} - \left(\frac{r - g_2}{\rho r} \right)^2 \Pi_{e,(\mathrm{II})} \Omega_{x,(\mathrm{II})} \right],$$

$$b = 2\varphi^2 \Pi_{e,(\mathrm{I})} \Pi_{e,(\mathrm{II})} \left[\left(\frac{r - g_1}{\rho r} \right)^2 \lambda_I \left(\lambda_1 \Omega_{\epsilon,(\mathrm{I})} + \lambda_K M_{K,(\mathrm{I})} \Pi_{e,(\mathrm{I})} \right) \Omega_{x,(\mathrm{I})} \right.$$
$$\left. + \left(\frac{r - g_2}{\rho r} \right)^2 \lambda_I \left(\lambda_1 \Omega_{\epsilon,(\mathrm{II})} + \lambda_I \varphi^2 \bar{M} \Pi_{e,(\mathrm{II})} \right) \Omega_{x,(\mathrm{II})} + 1 \right],$$

$$c = \Pi_{e,(\mathrm{II})} \left[\Omega_{y,(\mathrm{I})} + \left(\frac{r - g_1}{\rho r} \right)^2 \left(\lambda_1 \Omega_{\epsilon,(\mathrm{I})} + \lambda_K M_{K,(\mathrm{I})} \Pi_{e,(\mathrm{I})} \right)^2 \Omega_{x,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})} \right]$$
$$- \Pi_{e,(\mathrm{I})} \left[\Omega_{y,(\mathrm{II})} + \left(\frac{r - g_2}{\rho r} \right)^2 \left(\lambda_1 \Omega_{\epsilon,(\mathrm{II})} + \lambda_I \varphi^2 \bar{M} \Pi_{e,(\mathrm{II})} \right)^2 \Omega_{x,(\mathrm{II})} \right.$$
$$\left. + \left(\frac{r - g_1}{\rho r} \lambda_K z M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} \right)^2 \Omega_{x,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{II})} + \varphi^2 \bar{M} \Pi_{e,(\mathrm{II})} \right].$$

The linear case $a = 0$ occurs when $(r-g_1)^2 \Pi_{e,(\mathrm{I})} \Omega_{x,(\mathrm{I})} = (r-g_2)^2 \Pi_{e,(\mathrm{II})} \Omega_{x,(\mathrm{II})}$. The solution is then $M_I^{\text{int}} = -c/b$. When $a \neq 0$, the interior candidate is the positive root of the quadratic whenever it lies in $(0, \bar{M})$. Thus

$$M_{It}^* = \begin{cases} 0, & \text{if } c \geq 0, \\ \bar{M}, & \text{if } a = 0,\ -c/b \geq \bar{M}, \text{ or if } a \neq 0 \text{ and the positive} \\ & \text{root is at least } \bar{M}, \\ -c/b, & \text{if } a = 0 \text{ and } 0 < -c/b < \bar{M}, \\ (-b + \sqrt{b^2 - 4ac})/(2a), & \text{if } a \neq 0 \text{ and the positive root lies in } (0, \bar{M}). \end{cases}$$

In the closed-access equilibrium, the solution has the same form. The rescaled coefficients

become

$$b_{\text{cl}} = 2\varphi^2 \Pi_{e,(\text{I})} \Pi_{e,(\text{II})} \Bigg[\left(\frac{r - g_1}{\rho r} \right)^2 \lambda_I \lambda_1 \Omega_{\epsilon,(\text{I})} \Omega_{x,(\text{I})} + \left(\frac{r - g_2}{\rho r} \right)^2 \lambda_I \left(\lambda_1 \Omega_{\epsilon,(\text{II})} + \lambda_I \varphi^2 \bar{M} \Pi_{e,(\text{II})} \right) \Omega_{x,(\text{II})} + 1 \Bigg],$$

$$c_{\text{cl}} = \Pi_{e,(\text{II})} \left[\Omega_{y,(\text{I})} + \left(\frac{r - g_1}{\rho r} \right)^2 \lambda_1^2 \Omega_{\epsilon,(\text{I})}^2 \Omega_{x,(\text{I})} + \Omega_{\epsilon,(\text{I})} \right] - \Pi_{e,(\text{I})} \left[\Omega_{y,(\text{II})} + \left(\frac{r - g_2}{\rho r} \right)^2 \left(\lambda_1 \Omega_{\epsilon,(\text{II})} + \lambda_I \varphi^2 \bar{M} \Pi_{e,(\text{II})} \right)^2 \Omega_{x,(\text{II})} + \Omega_{\epsilon,(\text{II})} + \varphi^2 \bar{M} \Pi_{e,(\text{II})} \right].$$

Hence $M_{I,(\text{I});\text{cl}}$ is directly expressed in terms of the effective-precision primitives.

Proof of Corollary 1. 1) By the closed-market data-choice characterization, $m_{e,(\text{I});\text{cl}}(0) \geq m_{e,(\text{II});\text{cl}}(\bar{M})$ is equivalent to $c_{\text{cl}} \geq 0$, which implies $M^*_{I,(\text{I});\text{cl}} = 0$.

2) Under partial access, subtracting the closed-market coefficient c_{cl} from the rescaled coefficient c and collecting positive factors yields a quadratic in $M_{K,(\text{II})}$. Its slope over the feasible region is weakly positive whenever $\Pi_{e,(\text{I})} - z^2 \Pi_{e,(\text{II})} \leq 0$, equivalently $\Pi_{e,(\text{I})} \leq z^2 \Pi_{e,(\text{II})}$. The quadratic is strictly positive at $M_{K,(\text{II})} = 0$, so this condition implies $c - c_{\text{cl}} > 0$. Hence $c_{\text{cl}} \geq 0$ implies $c > 0$, and the partial-access optimum also satisfies $M^*_{I,(\text{I})} = 0$. □

IA.A.5 Proofs for the impacts of partial access

For reference, let Δ denote the partial-access value minus the closed-access value, let $\varphi_I \equiv \varphi$ and $\varphi_K \equiv 1$, and define

$$\omega_{L,(N)}(M) \equiv \lambda_L \varphi_L^2 M \Pi_{e,(N)}.$$

The own-asset SNR changes and the signed cross-asset loading are

$$\Delta \left| \zeta_{(\text{I})} \right| = \frac{r - g_1}{\rho r} \left[\omega_{I,(\text{I})} \left(M_{I,(\text{I})} - M_{I,(\text{I});\text{cl}} \right) + \omega_{K,(\text{I})} \left(M_{K,(\text{I})} \right) \right],$$

$$\Delta \left| \zeta_{(\text{II})} \right| = \frac{r - g_2}{\rho r} \omega_{I,(\text{II})} \left(M_{I,(\text{II})} - M_{I,(\text{II});\text{cl}} \right),$$

$$\Delta \zeta_{(\text{I} \leftarrow \text{II})} = \frac{r - g_1}{\rho r} z \, \omega_{K,(\text{II})} \left(M_{K,(\text{II})} \right).$$

Because $z < 0$, foreign Type-II processing creates a negative signed cross-asset SNR loading and therefore

$$\Delta\left|\zeta_{(\mathrm{I}\leftarrow\mathrm{II})}\right| = -\frac{r-g_1}{\rho r} z\, \omega_{K,(\mathrm{II})}\left(M_{K,(\mathrm{II})}\right) > 0.$$

Proof of Corollary 2. Under the maintained condition $\kappa(0) \leq 1$, foreign institutional investors concentrate their data processing on the type-I asset, so $M_{K,(\mathrm{I})} = \bar{M}$ and $M_{K,(\mathrm{II})} = 0$. Corollary 8 gives one primitive sufficient condition for this corner. The change in the type-I own-asset SNR is

$$\Delta\left|\zeta_{t,(\mathrm{I})}\right| = \frac{r-g_1}{\rho r}\left[\lambda_I \varphi^2\left(M_{I,(\mathrm{I})} - M_{I,(\mathrm{I});\mathrm{cl}}\right) + \lambda_K \bar{M}\right]\Pi_{e,(\mathrm{I})}.$$

Because $\Pi_{e,(\mathrm{I})} > 0$ and data choices lie in $[0, \bar{M}]$, a sufficient condition for $\Delta\left|\zeta_{t,(\mathrm{I})}\right| > 0$ is $\lambda_K > \lambda_I \varphi^2$.

For $\Delta\left|\zeta_{t,(\mathrm{I})}\right| < 0$, domestic institutions must reduce type-I learning, $M_{I,(\mathrm{I})} < M_{I,(\mathrm{I});\mathrm{cl}}$, and the foreign mass must satisfy

$$\lambda_K < \frac{(M_{I,(\mathrm{I});\mathrm{cl}} - M_{I,(\mathrm{I})})\varphi^2}{\bar{M}}\lambda_I.$$

□

Proof of Corollary 9. Under $\kappa(\bar{M}) \geq 1$, foreign institutional investors allocate all processing capacity to type-II data,

$$M_{K,(\mathrm{I})} = 0, \qquad M_{K,(\mathrm{II})} = \bar{M}.$$

Hence foreign information contributes to the cross-asset SNR but not directly to either own-asset SNR. Domestic institutions have the same fixed total processing capacity before and after partial access:

$$M_{I,(\mathrm{I})} + M_{I,(\mathrm{II})} = M_{I,(\mathrm{I});\mathrm{cl}} + M_{I,(\mathrm{II});\mathrm{cl}} = \bar{M}.$$

Therefore,

$$M_{I,(\mathrm{II})} - M_{I,(\mathrm{II});\mathrm{cl}} = -\left[M_{I,(\mathrm{I})} - M_{I,(\mathrm{I});\mathrm{cl}}\right]. \tag{IA.24}$$

Using effective data precision, the two own-asset SNR changes are

$$\Delta\left|\zeta_{t,(\mathrm{I})}\right| = \frac{\lambda_I \varphi^2}{\rho r}(r-g_1)\Pi_{e,(\mathrm{I})}\left[M_{I,(\mathrm{I})} - M_{I,(\mathrm{I});\mathrm{cl}}\right],$$
$$\Delta\left|\zeta_{t,(\mathrm{II})}\right| = \frac{\lambda_I \varphi^2}{\rho r}(r-g_2)\Pi_{e,(\mathrm{II})}\left[M_{I,(\mathrm{II})} - M_{I,(\mathrm{II});\mathrm{cl}}\right].$$

Substituting (IA.24) yields

$$\frac{\Delta\left|\zeta_{t,(\mathrm{I})}\right|}{(r-g_1)\Pi_{e,(\mathrm{I})}} = -\frac{\Delta\left|\zeta_{t,(\mathrm{II})}\right|}{(r-g_2)\Pi_{e,(\mathrm{II})}}.$$

Equivalently,

$$\Delta\left|\zeta_{t,(\mathrm{II})}\right| = -\frac{(r-g_2)\Pi_{e,(\mathrm{II})}}{(r-g_1)\Pi_{e,(\mathrm{I})}}\Delta\left|\zeta_{t,(\mathrm{I})}\right|.$$

Thus the two raw own-asset SNR changes have opposite signs whenever domestic institutions reallocate processing capacity. They are exact additive inverses only under the special restriction

$$(r-g_1)\Pi_{e,(\mathrm{I})} = (r-g_2)\Pi_{e,(\mathrm{II})}.$$

If there is no domestic reallocation, both own-asset SNR changes are zero. □

Proof of Proposition 3. Part (i) and Part (ii) are proved together in Appendix A of the manuscript. □

Proof of Corollary 4. Relative to the partial-access equilibrium, the change in the absolute type-II own-asset SNR can be written directly in effective precision units as

$$\Delta\left|\zeta_{t,(\mathrm{II})}\right| = \frac{r-g_2}{\rho r}\left[\lambda_I\left(M_{I,(\mathrm{II});\mathrm{fu}} - M_{I,(\mathrm{II})}\right)\varphi^2\Pi_{e,(\mathrm{II})} + \lambda_K M_{K,(\mathrm{II});\mathrm{fu}}\Pi_{e,(\mathrm{II})}\right].$$

Because $r > g_2$, the prefactor is positive. Hence the own-asset SNR declines if and only if

$$\lambda_I\left(M_{I,(\mathrm{II})} - M_{I,(\mathrm{II});\mathrm{fu}}\right)\varphi^2\Pi_{e,(\mathrm{II})} > \lambda_K M_{K,(\mathrm{II});\mathrm{fu}}\Pi_{e,(\mathrm{II})}.$$

The left-hand side is the lost domestic effective precision and the right-hand side is the newly produced foreign effective precision. □

Proof of Corollary 11. Under partial access, the type-II eligible-investor mass remains λ_1, so foreign entry creates neither mechanical dilution of type-II soft information nor a foreign Schur-complement discount. Relative to the closed-market counterfactual,

$$\begin{aligned}\Delta\bar{\hat{\Omega}}_{y,(\mathrm{II})} = {} & (\zeta^2_{t,(\mathrm{II})} - \zeta^2_{t,(\mathrm{II});\mathrm{cl}})\Omega_{x,(\mathrm{II})} + \zeta^2_{t,(\mathrm{I}\leftarrow\mathrm{II})}\Omega_{x,(\mathrm{I})} \\ & + \frac{\lambda_I}{\lambda_1}\varphi^2\big(M_{(\mathrm{II})} - M_{(\mathrm{II});\mathrm{cl}}\big)\Pi_{e,(\mathrm{II})}. \end{aligned} \tag{IA.25}$$

The second term is the additional type-II price-signal precision carried through the type-I price. It is nonnegative and is absent in the closed-market counterfactual because $\zeta_{t,(\mathrm{I}\leftarrow\mathrm{II});\mathrm{cl}} = 0$.

Define

$$\mathcal{H}_{\text{II}} = \lambda_I \varphi^2 \big(M_{(\text{II})} - M_{(\text{II});\text{cl}}\big)\Pi_{e,(\text{II})}.$$

Using $r > g_2$,

$$\Delta\big|\zeta_{t,(\text{II})}\big| = \frac{r - g_2}{\rho r}\mathcal{H}_{\text{II}}.$$

Therefore

$$\begin{aligned}(\zeta^2_{t,(\text{II})} - \zeta^2_{t,(\text{II});\text{cl}})\Omega_{x,(\text{II})} \\ = \frac{r - g_2}{\rho r}\left(\big|\zeta_{2,(\text{II})}\big| + \big|\zeta_{t,(\text{II});\text{cl}}\big|\right)\Omega_{x,(\text{II})}\mathcal{H}_{\text{II}},\end{aligned}$$

while the directly held hard-information term in (IA.25) equals $\mathcal{H}_{\text{II}}/\lambda_1$. Hence

$$\Delta\bar{\hat{\Omega}}_{y,(\text{II})} = A_{\text{II}}\mathcal{H}_{\text{II}} + \zeta^2_{t,(\text{I}\leftarrow\text{II})}\Omega_{x,(\text{I})},$$

which proves (IA.90). Because $A_{\text{II}} > 0$ and the cross-price term is nonnegative, $\mathcal{H}_{\text{II}} \geq 0$ implies a weak increase in average posterior precision, strict whenever $\mathcal{H}_{\text{II}} > 0$ or the cross-price coefficient is nonzero. If $\mathcal{H}_{\text{II}} < 0$, the same decomposition shows that average posterior precision decreases if and only if

$$\zeta^2_{t,(\text{I}\leftarrow\text{II})}\Omega_{x,(\text{I})} < -A_{\text{II}}\mathcal{H}_{\text{II}},$$

which proves part (2). □

IA.B Closed Access

The closed-access economy is the domestic-only specialization of the unrestricted partial-access system. Set the foreign masses to zero and retain the domestic information-allocation problem. Only domestic institutions produce hard information, and the two prices aggregate domestic information alone. In particular, the cross-asset price channel generated by foreign Type-II learning is absent:

$$\zeta_{(\text{I}\leftarrow\text{II});\text{cl}} = 0.$$

The domestic allocation rule in the main text, evaluated at $\lambda_K = 0$, determines the closed-access data allocation, price signal, and posterior precisions used in the partial- and full-access analyses.

IA.C Full Access: Solutions and Proofs

IA.C.1 Equilibrium

In this extension, a measure of λ_H foreign individual investors also enter the domestic market, and both types of risky assets are available for foreign investment. At the portfolio choice stage, foreign individual investor h only has the price signal $\boldsymbol{s}_p$, thus updating the conditional expectation and variance about $\widetilde{\boldsymbol{y}}$ following

$$\begin{aligned}\hat{\boldsymbol{\Sigma}}_{Hyt;\text{fu}} &\equiv \mathbb{V}ar(\widetilde{\boldsymbol{y}} \mid \mathcal{I}_{ht;\text{fu}}) = \mathbb{V}ar(\widetilde{\boldsymbol{y}} \mid \boldsymbol{p}_{t;\text{fu}}) = (\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_{pt;\text{fu}})^{-1},\\ \hat{\boldsymbol{\mu}}_{Hyt;\text{fu}} &\equiv \mathbb{E}[\widetilde{\boldsymbol{y}} \mid \mathcal{I}_{ht;\text{fu}}] = \mathbb{E}[\widetilde{\boldsymbol{y}} \mid \boldsymbol{p}_{t;\text{fu}}] = \hat{\boldsymbol{\Sigma}}_{Hyt;\text{fu}}\boldsymbol{\Omega}_{pt;\text{fu}}\boldsymbol{s}_{pt;\text{fu}}.\end{aligned}$$

The SNR becomes

$$\boldsymbol{\zeta}_{t;\text{fu}} = -\frac{1}{\rho r}(r\boldsymbol{I} - \boldsymbol{G})[(\lambda_I\varphi^2\boldsymbol{M}_{I;\text{fu}} + \lambda_K\boldsymbol{M}_{K;\text{fu}})\boldsymbol{\Pi}_e + \lambda_1\boldsymbol{\Omega}_\epsilon],$$

and the average posterior precision of dividend innovation is

$$\begin{aligned}\bar{\hat{\boldsymbol{\Omega}}}_{yt;\text{fu}} &= \frac{1}{\lambda_3}\left(\lambda_I\hat{\boldsymbol{\Omega}}_{Iyt;\text{fu}} + \lambda_J\hat{\boldsymbol{\Omega}}_{Jyt;\text{fu}} + \lambda_K\hat{\boldsymbol{\Omega}}_{Kyt;\text{fu}} + \lambda_H\hat{\boldsymbol{\Omega}}_{Hyt;\text{fu}}\right)\\ &= \frac{1}{\lambda_3}\left[\lambda_I(\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_{pt;\text{fu}} + \varphi^2\boldsymbol{M}_{I;\text{fu}}\boldsymbol{\Pi}_e + \boldsymbol{\Omega}_\epsilon) + \lambda_J(\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_{pt;\text{fu}} + \boldsymbol{\Omega}_\epsilon)\right]\\ &+ \frac{1}{\lambda_3}\left[\lambda_K(\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_{pt;\text{fu}} + \boldsymbol{M}_{K;\text{fu}}\boldsymbol{\Pi}_e) + \lambda_H(\boldsymbol{\Omega}_y + \boldsymbol{\Omega}_{pt;\text{fu}})\right]\\ &= \boldsymbol{\Omega}_y + \boldsymbol{\Omega}_{pt;\text{fu}} + \frac{\lambda_I\varphi^2\boldsymbol{M}_{I;\text{fu}} + \lambda_K\boldsymbol{M}_{K;\text{fu}}}{\lambda_3}\boldsymbol{\Pi}_e + \frac{\lambda_1}{\lambda_3}\boldsymbol{\Omega}_\epsilon,\end{aligned}$$

where $\lambda_3 = \lambda_I + \lambda_J + \lambda_K + \lambda_H$. The price coefficients update to

$$\boldsymbol{A}_{\text{fu}} = \frac{1}{r}\boldsymbol{\mu}_v + (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - \rho r(r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_{yt;\text{fu}}(r\boldsymbol{I} - \boldsymbol{G})^{-1}\frac{1}{\lambda_3}\bar{\boldsymbol{x}}, \tag{IA.26}$$

$$\boldsymbol{B}_{\text{fu}} = (r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}, \tag{IA.27}$$

$$\boldsymbol{C}_{\text{fu}} = (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}(\boldsymbol{I} - \bar{\hat{\boldsymbol{\Sigma}}}_{yt;\text{fu}}\boldsymbol{\Omega}_{y;\text{fu}}), \tag{IA.28}$$

$$\boldsymbol{D}_{\text{fu}} = (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}(\boldsymbol{I} - \bar{\hat{\boldsymbol{\Sigma}}}_{yt;\text{fu}}\boldsymbol{\Omega}_{y;\text{fu}})\boldsymbol{\zeta}_{t;\text{fu}}^{-1}. \tag{IA.29}$$

Foreign institutional investors face a distorted data-choice problem under partial access. Under full access, they face an optimization problem similar to that of domestic institutional in-

vestors. Their ex ante expected utility is as follows.

LEMMA 1 (Foreign ex-ante utility under full access). *At the trading date, before choosing a data allocation, foreign institutional investor k has ex ante utility*

$$\mathbb{E}\left[-\exp\left(-\rho w_k^+\right) \mid \mathcal{I}_{kt^-}\right] = -\exp\left(-\rho r w_k\right)\left|\boldsymbol{\Sigma}_{kt}\hat{\boldsymbol{\Sigma}}_{k2}^{-1}\right|^{-1/2}\exp\left(-\frac{1}{2}\boldsymbol{\mu}_{kt}'\boldsymbol{\Sigma}_{k2}^{-1}\boldsymbol{\mu}_{kt}\right), \tag{IA.30}$$

where $\boldsymbol{\Sigma}_{kt}$ and $\boldsymbol{\mu}_{kt}$ are, respectively, the pre-data-choice conditional variance and conditional mean of the current excess payoff, defined by (IA.4) and (IA.5).

Proof. At the portfolio-choice stage, optimal CARA-normal demand gives

$$\mathbb{E}\left[-\exp\left(-\rho w_k^+\right) \mid \mathcal{I}_{kt}\right] = -\exp\left(-\rho r w_k\right)\exp\left(-\frac{1}{2}\hat{\boldsymbol{\mu}}_{kt}'\hat{\boldsymbol{\Sigma}}_{kt}^{-1}\hat{\boldsymbol{\mu}}_{kt}\right).$$

Conditional on $\mathcal{I}_{kt^-}$, Gaussian updating implies

$$\hat{\boldsymbol{\mu}}_{kt} \sim N\Big(\boldsymbol{\mu}_{kt}, \boldsymbol{\Sigma}_{kt} - \hat{\boldsymbol{\Sigma}}_{kt}\Big).$$

Applying the standard Gaussian quadratic-form identity therefore yields

$$\begin{aligned}\mathbb{E}&\left[\exp\left(-\frac{1}{2}\hat{\boldsymbol{\mu}}_{kt}'\hat{\boldsymbol{\Sigma}}_{kt}^{-1}\hat{\boldsymbol{\mu}}_{kt}\right)\middle|\mathcal{I}_{kt^-}\right]\\ &= \left|\boldsymbol{\Sigma}_{kt}\hat{\boldsymbol{\Sigma}}_{kt}^{-1}\right|^{-1/2}\exp\left(-\frac{1}{2}\boldsymbol{\mu}_{kt}'\boldsymbol{\Sigma}_{kt}^{-1}\boldsymbol{\mu}_{kt}\right).\end{aligned}$$

Multiplying by $-\exp(-\rho r w_k)$ gives (IA.30). □

Conditional on the equilibrium price-information environment, which an atomistic investor takes as given, $\boldsymbol{\mu}_{kt}$ and $\boldsymbol{\Sigma}_{kt}$ do not depend on investor k's data allocation. Hence $(M_{K,(\mathrm{I})}, M_{K,(\mathrm{II})})$ affects ex-ante utility only through the posterior covariance $\hat{\boldsymbol{\Sigma}}_{kt}$. Maximizing ex-ante utility is therefore equivalent to maximizing $\left|\hat{\boldsymbol{\Sigma}}_{kt}^{-1}\right|$. Because

$$\hat{\boldsymbol{\Sigma}}_{kt}^{-1} = \frac{1}{r^2}(r\boldsymbol{I} - \boldsymbol{G})'\hat{\boldsymbol{\Omega}}_{kyt}(r\boldsymbol{I} - \boldsymbol{G}),$$

the transformation contributes only a data-choice-independent determinant factor, so the problem is equivalent to maximizing $\left|\hat{\boldsymbol{\Omega}}_{kyt}\right|$. Since the off-diagonal element of $\hat{\boldsymbol{\Omega}}_{kyt}$ is fixed with respect to the allocation, the objective further reduces to maximizing the product of the two

diagonal posterior precisions:

$$\max_{\{M_{K,(\mathrm{I})},M_{K,(\mathrm{II})}\}} \quad (\Omega_{y,(\mathrm{I})}+\hat{\Omega}_{p,(\mathrm{I})}+M_{K,(\mathrm{I})}\Pi_{e,(\mathrm{I})})(\Omega_{y,(\mathrm{II})}+\hat{\Omega}_{p,(\mathrm{II})}+M_{K,(\mathrm{II})}\Pi_{e,(\mathrm{II})}) \tag{IA.31}$$

$$\text{s.t.} \quad M_{K,(\mathrm{I})}+M_{K,(\mathrm{II})}\le \bar{M}. \tag{IA.32}$$

IA.C.2 Proofs for the impacts of full access

Proof of Corollary 5. The change in the type-I own-asset SNR from partial to full access is

$$\Delta\left|\zeta_{(\mathrm{I});\mathrm{fu}}\right| = \frac{r-g_1}{\rho r}\left[\lambda_I\varphi^2\left(M_{I,(\mathrm{I});\mathrm{fu}}-M_{I,(\mathrm{I})}\right)+\lambda_K\left(M_{K,(\mathrm{I});\mathrm{fu}}-\bar{M}\right)\right]\Pi_{e,(\mathrm{I})}.$$

Because $M_{K,(\mathrm{I});\mathrm{fu}}\le\bar{M}$, a sufficient condition for the change to be negative is $M_{I,(\mathrm{I});\mathrm{fu}}<M_{I,(\mathrm{I})}$.

For the change to be positive, domestic institutions must increase type-I learning. Conditional on that increase, positivity requires

$$\lambda_K < \frac{\left(M_{I,(\mathrm{I});\mathrm{fu}}-M_{I,(\mathrm{I})}\right)\varphi^2}{\bar{M}-M_{K,(\mathrm{I});\mathrm{fu}}}\lambda_I.$$

□

To keep the portfolio-constraint adjustment separate from private information, define

$$\begin{aligned}
\mathcal{P}_{(N);\mathrm{fu}} &\equiv \frac{\lambda_I\varphi^2M_{I,(N);\mathrm{fu}}+\lambda_KM_{K,(N);\mathrm{fu}}}{\lambda_3}\Pi_{e,(N)}+\frac{\lambda_1}{\lambda_3}\Omega_{\epsilon,(N)},\\
\mathcal{P}_{(\mathrm{I})} &\equiv \frac{\lambda_I\varphi^2M_{I,(\mathrm{I})}+\lambda_KM_{K,(\mathrm{I})}}{\lambda_2}\Pi_{e,(\mathrm{I})}+\frac{\lambda_1}{\lambda_2}\Omega_{\epsilon,(\mathrm{I})},\\
\mathcal{P}_{(\mathrm{II})} &\equiv \frac{\lambda_I}{\lambda_1}\varphi^2M_{I,(\mathrm{II})}\Pi_{e,(\mathrm{II})}+\Omega_{\epsilon,(\mathrm{II})}.
\end{aligned}$$

For any equilibrium object X, let $\Delta_{\mathrm{fu}}X\equiv X_{\mathrm{fu}}-X$, where the untagged object denotes the partial-access equilibrium.

Proof of Corollary 3. Under full access, the eligible-investor private-information density for asset N is

$$\mathcal{P}_{(N);\mathrm{fu}} = \frac{\lambda_I\varphi^2M_{I,(N);\mathrm{fu}}+\lambda_KM_{K,(N);\mathrm{fu}}}{\lambda_3}\Pi_{e,(N)}+\frac{\lambda_1}{\lambda_3}\Omega_{\epsilon,(N)}.$$

Under partial access, the corresponding private-information densities are

$$\mathcal{P}_{(\mathrm{I})} = \frac{\lambda_I \varphi^2 M_{I,(\mathrm{I})} + \lambda_K M_{K,(\mathrm{I})}}{\lambda_2} \Pi_{e,(\mathrm{I})} + \frac{\lambda_1}{\lambda_2} \Omega_{\epsilon,(\mathrm{I})},$$
$$\mathcal{P}_{(\mathrm{II})} = \frac{\lambda_I}{\lambda_1} \varphi^2 M_{I,(\mathrm{II})} \Pi_{e,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})}.$$

The asset-specific average posterior precisions among investors eligible to hold each asset are

$$\bar{\hat{\Omega}}_{y,(\mathrm{I})} = \Omega_{y,(\mathrm{I})} + \hat{\Omega}_{p,(\mathrm{I})} + \frac{\lambda_I \varphi^2 M_{I,(\mathrm{I})} + \lambda_K M_{K,(\mathrm{I})}}{\lambda_2} \Pi_{e,(\mathrm{I})} + \frac{\lambda_1}{\lambda_2} \Omega_{\epsilon,(\mathrm{I})} - \frac{\lambda_K}{\lambda_2} z^2 \hat{\Omega}_{Ky,(\mathrm{II})}, \tag{IA.33}$$
$$\bar{\hat{\Omega}}_{y,(\mathrm{II})} = \Omega_{y,(\mathrm{II})} + \hat{\Omega}_{p,(\mathrm{II})} + \frac{\lambda_I}{\lambda_1} \varphi^2 M_{I,(\mathrm{II})} \Pi_{e,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})}. \tag{IA.34}$$

Using (IA.33) and (13), the type-I change is

$$\Delta_{\mathrm{fu}} \bar{\hat{\Omega}}_{y,(\mathrm{I})} = \Delta_{\mathrm{fu}} \hat{\Omega}_{p,(\mathrm{I})} + \mathcal{P}_{(\mathrm{I});\mathrm{fu}} - \mathcal{P}_{(\mathrm{I})} + \mathcal{Q}_{\mathrm{I}},$$

which gives (IA.85). Since both the partial- and full-access type-I price precisions are represented by their own-asset SNRs,

$$\Delta_{\mathrm{fu}} \hat{\Omega}_{p,(\mathrm{I})} = \left(\zeta^2_{(\mathrm{I});\mathrm{fu}} - \zeta^2_{(\mathrm{I})} \right) \Omega_{x,(\mathrm{I})}.$$

For type-II, (IA.34) and (13) imply

$$\Delta_{\mathrm{fu}} \bar{\hat{\Omega}}_{y,(\mathrm{II})} = \Delta_{\mathrm{fu}} \hat{\Omega}_{p,(\mathrm{II})} + \mathcal{P}_{(\mathrm{II});\mathrm{fu}} - \mathcal{P}_{(\mathrm{II})},$$

which gives (IA.86). Under partial access, (IA.35) gives

$$\hat{\Omega}_{p,(\mathrm{II})} = \zeta^2_{(\mathrm{II})} \Omega_{x,(\mathrm{II})} + \zeta^2_{(\mathrm{I} \leftarrow \mathrm{II})} \Omega_{x,(\mathrm{I})}, \tag{IA.35}$$

whereas under full access

$$\hat{\Omega}_{p,(\mathrm{II});\mathrm{fu}} = \zeta^2_{(\mathrm{II});\mathrm{fu}} \Omega_{x,(\mathrm{II})}.$$

Subtracting yields (IA.87). The decomposition therefore separates three distinct forces: changes in price information, changes in eligible-investor private-information density, and, for type-I, recovery of the partial-access Schur-complement discount. For type-II, the price-information change additionally subtracts the cross-price information available under partial access. Hence neither the own-asset SNR nor price informativeness alone determines the change in average posterior precision. □

IA.D Unrestricted-Theory Proofs and Corollaries

This section collects proofs of the supporting lemmas and corollaries used in the main text.

IA.D.1 Proof of Lemma 1

Proof. Step 1: Calculate $\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_i\right]$

The interim utility for institutional investor i is

$$\begin{aligned}\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_i\right] &= -\exp\left(-\rho r w_i\right)\mathbb{E}\Big[\exp\Big(-(\mathbb{E}[\widetilde{\boldsymbol{d}}+\widetilde{\boldsymbol{v}}-\boldsymbol{p}r \mid \mathcal{I}_i])' \\ &\qquad \times \mathbb{V}ar^{-1}(\widetilde{\boldsymbol{d}}+\widetilde{\boldsymbol{v}} \mid \mathcal{I}_i)(\widetilde{\boldsymbol{d}}+\widetilde{\boldsymbol{v}}-\boldsymbol{p}r)\Big) \mid \mathcal{I}_i\Big] \\ &= -\exp\left(-\rho r w_i\right) \times \exp\left(-\frac{1}{2}\hat{\boldsymbol{\mu}}_i'\hat{\boldsymbol{\Sigma}}_i^{-1}\hat{\boldsymbol{\mu}}_i\right),\end{aligned}$$

where $\hat{\boldsymbol{\mu}}_i \equiv \mathbb{E}[\widetilde{\boldsymbol{d}}+\widetilde{\boldsymbol{v}}-\boldsymbol{p}r \mid \mathcal{I}_i] = \boldsymbol{\mu}_v + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}(\boldsymbol{\mu}+\boldsymbol{G}\boldsymbol{d}) + r(r\boldsymbol{I}-\boldsymbol{G})^{-1}\hat{\boldsymbol{\mu}}_{iy} - \boldsymbol{p}r$ and $\hat{\boldsymbol{\Sigma}}_i$ is a function of known model parameters. Note that before making data choice at the trading date, $-\frac{1}{2}\hat{\boldsymbol{\mu}}_i'\hat{\boldsymbol{\Sigma}}_i^{-1}\hat{\boldsymbol{\mu}}_i$ is a quadratic function of normal variables $\boldsymbol{p}$ (also $\boldsymbol{s}_i$ and $\boldsymbol{\xi}_i$ for domestic institutional investors). It is useful to write them in terms of the same zero-mean normal variable. Before that, we need an intermediate step to take the expectation over private signals $\boldsymbol{s}_i$ and $\boldsymbol{\xi}_i$.

Step 2: Calculate $\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right]$

Conditional on $\mathcal{I}_{i^-}$ and $\boldsymbol{p}$, w_i is non-random, and $\hat{\boldsymbol{\mu}}_i$ is random with $\boldsymbol{e}_i$. (Note that the inalienable information $\boldsymbol{\xi}_i$ is known to domestic institutional investors under $\mathcal{I}_{i^-}$). By law of iterated expectations, one has

$$\begin{aligned}\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right] &= \mathbb{E}\left[\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_i\right] \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right] \\ &= -\exp\left(-\rho r w_i\right)\mathbb{E}\left[\exp\left(-\frac{1}{2}\hat{\boldsymbol{\mu}}_i'\hat{\boldsymbol{\Sigma}}_i^{-1}\hat{\boldsymbol{\mu}}_i\right) \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right].\end{aligned}$$

Let $\boldsymbol{\Sigma}_i^p \equiv \mathbb{V}ar(\widetilde{\boldsymbol{d}}+\widetilde{\boldsymbol{v}}-\boldsymbol{p}r \mid \mathcal{I}_{i^-}, \boldsymbol{p})$. Conditional on $\mathcal{I}_{i^-}$ and $\boldsymbol{p}$, $\hat{\boldsymbol{\mu}}_i$ is still normally distributed. By law of total variance, $\boldsymbol{\Sigma}_i^p = \mathbb{E}\left[\hat{\boldsymbol{\Sigma}}_i \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right] + \mathbb{V}ar\left(\hat{\boldsymbol{\mu}}_i \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right)$. Since $\hat{\boldsymbol{\Sigma}}_i$ is $\mathcal{I}_{i^-}$-measurable, one has that $\mathbb{V}ar(\hat{\boldsymbol{\mu}}_i - \boldsymbol{\mu}_i^p \mid \mathcal{I}_{i^-}, \boldsymbol{p}) = \mathbb{V}ar\left(\hat{\boldsymbol{\mu}}_i \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right) = \boldsymbol{\Sigma}_i^p - \hat{\boldsymbol{\Sigma}}_i$. To work out $\mathbb{E}\left[\exp\left(-\frac{1}{2}\hat{\boldsymbol{\mu}}_i'\hat{\boldsymbol{\Sigma}}_i^{-1}\hat{\boldsymbol{\mu}}_i\right) \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right]$, it is useful to write it in terms of the zero-mean random variable

$\hat{\boldsymbol{\mu}}_i - \boldsymbol{\mu}_i^p$:

$$\mathbb{E}\left[\exp\left(-\frac{1}{2}\hat{\boldsymbol{\mu}}_i'\hat{\boldsymbol{\Sigma}}_i^{-1}\hat{\boldsymbol{\mu}}_i\right) \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right] = \mathbb{E}\Big[\exp\Big(-\frac{1}{2}\left(\hat{\boldsymbol{\mu}}_i - \boldsymbol{\mu}_i^p\right)'\hat{\boldsymbol{\Sigma}}_i^{-1}\left(\hat{\boldsymbol{\mu}}_i - \boldsymbol{\mu}_i^p\right) - \boldsymbol{\mu}_i^{p\prime}\hat{\boldsymbol{\Sigma}}_i^{-1}\left(\hat{\boldsymbol{\mu}}_i - \boldsymbol{\mu}_i^p\right) - \frac{1}{2}\boldsymbol{\mu}_i^{p\prime}\hat{\boldsymbol{\Sigma}}_i^{-1}\boldsymbol{\mu}_i^p\Big) \mid \mathcal{I}_{i^-}, \boldsymbol{p}\Big].$$

Therefore,

$$\mathbb{E}\left[\exp\left(-\frac{1}{2}\hat{\boldsymbol{\mu}}_i'\hat{\boldsymbol{\Sigma}}_i^{-1}\hat{\boldsymbol{\mu}}_i\right) \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right] = \left|\boldsymbol{\Sigma}_i^p\hat{\boldsymbol{\Sigma}}_i^{-1}\right|^{-1/2}\exp\left(-\frac{1}{2}\boldsymbol{\mu}_i^{p\prime}\boldsymbol{\Sigma}_i^{p2}\boldsymbol{\mu}_i^p\right).$$

Step 3: Calculate $\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_{i^-}\right]$

Notice that $\boldsymbol{\mu}_i^p$ is random with respect to $\mathcal{I}_{i^-}$ because of the random component $\boldsymbol{p}$. By law of iterated expectations, one has

$$\begin{aligned}\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_{i^-}\right] &= \mathbb{E}\left[\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_{i^-}, \boldsymbol{p}\right] \mid \mathcal{I}_{i^-}\right] \\ &= -\exp\left(-\rho r w_i\right) \times \left|\boldsymbol{\Sigma}_i^p\hat{\boldsymbol{\Sigma}}_i^{-1}\right|^{-1/2}\mathbb{E}\left[\exp\left(-\frac{1}{2}\boldsymbol{\mu}_i^{p\prime}(\boldsymbol{\Sigma}_i^p)^{-1}\boldsymbol{\mu}_i^p\right) \mid \mathcal{I}_{i^-}\right].\end{aligned}$$

Let $\boldsymbol{\Sigma}_i \equiv \mathbb{V}ar(\widetilde{\boldsymbol{d}} + \widetilde{\boldsymbol{v}} - \boldsymbol{p}r \mid \mathcal{I}_{i^-})$. Conditional on $\mathcal{I}_{i^-}$, $\boldsymbol{\mu}_i^p$ is still normally distributed. By law of total variance, $\boldsymbol{\Sigma}_i = \mathbb{E}\left[\boldsymbol{\Sigma}_i^p \mid \mathcal{I}_{i^-}\right] + \mathbb{V}ar\left(\boldsymbol{\mu}_i^p \mid \mathcal{I}_{i^-}\right)$. Since $\boldsymbol{\Sigma}_i^p$ is $\mathcal{I}_{i^-}$-measurable, one has that $\mathbb{V}ar\left(\boldsymbol{\mu}_i^p \mid \mathcal{I}_{i^-}\right) = \boldsymbol{\Sigma}_i - \boldsymbol{\Sigma}_i^p$, which is the variance of $\boldsymbol{\mu}_i^p - \boldsymbol{\mu}_i$ conditional on $\mathcal{I}_{i^-}$. To work out $\mathbb{E}\left[\exp\left(-\frac{1}{2}\boldsymbol{\mu}_i^{p\prime}(\boldsymbol{\Sigma}_i^p)^{-1}\boldsymbol{\mu}_i^p\right) \mid \mathcal{I}_{i^-}\right]$, it is useful to write it in terms of the zero-mean random variable $\boldsymbol{\mu}_i^p - \boldsymbol{\mu}_i$. Recall that $\boldsymbol{\mu}_i^p = \boldsymbol{\mu}_v + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G}\boldsymbol{d} + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu}_{iy}^p - \boldsymbol{p}r$ and $\boldsymbol{\mu}_i = \boldsymbol{\mu}_v + r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\mu} - r\boldsymbol{A} + (r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G} - r\boldsymbol{B})\boldsymbol{d} + (r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G} - r\boldsymbol{C})\boldsymbol{\mu}_{iyt}$. The constructed zero-mean normal variable is $\boldsymbol{\mu}_i^p - \boldsymbol{\mu}_i = r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{\Sigma}_{Ly}^p\boldsymbol{\Omega}_p\boldsymbol{C}^{-1}(\boldsymbol{p} - \boldsymbol{A} - \boldsymbol{B}\boldsymbol{d}) - \boldsymbol{p}r + r\boldsymbol{A} + r\boldsymbol{B}\boldsymbol{d} - (r(r\boldsymbol{I} - \boldsymbol{G})^{-1}\boldsymbol{G} - r\boldsymbol{C})\boldsymbol{\mu}_{iyt}$.

$$\boldsymbol{\mu}_i^p - \boldsymbol{\mu}_i = \mathcal{M}\boldsymbol{p} - \mathcal{M}(\boldsymbol{A} + \boldsymbol{B}\boldsymbol{d}),$$

where $\mathcal{M} = r(r\boldsymbol{I} - \boldsymbol{G})^{-1}(\boldsymbol{\Sigma}_y + \boldsymbol{\Sigma}_p)^{-1}\boldsymbol{\Sigma}_p\boldsymbol{C}^{-1} - rI$. Therefore

$$\begin{aligned}
&\mathbb{E}\left[\exp\left(-\frac{1}{2}\boldsymbol{\mu}_i^{p\prime}(\boldsymbol{\Sigma}_i^p)^{-1}\boldsymbol{\mu}_i^p\right) \mid \mathcal{I}_{i^-}\right] \\
&= \mathbb{E}\left[\exp\left(-\frac{1}{2}\left(\boldsymbol{\mu}_i^p - \boldsymbol{\mu}_i\right)'\left(\boldsymbol{\Sigma}_i^p\right)^{-1}\left(\boldsymbol{\mu}_i^p - \boldsymbol{\mu}_i\right) - \boldsymbol{\mu}_i'(\boldsymbol{\Sigma}_i^p)^{-1}\left(\boldsymbol{\mu}_i^p - \boldsymbol{\mu}_i\right) - \frac{1}{2}\boldsymbol{\mu}_i'(\boldsymbol{\Sigma}_i^p)^{-1}\boldsymbol{\mu}_i\right) \mid \mathcal{I}_{i^-}\right] \\
&= \left|\boldsymbol{\Sigma}_i(\boldsymbol{\Sigma}_i^p)^{-1}\right|^{-1/2}\exp\left(-\frac{1}{2}\boldsymbol{\mu}_i'\boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i\right),
\end{aligned}$$

and thus the ex ante utility is

$$\mathbb{E}\left[-\exp\left(-\rho w_i^+\right) \mid \mathcal{I}_{i^-}\right] = -\left|\boldsymbol{\Sigma}_i\hat{\boldsymbol{\Sigma}}_i^{-1}\right|^{-1/2}\exp\left(-\rho r w_i - \frac{1}{2}\boldsymbol{\mu}_i'\boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i\right).$$

□

The ex ante utility obtained above depends on the data choice only through the posterior variance $\hat{\boldsymbol{\Sigma}}_i$, so the data-choice problem reduces to maximizing $\left|\hat{\boldsymbol{\Omega}}_{iy}\right|$ subject to the binding capacity constraint:

Since the data choice $(M_{i,(\mathrm{I})}, M_{i,(\mathrm{II})})$ only enter the ex ante utility function through the posterior variance of payoff ($\hat{\boldsymbol{\Sigma}}_i^{-1}$), the data choice problem reduces to allocating data points to maximize $\left|\hat{\boldsymbol{\Omega}}_{iy}\right|$. Since $\hat{\boldsymbol{\Omega}}_{iy} = \boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \varphi^2\boldsymbol{M}_i\boldsymbol{\Pi}_e + \boldsymbol{\Omega}_\epsilon$, the data choices only enter through the $\varphi^2\boldsymbol{M}_i\boldsymbol{\Pi}_e$ term. In fact, the objective function can be further reduced to the product of the diagonal elements of the posterior precision matrix for dividend innovation, that is $\hat{\Omega}_{iy,(\mathrm{I})}\hat{\Omega}_{iy,(\mathrm{II})}$. To see this, let us write out the elements of $\hat{\boldsymbol{\Omega}}_{iy}$.

$$\hat{\boldsymbol{\Omega}}_{iy} = \begin{pmatrix} \Omega_{y,(\mathrm{I})} + \hat{\Omega}_{pt,(\mathrm{I})} + \varphi^2 M_{i,(\mathrm{I})}\Pi_{e,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})} & \mathcal{C}_y + \hat{\mathcal{C}}_p \\ \mathcal{C}_y + \hat{\mathcal{C}}_p & \Omega_{y,(\mathrm{II})} + \hat{\Omega}_{pt,(\mathrm{II})} + \varphi^2 M_{i,(\mathrm{II})}\Pi_{e,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})} \end{pmatrix}.$$

Then $\left|\hat{\boldsymbol{\Omega}}_{iy}\right|$ evaluates to $(\Omega_{y,(\mathrm{I})} + \hat{\Omega}_{pt,(\mathrm{I})} + \varphi^2 M_{i,(\mathrm{I})}\Pi_{e,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})})(\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{pt,(\mathrm{II})} + \varphi^2 M_{i,(\mathrm{II})}\Pi_{e,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})}) - (\mathcal{C}_y + \hat{\mathcal{C}}_p)^2$. The second term is irrelevant to investor i's data choice, which finally reduces to:

$$\max_{\{M_{i,(\mathrm{I})}, M_{i,(\mathrm{II})}\}} \quad (\Omega_{y,(\mathrm{I})} + \hat{\Omega}_{pt,(\mathrm{I})} + \varphi^2 M_{i,(\mathrm{I})}\Pi_{e,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})})(\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{pt,(\mathrm{II})} + \varphi^2 M_{i,(\mathrm{II})}\Pi_{e,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})}) \tag{IA.36}$$

$$\text{s.t.} \qquad M_{i,(\mathrm{I})} + M_{i,(\mathrm{II})} \leq \bar{M}. \tag{IA.37}$$

Because it is always optimal to acquire and process additional data points, the data-capacity constraint (IA.37) binds. Substituting $M_{i,(\mathrm{II})} = \bar{M} - M_{i,(\mathrm{I})}$ into the objective function (IA.36) yields the first-order condition

$$\frac{\varphi^2 \Pi_{e,(\mathrm{I})}}{\hat{\Omega}_{iy,(\mathrm{I})}} = \frac{\varphi^2 \Pi_{e,(\mathrm{II})}}{\hat{\Omega}_{iy,(\mathrm{II})}}.$$

Equivalently, by Definition 3,

$$m_{e,(\mathrm{I})}(M_{i,(\mathrm{I})}) = m_{e,(\mathrm{II})}(M_{i,(\mathrm{II})}).$$

Conditional on the equilibrium price signal, the individual investor takes $\hat{\Omega}_{pt,(N)}$ as given, so

$$m_{e,(N)}(M) = m_{e,(N)}(0) + M.$$

Using the binding capacity constraint, the interior Type-I allocation is therefore

$$\begin{aligned} M_i^{\text{int}} &= \frac{1}{2}\left[m_{e,(\mathrm{II})}(\bar{M}) - m_{e,(\mathrm{I})}(0)\right] \\ &= \frac{1}{2}\left[\frac{\Omega_{y,(\mathrm{II})} + \hat{\Omega}_{pt,(\mathrm{II})} + \Omega_{\epsilon,(\mathrm{II})}}{\varphi^2 \Pi_{e,(\mathrm{II})}} + \bar{M} - \frac{\Omega_{y,(\mathrm{I})} + \hat{\Omega}_{pt,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})}}{\varphi^2 \Pi_{e,(\mathrm{I})}}\right]. \end{aligned}$$

The boundary comparisons can be written compactly as

$$\begin{aligned} M_i^{\text{int}} \le 0 &\iff m_{e,(\mathrm{I})}(0) \ge m_{e,(\mathrm{II})}(\bar{M}), \\ M_i^{\text{int}} \ge \bar{M} &\iff m_{e,(\mathrm{II})}(0) \ge m_{e,(\mathrm{I})}(\bar{M}). \end{aligned}$$

This gives a water-filling interpretation of the allocation problem: each processed data point raises the corresponding equivalent data amount by one, so the investor directs capacity toward the asset with the lower current equivalent data amount until the two are equalized or a capacity corner is reached. This affine identity is conditional on the equilibrium price signal and should not be read as a comparative-equilibrium identity, because $\hat{\Omega}_{pt,(N)}$ changes endogenously when aggregate information production or the access regime changes.

IA.D.2 Proof of Lemma 2

Foreign institutional investor k trades only the Type-I risky asset, so her portfolio choice is determined by the posterior expectation and variance of that payoff. Applying the same three-step

argument as in the proof of Lemma 1, the ex ante utility reduces to

$$\mathbb{E}[-\exp(-\rho w_k^+) \mid \mathcal{I}_{kt^-}] = -\exp(-\rho r w_k) \left|\sigma^2_{K,(\mathrm{I})}/\hat{\sigma}^2_{kt,(\mathrm{I})}\right|^{-1/2} \exp\left(-\frac{\mu^2_{Kt,(\mathrm{I})}}{2\sigma^2_{K,(\mathrm{I})}}\right).$$

Because $\boldsymbol{G}$ is diagonal, $\hat{\sigma}^2_{kt,(\mathrm{I})} = (r/(r-g_1))^2\hat{\sigma}^2_{kyt,(\mathrm{I})}$, so minimizing the posterior variance of the tradable payoff is equivalent to minimizing the posterior variance of its dividend innovation. The data choice therefore maximizes the Type-I Schur-complement precision $\hat{\Omega}^S_{kyt,(\mathrm{I})} = \hat{\Omega}_{kyt,(\mathrm{I})} - z^2\hat{\Omega}_{kyt,(\mathrm{II})}$ with $z \equiv \hat{\mathcal{C}}_{kyt}/\hat{\Omega}_{kyt,(\mathrm{II})}$, which is the claim of the lemma. The closed-form interior solution and the corner characterization are developed below in Section IA.A.4.2.

IA.E Numerical Illustrations of the Analytical Comparative Statics

This section illustrates the analytical mechanism in the unconstrained partial-access model. The exercises vary one information parameter at a time and resolve the equilibrium allocation of the fixed processing capacity $\bar{M}$ between Type-I and Type-II data. They are mechanism illustrations, not calibrated counterfactuals from the constrained quantitative model.

The numerical implementation uses $\boldsymbol{\Omega}_e = \mathrm{diag}(0.05, 2)$ at the baseline, so the hard-information precision matrix is positive definite. The parameter γ denotes the ordinary covariance-space correlation of the two dividend innovations: the covariance matrix $\boldsymbol{\Sigma}_y = \boldsymbol{\Omega}_y^{-1}$ has off-diagonal element $\gamma\sqrt{\Sigma_{y,11}\Sigma_{y,22}}$. The baseline covariance matrix therefore implies $\gamma \simeq -0.64$. Each plotted point is obtained from a bounded fixed-point solution with an explicit iteration limit and convergence check. All plotted solutions satisfy the capacity identities

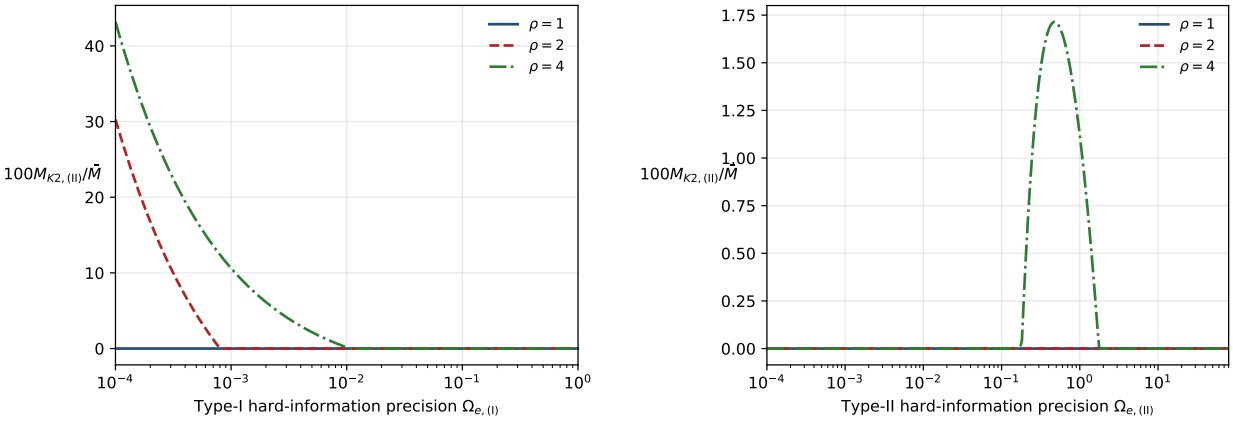
$$M_{I2,(\mathrm{I})} + M_{I2,(\mathrm{II})} = \bar{M}, \qquad M_{K2,(\mathrm{I})} + M_{K2,(\mathrm{II})} = \bar{M},$$

and the maximum fixed-point residual across the plotted grids is below 10^{-9}. Signal-to-noise ratios are reported in their original units; only the data-allocation panels express $M_{K2,(\mathrm{II})}/\bar{M}$ as a percentage.

IA.E.1 Hard-information precision

Figure IA.1 varies the unit precision of the hard data sources. Panel A changes Type-I precision while holding $\Omega_{e,(\mathrm{II})} = 3$. Panel B changes Type-II precision while holding $\Omega_{e,(\mathrm{I})} = 0.04$. The

second panel illustrates Proposition 2: the foreign institution assigns zero capacity to Type-II data when that source becomes either nearly uninformative or so precise that the equilibrium price already aggregates its information sufficiently. The nonmonotonic allocation is therefore a consequence of endogenous cross-asset learning rather than a mechanical preference for the inaccessible asset.



(a) Type-I hard-information precision

(b) Type-II hard-information precision

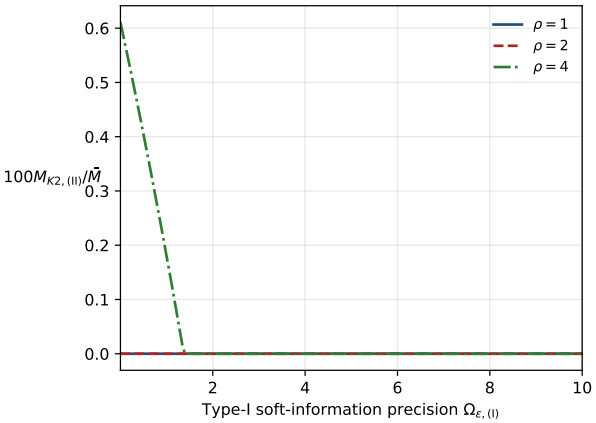
Figure IA.1: **Hard-information precision and cross-asset learning.** The vertical axis is the percentage of foreign processing capacity assigned to Type-II data. Each curve resolves the endogenous domestic and foreign data choices for the indicated coefficient of absolute risk aversion ρ.

IA.E.2 Domestic soft information

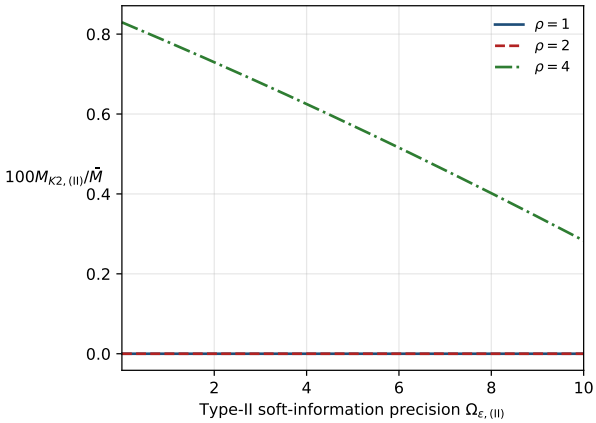
Figure IA.2 varies the precision of the soft information observed only by domestic investors. Panels A and B report the foreign allocation to Type-II hard data. Panels C and D report the own-asset price SNRs, $|\zeta_{2,11}|$ and $|\zeta_{2,22}|$, in their original units. This separation is important: changes in domestic soft information affect the foreign institution's hard-data choice through the price signal, while the own-SNR combines that endogenous reallocation with the direct contribution of domestic soft information to prices.

IA.E.3 Foreign technology in a high-correlation illustration

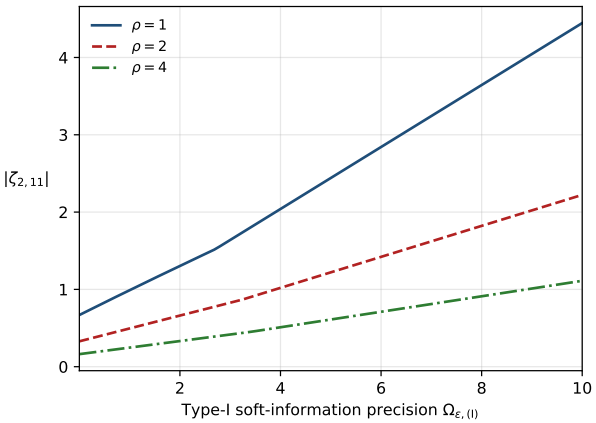
At the baseline correlation $|\gamma| \simeq 0.64$, the corrected numerical solution remains at the Type-I corner, $M_{K2,(\mathrm{II})} = 0$, over the plotted technology range. A scan over $|\gamma|$ shows that a nonzero Type-II allocation emerges only when the fundamental correlation is much stronger, between



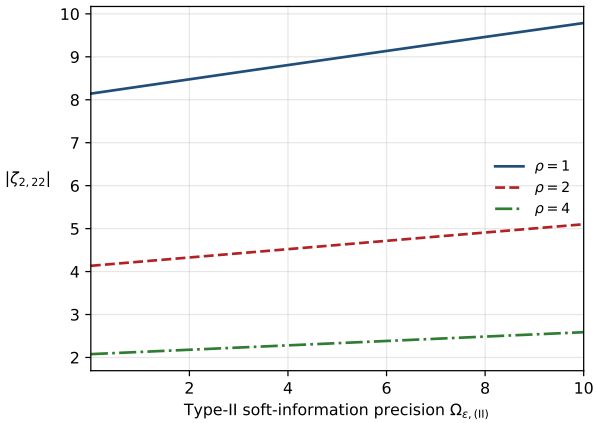
(a) Allocation response to Type-I soft precision



(b) Allocation response to Type-II soft precision



(c) Type-I own-SNR, $|\zeta_{2,11}|$



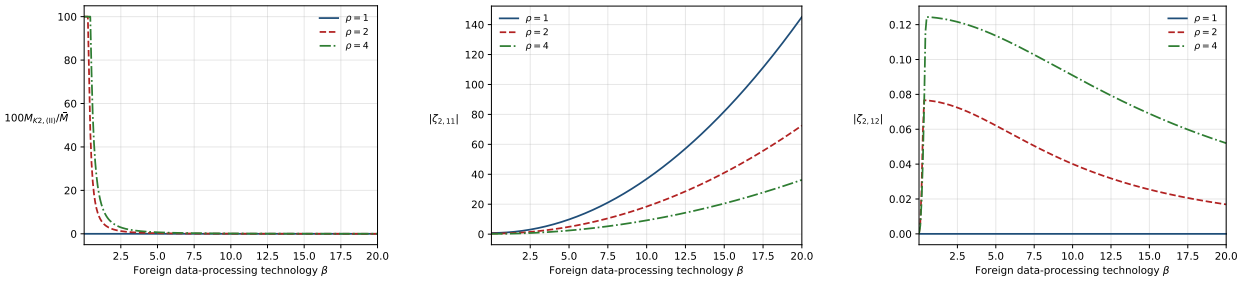
(d) Type-II own-SNR, $|\zeta_{2,22}|$

Figure IA.2: **Domestic soft information, foreign hard-data allocation, and own-asset price signals.** Panels A and C vary $\Omega_{\epsilon,(\mathrm{I})}$ while holding $\Omega_{\epsilon,(\mathrm{II})} = 5$. Panels B and D vary $\Omega_{\epsilon,(\mathrm{II})}$ while holding $\Omega_{\epsilon,(\mathrm{I})} = 0.1$. The allocation panels report a percentage of capacity; the SNR panels report unscaled magnitudes.

approximately 0.90 and 0.95 under this parameterization. Figure IA.3 therefore sets $\gamma = 0.95$ to illustrate the mechanism when cross-asset learning is quantitatively active. It must not be interpreted as a baseline or calibrated comparative static.

Panel A shows how better foreign technology changes the allocation of capacity to Type-II data. Panel B reports the own Type-I price SNR $|\zeta_{2,11}|$. Panel C reports $|\zeta_{2,12}|$, the component of the Type-I price signal generated by Type-II information. The last object is a cross-SNR, not the own SNR of the Type-II asset. Together, the panels display the mechanism

$$\beta \longrightarrow \text{foreign data allocation} \longrightarrow \text{own Type-I information and Type-II-to-Type-I information.}$$



(a) Type-II data allocation (b) Own Type-I SNR, $|\zeta_{2,11}|$ (c) Cross-SNR, $|\zeta_{2,12}|$

Figure IA.3: **Foreign technology and information aggregation at high correlation.** The panels vary foreign data-processing technology β with covariance-space correlation fixed at $\gamma = 0.95$. The curves are a high-correlation mechanism illustration. At the baseline $|\gamma| \simeq 0.64$, the corrected solution remains at the Type-I corner over the same technology range.

IA.F Partial Access with a Short-Sale Constraint: Complete Derivation

IA.F.1 Purpose and modeling convention

This section supplies the complete derivation underlying Proposition 4 in the main text. In particular, it derives the participation-weighted foreign-demand formula and its local response to posterior expected payoffs. It also characterizes the model when nonnegative risky-asset holdings are imposed structurally. The constraint is the weak inequality

$$q_{lt,(n)} \geq 0,$$

not the strict inequality $q_{lt,(n)} > 0$. A strict constraint can prevent an optimum from being attained when the optimal position is zero.

Domestic investors may trade both risky assets, so their feasible set is $\mathbb{R}^2_+$. Under partial access, foreign investors retain the paper's investability restriction on type-II equity, hence

$$\mathcal{Q}_I = \mathcal{Q}_J = \mathbb{R}^2_+, \qquad \mathcal{Q}_K = \mathbb{R}_+ \times \{0\}.$$

The zero foreign type-II position is an investability restriction and must not be counted as a binding no-short-sale event.

The central result is that exact constrained portfolio choice and exact continuum aggrega-

tion remain highly tractable. Globally, however, constrained market clearing generally produces a nonlinear price function, so the paper's Gaussian linear price-signal structure cannot simultaneously remain globally exact. The coherent extension developed here therefore uses exact constrained portfolios and aggregation together with a locally linear constrained equilibrium for price information.

IA.F.2 Individual portfolio choice

For investor l of type L, let

$$\boldsymbol{\mu} \equiv \widehat{\boldsymbol{\mu}}_{lt}, \qquad \boldsymbol{\Sigma} \equiv \widehat{\boldsymbol{\Sigma}}_L = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix},$$

with $\boldsymbol{\Sigma} \succ 0$. The constrained problem is

$$\max_{\boldsymbol{q} \geq 0} \left\{ \boldsymbol{q}'\boldsymbol{\mu} - \frac{\rho}{2}\boldsymbol{q}'\boldsymbol{\Sigma}\boldsymbol{q} \right\}.$$

The safest KKT representation is the complementarity system

$$\boldsymbol{q} \geq 0, \qquad \rho\boldsymbol{\Sigma}\boldsymbol{q} - \boldsymbol{\mu} \geq 0, \qquad \boldsymbol{q} \odot (\rho\boldsymbol{\Sigma}\boldsymbol{q} - \boldsymbol{\mu}) = 0. \tag{IA.38}$$

IA.F.3 KKT Characterization of Portfolio States

For computational purposes, the four portfolio states correspond to the active regions of the constrained quadratic program.

LEMMA 2 (Constrained two-asset portfolio choice). *Let $\Delta = \sigma_{11}\sigma_{22} - \sigma_{12}^2 > 0$. The constrained optimum falls into one of four portfolio states:*

$$\begin{aligned}
11: &\quad \boldsymbol{q}^{11} = \rho^{-1}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}, && \sigma_{22}\mu_1 - \sigma_{12}\mu_2 > 0, \quad \sigma_{11}\mu_2 - \sigma_{12}\mu_1 > 0, \\
10: &\quad \boldsymbol{q}^{10} = (\mu_1/(\rho\sigma_{11}), 0)', && \mu_1 > 0, \quad \mu_2 - \sigma_{12}\mu_1/\sigma_{11} \leq 0, \\
01: &\quad \boldsymbol{q}^{01} = (0, \mu_2/(\rho\sigma_{22}))', && \mu_2 > 0, \quad \mu_1 - \sigma_{12}\mu_2/\sigma_{22} \leq 0, \\
00: &\quad \boldsymbol{q}^{00} = \mathbf{0}, && \mu_1 \leq 0, \quad \mu_2 \leq 0.
\end{aligned}$$

When $\sigma_{12} \neq 0$, componentwise clipping of the unconstrained portfolio is generally incorrect: if one asset is excluded, the remaining position must be reoptimized.

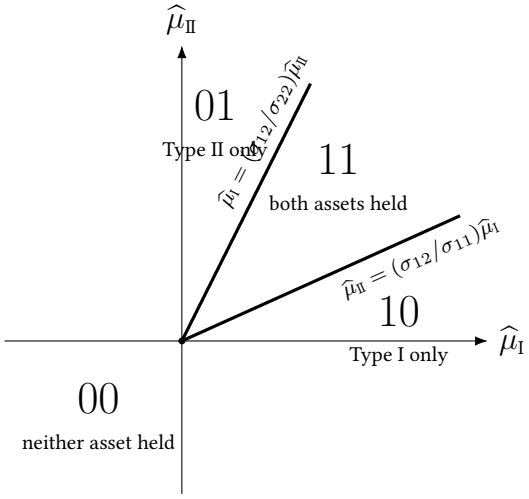


Figure IA.4: KKT portfolio-state regions for the illustrative case $\sigma_{12} > 0$. Lemma 2 gives the general inequalities.

IA.F.4 Foreign investor

The paper already imposes $q_{k,\text{II}} = 0$. With no short selling on the tradable asset,

$$q^*_{k,\text{I}} = \max\left\{0, \frac{\widehat{\mu}_{k,\text{I}}}{\rho\widehat{\Sigma}_{K,11}}\right\}, \qquad q^*_{k,\text{II}} = 0. \tag{IA.39}$$

If the posterior precision is

$$\widehat{\boldsymbol{\Omega}}_{Kyt} = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \qquad z \equiv c/b,$$

then

$$\frac{1}{\widehat{\Sigma}_{Kyt,11}} = a - \frac{c^2}{b}.$$

Thus the Schur complement remains the effective marginal precision of type I. The normalized cross-precision $z = c/b$ must be distinguished from the absolute Schur correction c^2/b and from the fractional precision reduction $c^2/(ab)$.

IA.F.5 Posterior distributions under a local Gaussian price signal

Let

$$\boldsymbol{T} \equiv r(r\boldsymbol{I} - \boldsymbol{G})^{-1},$$

and write the current excess payoff as

$$\widetilde{\boldsymbol{R}}_t = \boldsymbol{h} + \boldsymbol{T}\widetilde{\boldsymbol{y}}, \qquad \boldsymbol{h} = \boldsymbol{\mu}_v + \boldsymbol{T}\boldsymbol{\mu} + \boldsymbol{T}\boldsymbol{G}\boldsymbol{d} - r\boldsymbol{p}.$$

Then

$$\widehat{\boldsymbol{\mu}}_{lt} = \boldsymbol{h} + \boldsymbol{T}\widehat{\boldsymbol{\mu}}_{lyt}, \qquad \widehat{\boldsymbol{\Sigma}}_L = \boldsymbol{T}\widehat{\boldsymbol{\Sigma}}_{Lyt}\boldsymbol{T}'.$$

Under the locally linear price representation, use

$$\boldsymbol{s}_p = \boldsymbol{y} + \boldsymbol{\zeta}^{-1}\boldsymbol{x}, \qquad \boldsymbol{\Omega}_p = \boldsymbol{\zeta}'\boldsymbol{\Omega}_x\boldsymbol{\zeta}.$$

IA.F.6 Domestic institutional investors

Define

$$\boldsymbol{P}_I = \boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \varphi^2\boldsymbol{M}_I\boldsymbol{\Pi}_e + \boldsymbol{\Omega}_\epsilon, \qquad \boldsymbol{S}_I = \boldsymbol{P}_I^{-1}.$$

Then

$$\widehat{\boldsymbol{\Sigma}}_I = \boldsymbol{T}\boldsymbol{S}_I\boldsymbol{T}'.$$

Conditional on aggregate state $(\boldsymbol{y}, \boldsymbol{x})$, the cross-sectional posterior payoff mean is Gaussian,

$$\widehat{\boldsymbol{\mu}}_i \mid (\boldsymbol{y}, \boldsymbol{x}) \sim N(\boldsymbol{m}_I, \boldsymbol{V}_I),$$

where

$$\overline{\boldsymbol{\mu}}_{Iy} = \boldsymbol{S}_I\left[(\boldsymbol{P}_I - \boldsymbol{\Omega}_y)\boldsymbol{y} + \boldsymbol{\Omega}_p\boldsymbol{\zeta}^{-1}\boldsymbol{x}\right],$$

$$\boldsymbol{m}_I = \boldsymbol{h} + \boldsymbol{T}\overline{\boldsymbol{\mu}}_{Iy},$$

and, under the paper's independent signal-noise structure,

$$\boldsymbol{V}_{Iy} = \boldsymbol{S}_I\left(\varphi^2\boldsymbol{M}_I\boldsymbol{\Pi}_e + \boldsymbol{\Omega}_\epsilon\right)\boldsymbol{S}_I, \qquad \boldsymbol{V}_I = \boldsymbol{T}\boldsymbol{V}_{Iy}\boldsymbol{T}'.$$

IA.F.7 Domestic individual investors

Let

$$\boldsymbol{P}_J = \boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \boldsymbol{\Omega}_\epsilon, \qquad \boldsymbol{S}_J = \boldsymbol{P}_J^{-1}.$$

Then

$$\widehat{\boldsymbol{\Sigma}}_{Jt} = \boldsymbol{T}\boldsymbol{S}_J\boldsymbol{T}',$$

$$\overline{\boldsymbol{\mu}}_{Jy} = \boldsymbol{S}_J \left[(\boldsymbol{\Omega}_p + \boldsymbol{\Omega}_\epsilon)\boldsymbol{y} + \boldsymbol{\Omega}_p \boldsymbol{\zeta}^{-1} \boldsymbol{x} \right],$$

$$\boldsymbol{m}_J = \boldsymbol{h} + \boldsymbol{T}\overline{\boldsymbol{\mu}}_{Jy}, \qquad \boldsymbol{V}_J = \boldsymbol{T}\boldsymbol{S}_J \boldsymbol{\Omega}_\epsilon \boldsymbol{S}_J \boldsymbol{T}'.$$

Hence $\widehat{\boldsymbol{\mu}}_j \mid (\boldsymbol{y}, \boldsymbol{x}) \sim N(\boldsymbol{m}_J, \boldsymbol{V}_J)$. Unlike the unconstrained model, idiosyncratic signal noise does not disappear merely because it has cross-sectional mean zero: it changes portfolio-state probabilities, so in general $\mathbb{E}[Q(\widehat{\mu})] \neq Q(\mathbb{E}[\widehat{\mu}])$.

IA.F.8 Foreign institutional investors

Let

$$\boldsymbol{P}_K = \boldsymbol{\Omega}_y + \boldsymbol{\Omega}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e, \qquad \boldsymbol{S}_K = \boldsymbol{P}_K^{-1}.$$

Then

$$\widehat{\boldsymbol{\Sigma}}_K = \boldsymbol{T}\boldsymbol{S}_K \boldsymbol{T}',$$

$$\overline{\boldsymbol{\mu}}_{Ky} = \boldsymbol{S}_K \left[(\boldsymbol{\Omega}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e)\boldsymbol{y} + \boldsymbol{\Omega}_p \boldsymbol{\zeta}^{-1} \boldsymbol{x} \right],$$

$$\boldsymbol{m}_K = \boldsymbol{h} + \boldsymbol{T}\overline{\boldsymbol{\mu}}_{Ky}, \qquad \boldsymbol{V}_K = \boldsymbol{T}\boldsymbol{S}_K (\boldsymbol{M}_K \boldsymbol{\Pi}_e) \boldsymbol{S}_K \boldsymbol{T}'.$$

IA.F.9 Exact aggregation of constrained demand

IA.F.9.1. A reusable truncated-Gaussian identity

Let $\boldsymbol{X} \sim N(\boldsymbol{m}, \boldsymbol{V})$ and consider a region $\mathcal{R} = \{\boldsymbol{A}\boldsymbol{X} > 0\}$. Define

$$\boldsymbol{W} = \boldsymbol{A}\boldsymbol{X} \sim N(\boldsymbol{\nu}, \boldsymbol{S}), \qquad \boldsymbol{\nu} = \boldsymbol{A}\boldsymbol{m}, \qquad \boldsymbol{S} = \boldsymbol{A}\boldsymbol{V}\boldsymbol{A}'.$$

Write

$$\boldsymbol{S} = \begin{pmatrix} s_1^2 & \varrho s_1 s_2 \\ \varrho s_1 s_2 & s_2^2 \end{pmatrix}, \quad a = \nu_1/s_1, \quad b = \nu_2/s_2, \quad \delta = \sqrt{1-\varrho^2}.$$

Then

$$P(\mathcal{R}) = \Phi_2(a, b; \varrho),$$

and with

$$g_1 = \frac{1}{s_1}\phi(a)\Phi\left(\frac{b - \varrho a}{\delta}\right), \qquad g_2 = \frac{1}{s_2}\phi(b)\Phi\left(\frac{a - \varrho b}{\delta}\right),$$

$$\mathbb{E}[\boldsymbol{X} 1_{\{\boldsymbol{A}\boldsymbol{X} > 0\}}] = \boldsymbol{m}\Phi_2(a, b; \varrho) + \boldsymbol{V}\boldsymbol{A}'\boldsymbol{g}. \tag{IA.40}$$

IA.F.9.2. Portfolio-State Regions and Truncated Moments

For $L \in \{I, J\}$, suppress L temporarily and set

$$\boldsymbol{A}^{11} = \begin{pmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{12} & \sigma_{11} \end{pmatrix},$$

$$\boldsymbol{A}^{10} = \begin{pmatrix} 1 & 0 \\ \sigma_{12}/\sigma_{11} & -1 \end{pmatrix}, \qquad \boldsymbol{A}^{01} = \begin{pmatrix} 0 & 1 \\ -1 & \sigma_{12}/\sigma_{22} \end{pmatrix}.$$

These matrices transform the corresponding KKT inequalities into positive-quadrant events. Let

$$\boldsymbol{M}_L^a \equiv \mathbb{E}[\widehat{\boldsymbol{\mu}}_{lt} 1_{\mathcal{R}_L^a}], \qquad a \in \{11, 10, 01\},$$

computed from (IA.40). Then exact aggregate demand is

$$\overline{\boldsymbol{q}}_L = \frac{1}{\rho}\boldsymbol{\Sigma}_L^{-1}\boldsymbol{M}_L^{11} + \frac{1}{\rho\sigma_{11,L}}\boldsymbol{E}_{11}\boldsymbol{M}_L^{10} + \frac{1}{\rho\sigma_{22,L}}\boldsymbol{E}_{22}\boldsymbol{M}_L^{01}, \tag{IA.41}$$

where

$$\boldsymbol{E}_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \boldsymbol{E}_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The zero-holding region contributes no demand. Its probability is

$$P_L^{00} = \Phi_2\left(-\frac{m_{L1}}{\sqrt{V_{L11}}}, -\frac{m_{L2}}{\sqrt{V_{L22}}}; \frac{V_{L12}}{\sqrt{V_{L11}V_{L22}}}\right),$$

and a numerical identity that must hold is

$$P_L^{11} + P_L^{10} + P_L^{01} + P_L^{00} = 1.$$

The joint portfolio-state probabilities imply the asset-level participation probabilities

$$\Pr(q_{L,\mathrm{I}} > 0) = P_L^{11} + P_L^{10}, \qquad \Pr(q_{L,\mathrm{II}} > 0) = P_L^{11} + P_L^{01},$$

and the corresponding no-short-sale binding probabilities

$$B_{L,\mathrm{I}} = P_L^{01} + P_L^{00}, \qquad B_{L,\mathrm{II}} = P_L^{10} + P_L^{00}.$$

IA.F.9.3. Foreign Aggregate Demand

Let

$$m_{K1} = \boldsymbol{e}_1' \boldsymbol{m}_K, \qquad s_{K1}^2 = \boldsymbol{e}_1' \boldsymbol{V}_K \boldsymbol{e}_1, \qquad \alpha_K = m_{K1}/s_{K1}.$$

For $X \sim N(m, s^2)$, $\mathbb{E}[X_+] = s\phi(m/s) + m\Phi(m/s)$. Hence

$$\overline{q}_{K,\mathrm{I}} = \frac{s_{K1}\phi(\alpha_K) + m_{K1}\Phi(\alpha_K)}{\rho \widehat{\Sigma}_{K,11}}, \qquad \overline{q}_{K,\mathrm{II}} = 0. \tag{IA.42}$$

The economically meaningful foreign no-short binding probability is

$$B_{K,\mathrm{I}} = \mathbb{P}(q_{k,\mathrm{I}} = 0) = \Phi(-\alpha_K).$$

Differentiating (IA.42) yields

$$\frac{\partial \overline{q}_{K,\mathrm{I}}}{\partial m_{K1}} = \frac{\Phi(\alpha_K)}{\rho \widehat{\Sigma}_{K,11}},$$

which completes the derivation of Proposition 4.

IA.F.9.4. Aggregate Demand Responses and Participation

The derivative of expected constrained demand with respect to a shift in the mean posterior payoff has a simple form. For domestic L,

$$\boldsymbol{H}_L \equiv \frac{\partial \overline{\boldsymbol{q}}_L}{\partial \boldsymbol{m}_L'} = \frac{1}{\rho}\left[P_L^{11}\boldsymbol{\Sigma}_L^{-1} + \frac{P_L^{10}}{\sigma_{11,L}}\boldsymbol{E}_{11} + \frac{P_L^{01}}{\sigma_{22,L}}\boldsymbol{E}_{22}\right]. \tag{IA.43}$$

Define the participation-weighted effective risk-bearing matrix

$$\boldsymbol{\mathcal{H}}_L = P_L^{11}\widehat{\boldsymbol{\Omega}}_L + P_L^{10}\frac{\boldsymbol{E}_{11}}{\widehat{\Sigma}_{Lt,11}} + P_L^{01}\frac{\boldsymbol{E}_{22}}{\widehat{\Sigma}_{Lt,22}},$$

so that $\boldsymbol{H}_L = \rho^{-1}\boldsymbol{\mathcal{H}}_L$.

For foreigners,

$$\boldsymbol{H}_K = \frac{\Phi(\alpha_K)}{\rho \widehat{\Sigma}_{K,11}}\boldsymbol{E}_{11}. \tag{IA.44}$$

Thus the foreign type-I demand response equals the interior response multiplied by the endogenous participation probability $\Phi(\alpha_K)$. Locally one may interpret $\lambda_K\Phi(\alpha_K)$ as an effective foreign participation mass on the type-I demand margin.

The baseline model is nested as the slack-constraint limit: if $P_L^{11} \to 1$ for domestic investors and $\Phi(\alpha_K) \to 1$ for foreigners, then $\boldsymbol{\mathcal{H}}_L \to \widehat{\boldsymbol{\Omega}}_L$ and the foreign effective precision converges to the existing Schur complement.

IA.F.10 Market clearing and local price coefficients

The exact constrained stock-market-clearing condition is

$$\boldsymbol{F}(\boldsymbol{p}; \boldsymbol{y}, \boldsymbol{x}) \equiv \lambda_I \overline{\boldsymbol{q}}_I + \lambda_J \overline{\boldsymbol{q}}_J + \lambda_K \overline{\boldsymbol{q}}_K - \overline{\boldsymbol{x}} - \boldsymbol{x} = 0. \tag{IA.45}$$

Because the aggregate demands contain Gaussian CDF/PDF terms, the exact global price function is generally nonlinear.

At a reference state, the implicit-function theorem gives the local response matrices

$$\boldsymbol{C}_c = -\boldsymbol{F}_p^{-1} \boldsymbol{F}_y, \qquad \boldsymbol{D}_c = -\boldsymbol{F}_p^{-1} \boldsymbol{F}_x. \tag{IA.46}$$

Thus

$$\boldsymbol{p} \approx \boldsymbol{A}_c + \boldsymbol{B}_c \boldsymbol{d} + \boldsymbol{C}_c \boldsymbol{y} + \boldsymbol{D}_c \boldsymbol{x}.$$

Define the locally equivalent price SNR and precision by

$$\boldsymbol{\zeta}_c = (\boldsymbol{D}_c)^{-1} \boldsymbol{C}_c, \qquad \boldsymbol{\Omega}_{p;c} = (\boldsymbol{\zeta}_c)' \boldsymbol{\Omega}_x \boldsymbol{\zeta}_c. \tag{IA.47}$$

The price-information fixed point is therefore

$$\boldsymbol{\zeta}^{(n)} \to \boldsymbol{\Omega}_p^{(n)} \to \text{posterior beliefs} \to \boldsymbol{F}^{(n)} \to (\boldsymbol{C}^{(n+1)}, \boldsymbol{D}^{(n+1)}) \to \boldsymbol{\zeta}^{(n+1)},$$

with equilibrium condition

$$\boldsymbol{\zeta}^* = [\boldsymbol{D}(\boldsymbol{\zeta}^*)]^{-1} \boldsymbol{C}(\boldsymbol{\zeta}^*).$$

IA.F.11 Data choice under exact CARA utility

The conditional optimized portfolio value is

$$v(\boldsymbol{\mu}; \boldsymbol{\Sigma}) = \max_{\boldsymbol{q} \geq 0} \left\{ \boldsymbol{q}' \boldsymbol{\mu} - \frac{\rho}{2} \boldsymbol{q}' \boldsymbol{\Sigma} \boldsymbol{q} \right\},$$

with

$$v = \begin{cases} \frac{1}{2\rho}\boldsymbol{\mu}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}, & 11, \\ \frac{\mu_1^2}{2\rho\sigma_{11}}, & 10, \\ \frac{\mu_2^2}{2\rho\sigma_{22}}, & 01, \\ 0, & 00. \end{cases}$$

Optimized conditional CARA utility is proportional to $-\exp[-\rho v]$. Hence the exact ex-ante data problem minimizes

$$\mathcal{J}_I(M) = \mathbb{E}\left[e^{-\rho v(\widehat{\mu}(M);\widehat{\Sigma}(M))}\right]. \tag{IA.48}$$

This is the exact constrained replacement for the determinant-only criterion behind the paper's domestic data-choice lemma and its closed-form allocation rule.

IA.F.12 Gaussian tilting identity

If $\boldsymbol{X} \sim N(\boldsymbol{m}, \boldsymbol{V})$, then

$$\mathbb{E}\left[e^{-\frac{1}{2}\boldsymbol{X}'\boldsymbol{K}\boldsymbol{X}} 1_{\{\boldsymbol{A}\boldsymbol{X}>0\}}\right] = \mathcal{C}(\boldsymbol{K};\boldsymbol{m},\boldsymbol{V})\mathbb{P}_{\widetilde{X}}(\boldsymbol{A}\widetilde{\boldsymbol{X}} > 0),$$

where

$$\widetilde{\boldsymbol{V}} = (\boldsymbol{V}^{-1} + \boldsymbol{K})^{-1}, \qquad \widetilde{\boldsymbol{m}} = \widetilde{\boldsymbol{V}}\boldsymbol{V}^{-1}\boldsymbol{m},$$

and

$$\mathcal{C}(\boldsymbol{K};\boldsymbol{m},\boldsymbol{V}) = |\boldsymbol{I} + \boldsymbol{V}\boldsymbol{K}|^{-1/2} \exp\left[-\frac{1}{2}\boldsymbol{m}'\boldsymbol{K}(\boldsymbol{I} + \boldsymbol{V}\boldsymbol{K})^{-1}\boldsymbol{m}\right].$$

For the four domestic portfolio states use

$$\boldsymbol{K}^{11} = \boldsymbol{\Sigma}^{-1}, \qquad \boldsymbol{K}^{10} = \boldsymbol{E}_{11}/\sigma_{11}, \qquad \boldsymbol{K}^{01} = \boldsymbol{E}_{22}/\sigma_{22}, \qquad \boldsymbol{K}^{00} = 0.$$

Therefore

$$\mathcal{J}_I(M) = J^{11}(M) + J^{10}(M) + J^{01}(M) + P^{00}(M), \tag{IA.49}$$

where each J^a is a determinant/exponential factor times a bivariate-normal quadrant probability under the corresponding tilted Gaussian distribution. No investor simulation or numerical integration is necessary.

With $M \equiv M_{I,\mathrm{I}}$ and $M_{I,\mathrm{II}} = \bar{M} - M$,

$$M_I^* \in \arg\min_{M\in[0,\bar{M}]} \mathcal{J}_I(M).$$

If the objective is globally convex, the usual endpoint/interior derivative characterization applies; without a proof of convexity, the safe definition is the global minimizer on $[0, \bar{M}]$. The baseline closed form is recovered as the probability of the interior portfolio tends to one.

IA.F.13 Foreign data choice

Let $X = \widehat{\mu}_{K,\mathrm{I}} \sim N(m, s^2)$ and $\sigma^2 = \widehat{\Sigma}_{K,11}$. Then

$$\mathcal{J}_K = \mathbb{P}(X \leq 0) + \mathbb{E}\left[e^{-X^2/(2\sigma^2)} 1_{\{X>0\}}\right].$$

A one-dimensional Gaussian tilt yields

$$\mathcal{J}_K = \Phi\left(-\frac{m}{s}\right) + \frac{\sigma}{\sqrt{s^2+\sigma^2}} \exp\left[-\frac{m^2}{2(s^2+\sigma^2)}\right] \Phi\left(\frac{m\sigma}{s\sqrt{s^2+\sigma^2}}\right). \tag{IA.50}$$

Hence

$$M_K^* \in \arg\min_{M\in[0,\bar{M}]} \mathcal{J}_K(M).$$

The Schur complement remains a precision channel, but it is no longer the only sufficient statistic: data also change the probability $\pi_K = \Phi(m/s)$ that the foreign investor participates in the type-I market. The manuscript's existing spillover statistic

$$\kappa = \left(\frac{\widehat{\mathcal{C}}_{Kyt}}{\widehat{\Omega}_{Kyt,\mathrm{II}}}\right)^2 \frac{\Pi_{e,\mathrm{II}}}{\Pi_{e,\mathrm{I}}}$$

therefore remains interpretable as a precision spillover rate, while π_K captures the new participation margin. In general $\kappa = 1$ is no longer the complete interior data-choice FOC.

IA.F.14 Feasibility of no short selling and the supply process

If every investor must hold nonnegative risky positions, aggregate holdings are componentwise nonnegative. Therefore market clearing requires

$$\overline{\boldsymbol{x}} + \widetilde{\boldsymbol{x}} \geq 0 \quad \text{componentwise.} \tag{IA.51}$$

If the number of shares outstanding is modeled as a Gaussian level with unbounded support, then for every finite mean there is positive probability of negative total supply, in which case no equilibrium satisfying both market clearing and universal $q \geq 0$ exists in those states.

A structurally clean repair is positive-support supply, for example componentwise

$$X_n = \bar{x}_n \exp\left(u_n - \frac{1}{2}\mathbb{V}ar(u_n)\right), \qquad \boldsymbol{u} \sim N(0, \boldsymbol{\Sigma}_u). \tag{IA.52}$$

Then $X_n > 0$ and $\mathbb{E}[X_n] = \bar{x}_n$. Around the reference state,

$$\boldsymbol{X} \approx \overline{\boldsymbol{x}} + \operatorname{diag}(\overline{\boldsymbol{x}})\boldsymbol{u},$$

so the paper's Gaussian supply shock can be interpreted as the first-order local approximation. If

$$\boldsymbol{x} \equiv \operatorname{diag}(\overline{\boldsymbol{x}})\boldsymbol{u},$$

then

$$\boldsymbol{\Sigma}_x^{\text{loc}} = \operatorname{diag}(\overline{\boldsymbol{x}})\boldsymbol{\Sigma}_u \operatorname{diag}(\overline{\boldsymbol{x}}).$$

IA.F.15 Existence and uniqueness conditional on beliefs

For an investor define the indirect portfolio surplus

$$\psi_L(\boldsymbol{\mu}) = \max_{\boldsymbol{q} \in \mathcal{Q}_L} \left\{ \boldsymbol{q}'\boldsymbol{\mu} - \frac{\rho}{2}\boldsymbol{q}'\boldsymbol{\Sigma}_L\boldsymbol{q} \right\}.$$

It is convex in $\boldsymbol{\mu}$, and its gradient is the optimal constrained demand away from measure-zero portfolio-state boundaries. If $\boldsymbol{\mu}_L = \boldsymbol{a}_L - r\boldsymbol{p}$, define the aggregate potential

$$\mathcal{P}(\boldsymbol{p}) = \sum_L \lambda_L \mathbb{E}[\psi_L(\boldsymbol{a}_L - r\boldsymbol{p})] + r\boldsymbol{p}'\boldsymbol{X}. \tag{IA.53}$$

Then

$$\nabla_p \mathcal{P} = -r\boldsymbol{D}(\boldsymbol{p}) + r\boldsymbol{X},$$

so a stationary point is exactly a market-clearing price. The Hessian is

$$\nabla^2_{pp}\mathcal{P} = r^2 \sum_L \lambda_L \boldsymbol{H}_L.$$

For a positive-mass domestic type with nonsingular cross-sectional posterior-mean distribution, the interior portfolio-state region is an open nonempty set and has positive Gaussian probability, so $P_L^{11} > 0$. Since $\boldsymbol{\Sigma}_L^{-1} \succ 0$, (IA.43) then implies $\boldsymbol{H}_L \succ 0$. Consequently the aggregate po-

tential is strictly convex and the market-clearing price is unique conditional on the information environment and positive finite supply. Foreign demand alone is rank one because foreigners cannot hold type II; domestic participation is therefore essential for full two-dimensional price determination.

IA.F.16 Exact nonlinear REE versus locally linear constrained equilibrium

Exact constrained market clearing generally implies a nonlinear mapping

$$\boldsymbol{p} = P(\boldsymbol{y}, \boldsymbol{X}).$$

With a nonlinear price function, observing $\boldsymbol{p}$ no longer generally reveals a Gaussian signal of the form

$$\boldsymbol{y} + \boldsymbol{\zeta}^{-1}\boldsymbol{x}.$$

Hence the following three properties cannot generally all be globally exact:

$$\boldsymbol{q} \geq 0, \qquad \boldsymbol{p} = \boldsymbol{A} + \boldsymbol{B}\boldsymbol{d} + \boldsymbol{C}\boldsymbol{y} + \boldsymbol{D}\boldsymbol{x}, \qquad \boldsymbol{s}_p = \boldsymbol{y} + \boldsymbol{\zeta}^{-1}\boldsymbol{x} \text{ with Gaussian updating.}$$

One could solve a fully nonlinear rational-expectations equilibrium and update beliefs from the nonlinear price mapping, but posterior beliefs would generally cease to be Gaussian and much of the paper's SNR/precision machinery would have to be replaced. The less disruptive and internally coherent extension is therefore a locally linear constrained equilibrium: portfolio constraints and aggregation are exact, while the price mapping is linearized at the reference state and the resulting local signal is used for Gaussian updating. Comparative statics of $\boldsymbol{\zeta}_c$ are correspondingly local or obtained by resolving the constrained equilibrium at alternative parameter values.

IA.F.17 Formal definition of a locally linear no-short-sale equilibrium

A locally linear no-short-sale equilibrium consists of data strategies, portfolio strategies, reference prices, and local price-response matrices. Investors update beliefs using the locally equivalent price signal

$$\boldsymbol{s}_{pt;c} = \boldsymbol{y} + (\boldsymbol{\zeta}_c)^{-1}\boldsymbol{x}, \qquad \boldsymbol{\zeta}_c = (\boldsymbol{D}_c)^{-1}\boldsymbol{C}_c.$$

Portfolios solve the constrained quadratic problems with feasible sets $\mathcal{Q}_I = \mathcal{Q}_J = \mathbb{R}^2_+$ and $\mathcal{Q}_K = \mathbb{R}_+ \times \{0\}$, and institutional data choices maximize ex-ante terminal expected utility, taking the local price-information environment as given when evaluating an individual deviation. Aggregate constrained holdings equal positive share supply at the reference state, while $\boldsymbol{C}_c$ and $\boldsymbol{D}_c$ equal the derivatives of the constrained market-clearing price function there. Finally, local belief consistency requires

$$\boldsymbol{\Omega}_p = (\boldsymbol{\zeta}_c)' \boldsymbol{\Omega}_{xt} \boldsymbol{\zeta}_c.$$

IA.F.18 Binding probabilities and economic interpretation

For domestic type L, define

$$B_{L,\mathrm{I}} = P_L^{01} + P_L^{00}, \qquad B_{L,\mathrm{II}} = P_L^{10} + P_L^{00}.$$

These are the fractions with zero holdings of types I and II, respectively. For foreigners,

$$B_{K,\mathrm{I}} = \Phi(-\alpha_K).$$

The foreign type-II zero is not included because it is an exogenous investability restriction.

The constrained model separates two channels that are collapsed in the unconstrained equilibrium:

information precision and endogenous market participation.

Information changes posterior precision, posterior-mean dispersion, portfolio-state probabilities, aggregate risk-bearing capacity, and therefore local price informativeness. Type-II information can still affect foreign type-I demand through the Schur complement and posterior mean, but it can additionally alter the probability that the foreign investor holds any type-I shares.

IA.F.19 Nesting and validation results

Two limits are important.

Slack-constraint limit. If $P_L^{11} \to 1$ for domestic investors and $\Phi(\alpha_K) \to 1$ for foreigners, constrained portfolio responses, data-choice objectives, and effective precision objects converge to their baseline unconstrained counterparts.

Local-shock limit. For a differentiable exact constrained price function,

$$P(\boldsymbol{y}, \boldsymbol{x}) = \bar{\boldsymbol{p}} + \boldsymbol{C}\boldsymbol{y} + \boldsymbol{D}\boldsymbol{x} + o(\|(\boldsymbol{y}, \boldsymbol{x})\|),$$

so the locally linear constrained equilibrium is first-order exact around the reference state.

IA.F.20 Summary of what changes relative to the baseline

Object	Baseline	Constrained equilibrium
Portfolio	linear CARA-normal demand	exact KKT portfolio-state demand
Domestic aggregation	mean private noise integrates out	truncated bivariate-normal moments
Foreign demand	Schur-complement linear demand	positive-part Gaussian demand
Demand sensitivity	posterior precision	participation-weighted risk-bearing matrix
Data choice	determinant/effective-precision criterion	exact CARA objective plus participation margin
Price function	globally linear conjecture	generally nonlinear; locally linearized
Price SNR	global linear-Gaussian object	local equilibrium object
Supply	Gaussian level shock	positive support structurally; Gaussian locally
Calibration truncation	ex-post clipping possible	must solve exact constrained portfolio

IA.F.21 Bottom line

A structural no-short-sale extension is feasible without investor-level Monte Carlo. Constrained portfolios are closed form by portfolio state; continuum aggregation uses only univariate and bivariate Gaussian CDF/PDF evaluations; exact CARA data-choice objectives admit Gaussian-tilting representations. The substantive modeling change is that exact constrained market clearing is generally nonlinear, so the paper's Gaussian price signal should be interpreted through a

locally linear constrained equilibrium. Positive-support share supply is also required for global feasibility when all risky holdings are nonnegative.

The central economic mechanism is that information affects equilibrium through both posterior precision and endogenous participation. The original unconstrained model is recovered when the short-sale constraints are slack with probability approaching one.

IA.G Closed Access with a Short-Sale Constraint

The closed constrained economy is the domestic-only specialization of the partial-access constrained system. Set the foreign mass to zero, retain the same domestic portfolio-state aggregation and information-choice problem, and solve the locally linear constrained market-clearing system using domestic demand only. This is the constrained closed-access benchmark for the partial- and full-access counterfactuals reported in the main text.

IA.H Full Access with a Short-Sale Constraint: Complete Derivation

As in the derivation note, this appendix uses compact scalar notation for vectors and matrices; boldface in the main text denotes the same objects. The subscripts ; fu, c mark the full-access constrained equilibrium; fu in the main text refers to the unconstrained full-access equilibrium.

This appendix applies the constrained-equilibrium framework of Internet Appendix Section IA.F to full access. The changes are that foreign institutions may hold both risky assets and that a mass λ_H of foreign individual investors enters. As before, portfolio choice and continuum aggregation are exact, while price information is represented by a locally linear constrained equilibrium around a reference state.

Under full access, the feasible risky-asset set is

$$\mathcal{Q}_I = \mathcal{Q}_J = \mathcal{Q}_K = \mathcal{Q}_H = \mathbb{R}^2_+.$$

This is simpler than the partial-access constrained economy because the foreign institutional investor no longer has the exogenous restriction $q_{Kt,\text{II}} = 0$.

The full-access notation appends the subscript fu (with c for the no-short-sale restriction) when it is useful to distinguish this equilibrium from the partial-access economy.

IA.H.1 Assets and local price representation

Let

$$T \equiv r(rI - G)^{-1}.$$

The current excess payoff can be written

$$\widetilde{R} = h + T\widetilde{y}, \qquad h = \mu_v + T\mu + TGd_{-1} - rp.$$

Around the reference state, write the constrained price map as

$$p \simeq A_{\text{fu},c} + B_{\text{fu},c}d_{-1} + C_{\text{fu},c}y + D_{\text{fu},c}x.$$

The lagged-dividend coefficient is unchanged:

$$B_{\text{fu},c} = (rI - G)^{-1}G.$$

Define the locally equivalent price signal by

$$s_{p;\text{fu},c} = y + (\zeta_{\text{fu},c})^{-1}x, \qquad \zeta_{\text{fu},c} = (D_{\text{fu},c})^{-1}C_{\text{fu},c},$$

with price-signal precision

$$\Omega_{pt;\text{fu},c} = (\zeta_{\text{fu},c})'\Omega_x\zeta_{\text{fu},c}.$$

IA.H.2 Posterior beliefs

For compactness define the non-price information precision contributed to investor type L by

$$Q_I = \varphi^2 M_I \Pi_e + \Omega_\epsilon, \qquad Q_J = \Omega_\epsilon, \qquad Q_K = M_K \Pi_e, \qquad Q_H - X| = 0.$$

Then

$$P_L = \Omega_y + \Omega_p + Q_L, \qquad S_L = P_L^{-1},$$

and the posterior payoff covariance is

$$\Sigma_L = TS_LT'.$$

For $L \in \{I, J, K\}$, conditional on the aggregate state (y, x), private or soft signal noise creates

cross-sectional dispersion in posterior expected excess payoffs:

$$\widehat{\mu}_{lt} \mid (y, x) \sim N(m_L, V_L^{cs}),$$

where, at the reference state $d_{-1} = y = x - X| = 0$,

$$m_L = -rA_{\text{fu},c}$$

and

$$V_L^{cs} = TS_L Q_L S_L T'.$$

Foreign individual investors have only the equilibrium price signal. Hence there is no cross-sectional private-signal dispersion for type H:

$$V_H^{cs} - X| = 0.$$

At the reference state their posterior expected payoff is therefore the deterministic vector

$$m_H = -rA_{\text{fu},c}.$$

IA.H.3 Portfolio Choice and Exact Continuum Aggregation

Under full access, types I, J, K, H all have feasible set $\mathbb{R}^2_+$. Hence the generic two-asset constrained portfolio solution in Lemma 2 applies to every investor type. Relative to partial access, the key change is that the foreign institutional investor's Type-II position is endogenous rather than fixed at zero.

For $L \in \{I, J, K\}$, let $X_L \equiv \widehat{\mu}_{lt} \sim N(m_L, V_L^{cs})$. The portfolio-state regions, truncated moments, probabilities, and exact aggregation formula are those in Internet Appendix Section IA.F, evaluated at the full-access posterior covariance Σ_L. For foreign individuals, posterior expected payoff is homogeneous at the reference state, so $\bar{q}_H = Q(m_H; \Sigma_H)$, where $Q(\cdot)$ is the deterministic policy characterized in Lemma 2.

IA.H.4 Aggregate Demand Responses and Participation

For $L \in \{I, J, K\}$, aggregate demand responses follow the participation-weighted derivative in equation (IA.43), evaluated at Σ_L.

The binding probabilities are

$$B_{L,I} = P_L^{01} + P_L^{00}, \qquad B_{L,II} = P_L^{10} + P_L^{00}.$$

Unlike the partial-access constrained economy, these definitions apply to foreign institutions K for both assets.

For the homogeneous foreign-individual type H, the local Jacobian is the deterministic portfolio-state derivative:

$$H_H = \begin{cases} \rho^{-1}\Sigma_H^{-1}, & 11, \\ (\rho\sigma_{11,H})^{-1}E_{11}, & 10, \\ (\rho\sigma_{22,H})^{-1}E_{22}, & 01, \\ 0, & 00. \end{cases}$$

IA.H.5 Reference-state market clearing

The exact constrained reference-state market-clearing condition is

$$F_{\text{fu},c}(p; y, x) = \lambda_I \bar{q}_I + \lambda_J \bar{q}_J + \lambda_K \bar{q}_K + \lambda_H \bar{q}_H - X| = 0 - X| = 0.$$

At the reference state,

$$X = \bar{x}.$$

Because the aggregated demands contain Gaussian CDF/PDF terms, the exact global price map is generally nonlinear. The locally linear equilibrium uses the derivatives of this exact market-clearing system.

IA.H.6 Local price-information fixed point

Holding price fixed, the direct sensitivity of posterior expected payoff to the primitive dividend innovation is

$$\frac{\partial m_L}{\partial y'} = TS_L Q_L, \qquad L \in \{I, J, K\}.$$

For foreign individuals $Q_H - X| = 0$, so

$$\left.\frac{\partial m_H}{\partial y'}\right|_p - X| = 0.$$

Therefore

$$F_y = \lambda_I H_I T S_I Q_I + \lambda_J H_J T S_J Q_J + \lambda_K H_K T S_K Q_K.$$

Local signal consistency implies

$$\zeta_{\text{fu},c} = -F_y.$$

This is the constrained analogue of the paper's full-access SNR equation. The mass λ_H does not enter F_y directly because foreign individuals contribute no private or soft information.

To recover the price Jacobian, define

$$\mathcal{H} = \lambda_I H_I + \lambda_J H_J + \lambda_K H_K + \lambda_H H_H$$

and

$$\mathcal{K} = \lambda_I H_I T S_I \Omega_p + \lambda_J H_J T S_J \Omega_p + \lambda_K H_K T S_K \Omega_p + \lambda_H H_H T S_H \Omega_p.$$

Then

$$F_p = \left(I - \mathcal{K}\zeta^{-1}\right)^{-1} (-r\mathcal{H}).$$

Since the supply shock enters market clearing with derivative $-I$,

$$D_{\text{fu},c} = F_p^{-1}, \qquad C_{\text{fu},c} = D_{\text{fu},c}\zeta_{\text{fu},c}.$$

Hence the local price-information equilibrium is the fixed point

$$\zeta \longrightarrow \Omega_p \longrightarrow \{S_L, \Sigma_L, V_L^{cs}, H_L\} \longrightarrow F_y \longrightarrow \zeta.$$

IA.H.7 Exact constrained CARA data choice under full access

Under full access, both domestic institutions and foreign institutions can trade both assets. Therefore both solve a two-asset constrained information problem.

For a conditional expected payoff vector μ and posterior covariance Σ, define the optimized portfolio surplus

$$v(\mu;\Sigma) = \max_{q \geq 0} \left\{ q'\mu - \frac{\rho}{2} q'\Sigma q \right\}.$$

Piecewise,

$$v(\mu;\Sigma) = \begin{cases} \dfrac{1}{2\rho}\mu'\Sigma^{-1}\mu, & 11, \\ \dfrac{\mu_1^2}{2\rho\sigma_{11}}, & 10, \\ \dfrac{\mu_2^2}{2\rho\sigma_{22}}, & 01, \\ 0, & 00. \end{cases}$$

At the data-choice stage, an individual investor treats the local price-information environment as given. Parameterize

$$M_L(m) = \text{diag}(m, \bar{M} - m), \qquad m \in [0, \bar{M}].$$

For a candidate m, let

$$Q_L(m) = \begin{cases} \varphi^2 M_I(m)\Pi_e + \Omega_\epsilon, & L = I, \\ M_K(m)\Pi_e, & L = K. \end{cases}$$

The investor's posterior payoff covariance is

$$\Sigma_L(m) = T\left[\Omega_y + \Omega_p + Q_L(m)\right]^{-1} T'.$$

For the ex-ante distribution of posterior expected payoff define

$$A_{y,L} = TS_L(\Omega_p + Q_L) - rC,$$

$$A_{x,L} = TS_L\Omega_p\zeta^{-1} - rD,$$

and

$$V_L^{ex} = A_{y,L}\Sigma_y A'_{y,L} + A_{x,L}\Sigma_x A'_{x,L} + TS_L Q_L S_L T'.$$

At the reference state its mean is

$$m_L^{ex} = -rA.$$

The exact ex-ante CARA factor is

$$J_L(m) = \mathbb{E}\left[\exp\{-\rho v(X_L; \Sigma_L(m))\}\right], \qquad X_L \sim N(m_L^{ex}, V_L^{ex}). \tag{IA.54}$$

Using Gaussian tilting, $J_L(m)$ is the sum of four portfolio-state region terms and can be evaluated deterministically using Gaussian CDFs. No investor simulation is necessary.

The exact constrained full-access data choices are therefore

$$M_I^* \in \arg\min_{m\in[0,\bar{M}]} J_I(m), \qquad M_K^* \in \arg\min_{m\in[0,\bar{M}]} J_K(m).$$

This replaces the determinant-only full-access rule from the unconstrained model. In the slack-constraint limit, the exact CARA criterion converges back to the unconstrained determinant criterion.

IA.H.8 Joint price/data equilibrium

Let

$$m \equiv \begin{pmatrix} M_{I,I} \\ M_{K,I} \end{pmatrix}.$$

For every proposed aggregate m, solve the local constrained price equilibrium and then compute the individual exact data best response

$$BR(m) = \begin{pmatrix} BR_I(m) \\ BR_K(m) \end{pmatrix}.$$

A pure symmetric full-access data equilibrium satisfies

$$BR(m^*) = m^*.$$

Because the exact no-short-sale CARA objective is not globally guaranteed to be convex, the global best-response map can be discontinuous. Accordingly, the implementation does not treat a failed Newton root as proof of an equilibrium. It evaluates the global one-dimensional data objectives, iterates the best-response map with damping, and reports a nonzero data-choice gap when no pure symmetric fixed point is found.

IA.H.9 Average posterior precision

Let

$$\lambda_3 = \lambda_I + \lambda_J + \lambda_K + \lambda_H.$$

Portfolio constraints do not change the individual Bayesian posterior precision conditional on information. Hence the population-average posterior precision of dividend innovation remains

$$\bar{\widehat{\Omega}}_{yt;\mathrm{fu},c} = \Omega_y + \Omega_p + \frac{\lambda_I \varphi^2 M_I + \lambda_K M_K}{\lambda_3} \Pi_e + \frac{\lambda_I + \lambda_J}{\lambda_3} \Omega_\epsilon.$$

The no-short-sale constraint affects this object endogenously through the equilibrium choices M_I, M_K and through the local price precision Ω_p.

IA.H.10 Comparison with constrained partial access

The constrained partial-access economy has

$$\mathcal{Q}_K = \mathbb{R}_+ \times \{0\},$$

whereas under full access

$$\mathcal{Q}_K = \mathbb{R}_+^2.$$

This changes both the information-allocation problem and participation:

	Partial access	Full access
Foreign type-II holding	$q_{K,\mathrm{II}} - X] = 0$ exogenously	$q_{K,\mathrm{II}} \geq 0$ endogenously
Foreign participation states	one tradable margin	both Type-I and Type-II participation endogenous
Value of type-II information	indirect spillover to type I	direct trading value + spillover
Data-choice statistic	precision + type-I participation	two-asset precision + two-asset participation

This makes the partial-to-full-access counterfactual particularly useful for the paper's proposed "market access and information production" framing.

IA.H.11 Existence, regularity, and supply

Conditional on beliefs and positive finite supply, uniqueness of the reference-state price follows from the same aggregate-potential argument as in Internet Appendix Section IA.F when the aggregate demand Jacobian is positive definite.

A separate feasibility issue remains. If every investor must hold nonnegative risky positions, total share supply must be componentwise nonnegative. A Gaussian level supply shock has negative-support states. The structurally clean specification is positive-support supply (for example lognormal), while the Gaussian supply shock is interpreted as its local first-order approximation.

IA.H.12 Bottom line

The full-access no-short-sale extension is analytically tractable at the portfolio, aggregation, price, and individual data-choice stages. Relative to constrained partial access, the major structural

change is that foreign institutional investors become ordinary two-asset constrained investors, so information affects their participation in both asset markets.

The aggregate institutional data allocation is a fixed point of the exact constrained CARA best-response map; at the current calibration this fixed point exists, with foreign institutions allocating $M_{K,(\mathrm{I})} = 2.000$ and $M_{K,(\mathrm{II})} = 0.000$ of their processing capacity, and it is reported in the main text.

IA.I Welfare and Planner

IA.I.1 Planner problem and proofs

This section provides a welfare benchmark for the equilibrium allocation of information. The planner is constrained in two respects. First, it takes the market-access regime and the no-short-sale restrictions as given. Second, it chooses only the common data allocation of each institutional-investor type; investors retain their private signals and choose their own portfolios. The planner therefore does not pool private information or replace financial markets. Its sole additional ability is to internalize how one group's data allocation changes equilibrium prices and, through prices, the information and trading opportunities of all investors.

Let a index the access regime, with $a = \mathrm{cl}$, no tag, and $a = \mathrm{fu}$ denoting closed, partial, and full access, respectively. The no-short-sale tag c is fixed throughout this appendix and is therefore suppressed. The investor types present under each regime are

$$\mathcal{L}_{\mathrm{cl}} = \{I, J\}, \qquad \mathcal{L}_{c} = \{I, J, K\}, \qquad \mathcal{L}_{\mathrm{fu},c} = \{I, J, K, H\},$$

and the data-producing types are $\mathcal{D}_{\mathrm{cl}} = \{I\}$ and $\mathcal{D}_{c} = \mathcal{D}_{\mathrm{fu},c} = \{I, K\}$. For each $L \in \mathcal{D}_a$, write $m_L \equiv M_{L,(\mathrm{I})}$ and

$$\boldsymbol{M}_L(m_L) = \mathrm{diag}(m_L, \bar{M} - m_L), \qquad 0 \le m_L \le \bar{M}. \tag{IA.55}$$

Thus the planner reallocates a fixed processing capacity across the two data sources and cannot leave capacity idle. This fixed-capacity benchmark preserves the paper's private allocation problem and asks whether investors direct an exogenously committed processing budget to the socially efficient source. An alternative planner can choose total information production by replacing the equality in (IA.55) with $M_{L,(\mathrm{I})} + M_{L,(\mathrm{II})} \le \bar{M}$; the slack-capacity audit below reports that distinct benchmark.

For a vector of institutional data allocations $\boldsymbol{m} = (m_L)_{L \in \mathcal{D}_a}$, let $\mathcal{E}_a(\boldsymbol{m})$ denote the locally

linear rational-expectations equilibrium characterized above. In this equilibrium, prices aggregate the signals generated by $\boldsymbol{m}$, beliefs obey Bayes' rule, portfolios solve the constrained CARA problems, and markets clear. In particular, the portfolio sets are

$$\mathcal{Q}_I^a = \mathcal{Q}_J^a = \mathbb{R}_+^2, \qquad \mathcal{Q}_K = \mathbb{R}_+ \times \{0\}, \qquad \mathcal{Q}_K^F = \mathcal{Q}_H^F = \mathbb{R}_+^2.$$

For investor type L, define the optimized portfolio surplus under regime a by

$$v_L^a(\widehat{\boldsymbol{\mu}}_L, \widehat{\boldsymbol{\Sigma}}_L) = \max_{\boldsymbol{q} \in \mathcal{Q}_L^a} \left\{ \boldsymbol{q}' \widehat{\boldsymbol{\mu}}_L - \frac{\rho}{2} \boldsymbol{q}' \widehat{\boldsymbol{\Sigma}}_L \boldsymbol{q} \right\}. \tag{IA.56}$$

The function in (IA.56) is the same piecewise quadratic surplus used in the exact constrained data-choice problem. Let $\widetilde{w}_{L^+}^a(\boldsymbol{m})$ be terminal wealth after the equilibrium portfolio choice, with all prices, posterior moments, and binding probabilities evaluated at $\mathcal{E}_a(\boldsymbol{m})$. The ex ante certainty equivalent of type L is

$$\mathrm{CE}_{L;a}(\boldsymbol{m}) = -\frac{1}{\rho} \log \mathbb{E}[\exp\{-\rho \widetilde{w}_{L^+}^a(\boldsymbol{m})\} \mid \mathcal{I}_{Lt^-}]. \tag{IA.57}$$

Equation (IA.57) retains both the direct value of a type's private signal and the indirect effects of every type's information choice through the equilibrium price. It can be evaluated using the same Gaussian-tilting and portfolio-state region formulas as J_L in (IA.54).

Given nonnegative welfare weights θ_L, the constrained planner solves

$$\boldsymbol{m}_{\mathrm{pl},a} \in \boldsymbol{m}_{\mathrm{pl},a} \in \operatorname*{arg\,max}_{\boldsymbol{m} \in [0,\bar{M}]^{|\mathcal{D}_a|}} W_a(\boldsymbol{m}), \qquad W_a(\boldsymbol{m}) \equiv \sum_{L \in \mathcal{L}_a} \theta_L \lambda_L \, \mathrm{CE}_{L;a}(\boldsymbol{m}), \tag{IA.58}$$

subject to $\mathcal{E}_a(\boldsymbol{m})$. A global planner sets $\theta_L = 1$ for every participating type. A domestic regulator can be represented by setting the welfare weights of foreign investors to zero. These two benchmarks distinguish a global information externality from a transfer between domestic and foreign investors.

The decentralized allocation $\boldsymbol{m}_a$ is the fixed point of institutional investors' data best responses. Each atomistic investor takes the price law as given when reallocating one unit of data capacity. At an interior decentralized choice for type R,

$$\left. \frac{\partial \, \mathrm{CE}_{R;a}(\boldsymbol{m})}{\partial m_R} \right|_{\text{price law fixed}} = 0. \tag{IA.59}$$

By contrast, an interior planner allocation satisfies

$$0 = \theta_R \lambda_R \left. \frac{\partial \, \mathrm{CE}_{R;a}}{\partial m_R} \right|_{\text{price law fixed}} + \sum_{L \in \mathcal{L}_a} \theta_L \lambda_L \frac{\partial \, \mathrm{CE}_{L;a}}{\partial \boldsymbol{\Xi}_a} : \frac{\partial \boldsymbol{\Xi}_a}{\partial m_R}, \tag{IA.60}$$

where $\boldsymbol{\Xi}_a$ collects the endogenous price coefficients, price precision, posterior moments, and binding probabilities; ":" denotes the appropriate inner product. The second line is the equilibrium information externality omitted from the private problem. It includes changes in price levels, price informativeness, risk sharing, and the probability that portfolio constraints bind. Its sign is ambiguous in general: more private information can improve price discovery while reducing other investors' information rents.

To isolate this wedge, define the marginal equilibrium externality generated by type R as

$$\mathcal{X}_{R;a}(\boldsymbol{m}) \equiv \sum_{L \in \mathcal{L}_a} \theta_L \lambda_L \frac{\partial \, \mathrm{CE}_{L;a}}{\partial \boldsymbol{\Xi}_a} : \frac{\partial \boldsymbol{\Xi}_a}{\partial m_R}. \tag{IA.61}$$

At an interior decentralized allocation, the private term in (IA.60) is zero. Hence $dW_a/dm_R = \mathcal{X}_{R;a}$ at that allocation. The following result gives a direct analytical test for local inefficiency.

The sign condition in Proposition 1 (stated below) has a closed form in a fixed-active-set benchmark built from the model's Schur-complement learning channel. Let $n_K = M_{K,(\mathrm{II})}$ and hold the other investors' information choices and the local price-information environment at their reference values. Write the foreign posterior precision before adding foreign data as

$$\begin{pmatrix} a_0 & c \\ c & b_0 \end{pmatrix}, \qquad \Pi_{e,(\mathrm{I})} = \Pi_{e,(\mathrm{I})}, \qquad \Pi_{e,(\mathrm{II})} = \Pi_{e,(\mathrm{II})}.$$

After the foreign investor assigns n_K observations to type-II, the two diagonal precision terms are $a(n_K) = a_0 + \Pi_{e,(\mathrm{I})}(\bar{M} - n_K)$ and $b(n_K) = b_0 + \Pi_{e,(\mathrm{II})} n_K$. Because only type-I is tradable, the private effective precision is its Schur complement,

$$\mathcal{S}_K(n_K) = a(n_K) - \frac{c^2}{b(n_K)}. \tag{IA.62}$$

For the global planner, or any welfare criterion with $\theta_K > 0$, the social pass-through can be obtained from the model's welfare derivatives rather than treated as a free parameter. Define the

private marginal cross-learning benefit and its certainty-equivalent welfare weight by

$$\mathcal{B}_K(n_K) \equiv \frac{c^2 \Pi_{e,(\text{II})}}{b(n_K)^2}, \qquad \omega_K(n_K) \equiv \theta_K \lambda_K \frac{\partial \, \text{CE}_K}{\partial \mathcal{S}_K} > 0, \tag{IA.63}$$

and let the equilibrium externality with respect to type-II information be

$$\mathcal{X}_{K,n} \equiv \sum_{L \in \mathcal{L}} \theta_L \lambda_L \frac{\partial \, \text{CE}_L^P}{\partial \boldsymbol{\Xi}^P} : \frac{\partial \boldsymbol{\Xi}^P}{\partial n_K} = -\mathcal{X}_K. \tag{IA.64}$$

In the fixed-active-set benchmark, the total welfare derivative is

$$\begin{aligned} \frac{dW}{dn_K} &= \omega_K \left[-\Pi_{e,(\text{I})} + \mathcal{B}_K(n_K) \right] + \mathcal{X}_{K,n} \\ &= \omega_K \left[-\Pi_{e,(\text{I})} + \chi(n_K) \mathcal{B}_K(n_K) \right], \end{aligned} \tag{IA.65}$$

where the endogenous social pass-through is

$$\chi(n_K) = 1 + \frac{\mathcal{X}_{K,n}(n_K)}{\omega_K(n_K) \mathcal{B}_K(n_K)}. \tag{IA.66}$$

Thus $\chi < 1$ is equivalent to a negative equilibrium externality from additional foreign learning about the inaccessible asset. When $0 \leq \chi \leq 1$ and it is locally constant along the reallocation path, it has the interpretation of the fraction of the private marginal cross-learning benefit that remains in weighted social surplus. The planner's corresponding local certainty-equivalent information index is proportional to

$$\mathcal{S}_{K;\text{pl}}(n_K) = a(n_K) - \chi \frac{c^2}{b(n_K)}. \tag{IA.67}$$

At an interior decentralized allocation, $\mathcal{B}_K = \Pi_{e,(\text{I})}$, so $\chi = 1 + \mathcal{X}_{K,n} / (\omega_K \Pi_{e,(\text{I})})$.

Proposition 1 (Closed-form sufficient condition). *In the fixed-active-set benchmark, suppose* $0 \leq \chi \leq 1$ *is constant along the reallocation path, and let* $\Pi_{[0,\bar{M}]}(x) = \min\{\bar{M}, \max\{0, x\}\}$. *The decen-*

tralized and planner allocations to information about the inaccessible type-Ⅱ asset are

$$n_K = \Pi_{[0,\bar{M}]}\left(\frac{|c|\sqrt{\Pi_{e,(\mathrm{II})}/\Pi_{e,(\mathrm{I})}} - b_0}{\Pi_{e,(\mathrm{II})}}\right), \tag{IA.68}$$

$$n_{K;\mathrm{pl}} = \Pi_{[0,\bar{M}]}\left(\frac{|c|\sqrt{\chi\Pi_{e,(\mathrm{II})}/\Pi_{e,(\mathrm{I})}} - b_0}{\Pi_{e,(\mathrm{II})}}\right). \tag{IA.69}$$

Consequently, $n_{K;\mathrm{pl}} \leq n_K$*. If both choices are interior,* $c \neq 0$*, and* $\chi < 1$*, then*

$$n_K - n_{K;\mathrm{pl}} = \frac{|c|(1-\sqrt{\chi})}{\sqrt{\Pi_{e,(\mathrm{I})}\Pi_{e,(\mathrm{II})}}} > 0. \tag{IA.70}$$

The wedge is zero when fundamentals are uncorrelated. Conditional on the same local representation, any relaxation of market access that raises the social pass-through χ_a *toward one attenuates the wedge.*

Proof. The private index in (IA.62) is strictly concave in n_K when $c \neq 0$; the social index in (IA.67) is strictly concave when $\chi_c \neq 0$ and is decreasing and linear when $\chi = 0$. Their interior first-order conditions are, respectively,

$$\Pi_{e,(\mathrm{I})} = \frac{c^2\Pi_{e,(\mathrm{II})}}{b(n_K)^2}, \qquad \Pi_{e,(\mathrm{I})} = \chi\frac{c^2\Pi_{e,(\mathrm{II})}}{b(n_K)^2}.$$

Solving and imposing the capacity bounds gives (IA.68)–(IA.69). Because $\sqrt{\chi} \leq 1$ and projection onto an interval is monotone, $n_{K;\mathrm{pl}} \leq n_K$. Subtracting the two interior solutions gives (IA.70). If $c = 0$, both objectives decrease in n_K, so both allocations equal zero. □

Implication of the current calibration. The estimated constrained partial-opening equilibrium reported in Section 6.3 has $\bar{M} = 2$ and $\boldsymbol{M}_K = (2, 0)$. Hence $n_K = M_{K,(\mathrm{II})} = 0$. Proposition 1 then implies

$$n_{K;\mathrm{pl}} = n_K = 0, \qquad n_K - n_{K;\mathrm{pl}} = 0 \tag{IA.71}$$

for the fixed-active-set benchmark. Thus the present calibration does not support the claim that foreign investors produce excessive information about the inaccessible asset. This restriction is informative: the welfare case for misallocation must instead come from the endogenous response of domestic information production, from the price and participation externalities in (IA.61), or from parameter regions in which the foreign data choice is interior. The full planner calculation below evaluates the first two channels; neither should be inferred from the cross-learning

mechanism alone.

Numerical constrained-planner solution. For each candidate $(m_I, m_K) \in [0, \bar{M}]^2$, the computation resolves the local rational-expectations price fixed point, constrained portfolios, and binding probabilities. For investor type L, let J_L denote the exact Gaussian expectation of the optimized CARA utility factor, integrated over the relevant portfolio active sets. The reported trading certainty equivalent is $\mathrm{CE}_L = -\log(J_L)/\rho$. The reported levels include a common initial-wealth component; this component cancels in the welfare gains reported below, and the welfare comparisons do not depend on its normalization. The numerical search first evaluates a 21×21 grid on the price-equilibrium branch connected to the decentralized allocation and then locally refines the best grid point. Table IA.I reports the resulting solutions.

Table IA.I
Constrained information allocations under partial opening

	Decentralized	Global planner
$M_{I,(\mathrm{I})}$	2.000	0.000
$M_{I,(\mathrm{II})}$	0.000	2.000
$M_{K,(\mathrm{I})}$	2.000	0.128
$M_{K,(\mathrm{II})}$	0.000	1.872
CE_I	0.667801	0.658301
CE_J	0.643353	0.657850
CE_K	0.385070	0.319272
W_G	0.644000	0.651365

Notes: Each investor type has capacity $\bar{M} = 2$; the two allocation rows for a given type therefore sum to two. The global welfare gain is $0.651365 - 0.644000 = 0.007365$. A continuation method from the decentralized equilibrium converges at all grid points.

The decentralized data game converges with zero best-response gap: at its price law, both type I and type K choose $(2, 0)$—all domestic and foreign institutional processing capacity is allocated to the accessible asset. The full search over the grid nevertheless locates a welfare-improving allocation: the global planner selects $M_I = (0, 2)$ and $M_K = (0.128, 1.872)$, reallocating both domestic and foreign capacity toward the inaccessible asset, with a global welfare gain of 0.0074. The decentralized allocation is therefore a corner equilibrium of the data game in which informa-

tion about the inaccessible asset is underproduced relative to the planner optimum: the private value of inaccessible-asset information is limited when holdings of that asset are constrained, while the planner internalizes the price and participation effects in (IA.61). The full calculation differs from the local directional conditions at the decentralized corners because it includes those endogenous price and participation effects and because the welfare surface is not concave.

Welfare decomposition. To isolate those equilibrium effects, let $U_a(\boldsymbol{m}, \Xi)$ denote the planner-weighted certainty equivalent when portfolios are unconstrained, with private information allocation $\boldsymbol{m}$ and price law Ξ. Let $W_a(\boldsymbol{m}, \Xi)$ denote the corresponding constrained welfare. The direct data-allocation and equilibrium price-law components are the two-factor Shapley decomposition of the change in U_a:

$$\mathcal{D}_a = \frac{1}{2}\left[U_a(\boldsymbol{m}_{\text{pl}}, \Xi) - U_a(\boldsymbol{m}, \Xi) + U_a(\boldsymbol{m}_{\text{pl}}, \Xi_{\text{pl}}) - U_a(\boldsymbol{m}, \Xi_{\text{pl}})\right], \tag{IA.72}$$

$$\mathcal{P}^a = \frac{1}{2}\left[U_a(\boldsymbol{m}, \Xi_{\text{pl}}) - U_a(\boldsymbol{m}, \Xi) + U_a(\boldsymbol{m}_{\text{pl}}, \Xi_{\text{pl}}) - U_a(\boldsymbol{m}_{\text{pl}}, \Xi)\right]. \tag{IA.73}$$

The participation component is the change in the certainty-equivalent cost of the portfolio constraints,

$$\mathcal{C}^a = \left[W_a(\boldsymbol{m}_{\text{pl}}, \Xi_{\text{pl}}) - U_a(\boldsymbol{m}_{\text{pl}}, \Xi_{\text{pl}})\right] - \left[W_a(\boldsymbol{m}, \Xi) - U_a(\boldsymbol{m}, \Xi)\right]. \tag{IA.74}$$

By construction, the components add exactly to the constrained welfare gain: $\mathcal{L}_a = \mathcal{D}_a + \mathcal{P}^a + \mathcal{C}^a$.

For the global planner, the direct reallocation of private information lowers welfare by 0.068642 when the equilibrium-price channel is stripped out. The endogenous price-law response contributes 0.065161 and the change in the constraint wedge contributes 0.008252. The price externality and the constraint-wedge change therefore jointly offset the adverse direct effect, leaving a small positive global gain of 0.004771, or about 0.5% of the decentralized certainty equivalent. The planner barely changes participation: the probability that the type-I constraint binds moves from 56.04% to 55.66% for type I, from 56.03% to 55.66% for type J, and from 55.53% to 55.25% for type K. Domestic investors receive 0.011594 of the global gain while foreign investors lose 0.006824, so the global-planner allocation is primarily redistributive toward domestic types. For the domestic regulator, the price-law channel is the only positive component (0.057256); the regulator raises domestic certainty equivalents by 0.018087 at a foreign cost of 0.022769, so the domestic-regulator allocation lowers global welfare by 0.004682 relative to the decentralized outcome. These results identify the calibrated mechanism as a small equilibrium price-information externality whose main content is redistributive across investor types rather than a large aggregate efficiency gain.

Robustness and the role of fixed capacity. Figure IA.5 recomputes the decentralized equilibrium and the fixed-capacity global planner across fundamental correlation and foreign participation. Over correlations from zero to 0.30, the certainty-equivalent gain ranges from 0.004495 to 0.004836. Over foreign investor shares from 0.10 to 0.30, it ranges from 0.003241 to 0.010236. The decentralized allocation assigns no foreign capacity to the inaccessible asset throughout, while the planner assigns between 1.3333 and 1.3687 units to the accessible asset in the correlation sweep, and between 1.3433 and 1.4075 in the participation sweep. Thus the fixed-capacity welfare gain is quantitatively stable. It remains positive when fundamental correlation is zero, where the solved price and posterior matrices are diagonal. Consequently, this robustness is evidence for the equilibrium price-information externality, not for cross-asset learning by itself.

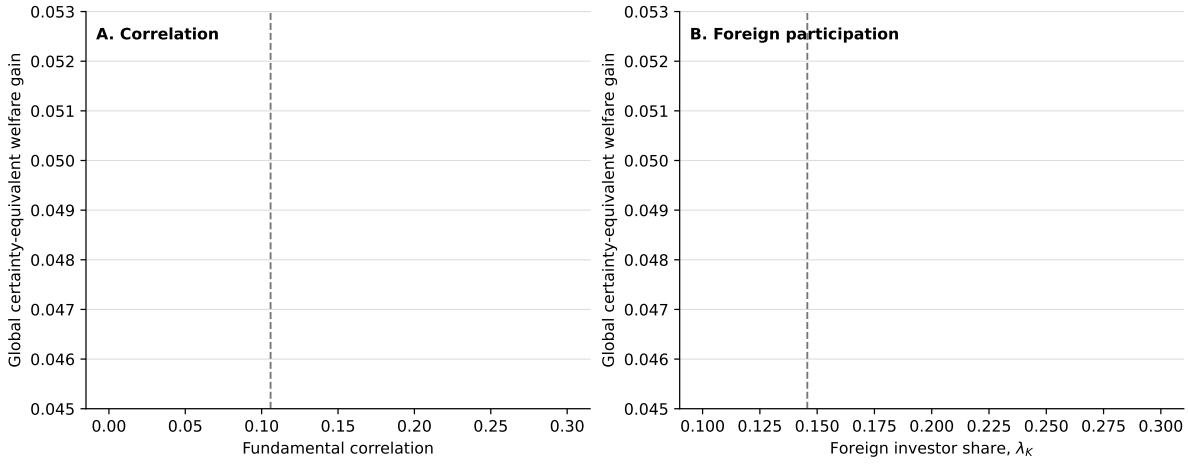


Figure IA.5: Fixed-capacity planner robustness

Notes: Each point recomputes the decentralized data equilibrium, the constrained price equilibrium, and the global planner. Dashed vertical lines mark the baseline calibration. Both sweeps have full 3×3 coarse-grid coverage at every reported point. The point $\lambda_K = 0.05$ is omitted because the joint data–price iteration does not converge and the calibrated corner has a best-response gap of 1.618.

The equality-capacity restriction is material for interpretation. At the decentralized equilibrium, neither domestic nor foreign institutions allocate processing capacity to the inaccessible asset, whereas the global planner allocates 1.872 units of foreign capacity and all domestic capacity to inaccessible-asset information, with a global welfare gain of 0.0074. The wedge therefore reflects underproduction of inaccessible-asset information at the decentralized data game: the private value of such information is limited by the trading restriction, but its social value operates through the equilibrium price and participation effects in (IA.61). Because the planner reallocation changes the equilibrium price law and the welfare surface is not concave, the decentralized corner equilibrium is not the constrained-efficient allocation even though both private

data choices are locally optimal.

Global versus domestic welfare. The global planner and a domestic regulator generally value the price and information effects differently. Define

$$W_{G;a}(\boldsymbol{m}) \equiv \sum_{L \in \mathcal{L}_a} \lambda_L \, \mathrm{CE}_{L;a}(\boldsymbol{m}), \tag{IA.75}$$

$$W_{D;a}(\boldsymbol{m}) \equiv \lambda_I \, \mathrm{CE}_{I;a}(\boldsymbol{m}) + \lambda_J \, \mathrm{CE}_{J;a}(\boldsymbol{m}). \tag{IA.76}$$

The corresponding second-best allocations and welfare losses are

$$\boldsymbol{m}_{G;\mathrm{pl},a} \in \arg\max_{\boldsymbol{m}} W_{G;a}(\boldsymbol{m}), \qquad \mathcal{L}_G^a \equiv W_{G;a}(\boldsymbol{m}_{G;\mathrm{pl},a}) - W_{G;a}(\boldsymbol{m}_a) \geq 0, \tag{IA.77}$$

$$\boldsymbol{m}_{D;\mathrm{pl},a} \in \arg\max_{\boldsymbol{m}} W_{D;a}(\boldsymbol{m}), \qquad \mathcal{L}_D^a \equiv W_{D;a}(\boldsymbol{m}_{D;\mathrm{pl},a}) - W_{D;a}(\boldsymbol{m}_a) \geq 0. \tag{IA.78}$$

Both optimizations are subject to the same equilibrium, market-access, and portfolio constraints as (IA.58). Under partial opening, the difference between their marginal assessments of type R's information choice is

$$\frac{dW_G}{dm_R} - \frac{dW_D}{dm_R} = \lambda_K \frac{d\,\mathrm{CE}_K}{dm_R}. \tag{IA.79}$$

Under full opening, the right-hand side additionally contains $\lambda_H d\,\mathrm{CE}_H^F / dm_R$. Equation (IA.79) isolates the foreign-incidence component of the welfare wedge. If the decentralized allocation differs from both planner benchmarks and $\mathcal{L}_G^a > 0$, information is globally misallocated. If the domestic loss is economically meaningful while the global loss is negligible, the main effect is incidence across domestic and foreign investors rather than an aggregate welfare loss. Accordingly, the quantitative analysis should report $\boldsymbol{m}_a$, $\boldsymbol{m}_{G;\mathrm{pl},a}$, $\boldsymbol{m}_{D;\mathrm{pl},a}$, $\mathcal{L}_G^a$, and $\mathcal{L}_D^a$ separately.

For partial opening, the model supports the global statement that foreign investors produce too much information about the inaccessible type-II asset only if the solution satisfies

$$M_{K,(\mathrm{II})} > M_{K,(\mathrm{II});\mathrm{pl}}. \tag{IA.80}$$

More generally, $\boldsymbol{m} \neq \boldsymbol{m}_{\mathrm{pl}}$ establishes social misallocation, while the direction and economic source of the wedge follow from (IA.60). The quantitative exercise can therefore report three objects for each access regime: the decentralized allocation, the planner allocation, and the certainty-equivalent welfare loss. This comparison turns the paper's information-allocation mechanism into a welfare result without changing the underlying trading environment.

IA.J Quantitative Analysis

IA.J.1 Institutional Background

IA.J.1.1. Market

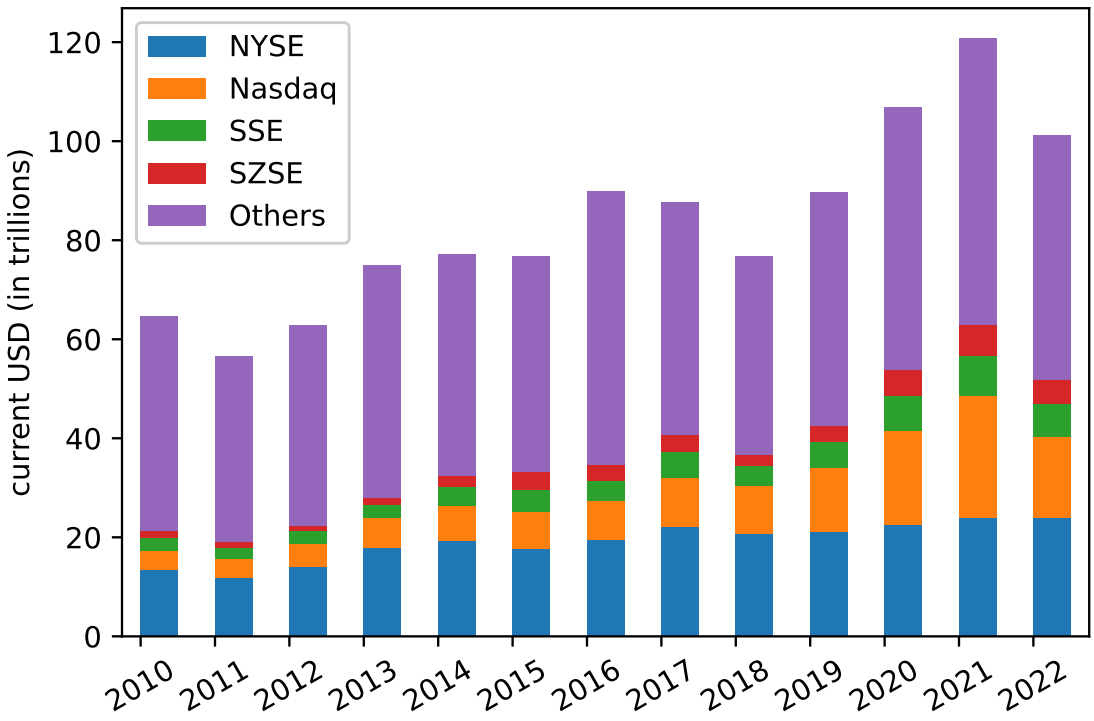


Figure IA.6: Total equity market - market capitalization in current $trillion

IA.J.1.2. China's Stock Market Liberalization Timeline

Following the Shanghai-Hong Kong Stock Connect, more programs are being launched by China to liberalize its capital market, including the Shenzhen-Hong Kong Stock Connect (December 5, 2016) and the Shanghai-London Stock Connect (June 17, 2019). A large scale capital reallocation will be involved as China furthers its stock market liberalization. As of November 22, a total of ¥273 billion capital net flow has been directed to China within one year. Furthermore, on Nov 7, 2019, the State Council of China announced a new policy that will "*eliminate all restrictions on the scope of business for foreign banks, securities companies and fund management companies, and*

Table IA.II
Timeline of China's Stock Market Liberalization[a]

Date	Event
Nov 17, 2014	The Shanghai-Hong Kong Stock Connect was launched.
Dec 5, 2016	The Shenzhen-Hong Kong Stock Connect was launched.
2018	China would relax the limit on foreign investment in joint-venture securities, fund management and futures companies to 51 percent.
June 17, 2019	The Shanghai-London Stock Connect was launched.
July 20, 2019	The State Council of China announced that the plan of lifting the foreign ownership limits on securities companies, fund management companies and futures companies would be advanced by one year to 2020.
Jan 1, 2020	The foreign ownership limit on futures companies was lifted. This limit was 51% for eligible foreign investors.
April 1, 2020	The foreign ownership limit on securities companies was lifted. The new rule was consistent with implementation of the Sino–US Phase One trade deal.

[a] The information was collected from the official website of various Chinese government agencies, including the State Council (http://english.www.gov.cn/statecouncil/).

remove requirements on total assets for the establishment of foreign-funded banks" (State Council of China, No. 23, 2019). Specifically in 2020, foreign investors will be allowed to hold more than 51% of shares in securities companies, fund management companies, futures companies, and life insurance companies. This model provides a new perspective from informational frictions resulting from imperfect data processing technologies and helps predict the impact of China's new policy of the full extension of business scope of foreign financial institutions.

IA.J.1.3. Details of the Stock Connect Programs

Table IA.III
Details of the Stock Connect Programs

Panel A: The Shanghai-Hong Kong Stock Connect

Category	Restrictions
Northbound	
Eligible Shares	Constituents of the SSE 180 Index and SSE 380 Index, and shares of all companies listed on both SSE and SEHK.

continued on next page

Details of the Stock Connect Programs - continued

Quotas[a]	Aggregate quota of RMB300 billion and a daily quota of RMB13 billion
Eligible Investors	No restrictions.
Southbound	
Eligible Shares	Constituents of the Hang Seng Composite LargeCap Index and Hang Seng Composite MidCap Index, and shares of all companies listed on both SSE and SEHK.
Quotas[a]	Aggregate quota of RMB250 billion and a daily quota of RMB10.5 billion.
Eligible Investors	Institutional investors, individual investors with an aggregate balance of not less than RMB 500,000 in their securities and cash accounts.

Panel B: The Shenzhen-Hong Kong Stock Connect

Category	Restrictions
Northbound	
Eligible Shares	Constituents of the SZSE Component Index and SZSE Small/Mid Cap Innovation Index which has a market capitalisation of RMB6 billion or above, and all SZSE-listed shares of companies which have issued both A shares and H shares.
Quotas	Daily quota of RMB13 billion.
Eligible Investors	Investors eligible to trade shares that are listed on the ChiNext Board of SZSE under the Northbound Shenzhen Trading Link will be limited to institutional professional investors as defined in the relevant Hong Kong rules and regulations.
Southbound	
Eligible Shares	Constituent stocks of the Hang Seng Composite LargeCap Index and Hang Seng Composite MidCap Index, any constituent stock of the Hang Seng Composite SmallCap Index which has a market capitalisation of HK$5 billion or above, and all SEHK-listed shares of companies which have issued both A shares and H shares.

continued on next page

Details of the Stock Connect Programs - continued

Quotas	Daily quota of RMB10.5 billion.
Eligible Investors	Institutional investors, individual investors with an aggregate balance of not less than RMB 500,000 in their securities and cash accounts.

Panel C: The Shanghai-London Stock Connect

Category	Restrictions
Eastbound	
Eligible Shares	Eligible companies listed on the London Stock Exchange will be able to issue Chinese Depository Receipts (CDRs) to Chinese investors and apply for them to be listed on the Main Board of the Shanghai Stock Exchange.
Quotas	Aggregate quota of RMB 250 billion.
Eligible Investors	Only qualified securities institutions in each of the two markets may conduct cross-border conversion business in relation to CDRs and GDRs.
Westbound	
Eligible Shares	Eligible companies listed on the Shanghai Stock Exchange will be able to issue Global Depository Receipts (GDRs) to UK and global investors and apply for them to be listed on the London Stock Exchange's Main Market.
Quotas	Aggregate quota of RMB 300 billion.
Eligible Investors	Only qualified securities institutions in each of the two markets may conduct cross-border conversion business in relation to CDRs and GDRs.

[a] The aggregate quota for the Shanghai-Hong Kong Stock Connect was abolished on Aug 16, 2016.

A-shares are issued to domestic investors and qualified foreign institutional investors (QFII). Launched on November 17, 2014, this program allows investors in Shanghai to trade stocks listed in the Hong Kong Exchange (HKEX, southbound trades), and allows investors in Hong Kong to trade those in the SSE (northbound trades). The Shanghai-Hong Kong Connect mainly covers index component stocks with large market cap listed in the SSE and HKEX. All Hong Kong and

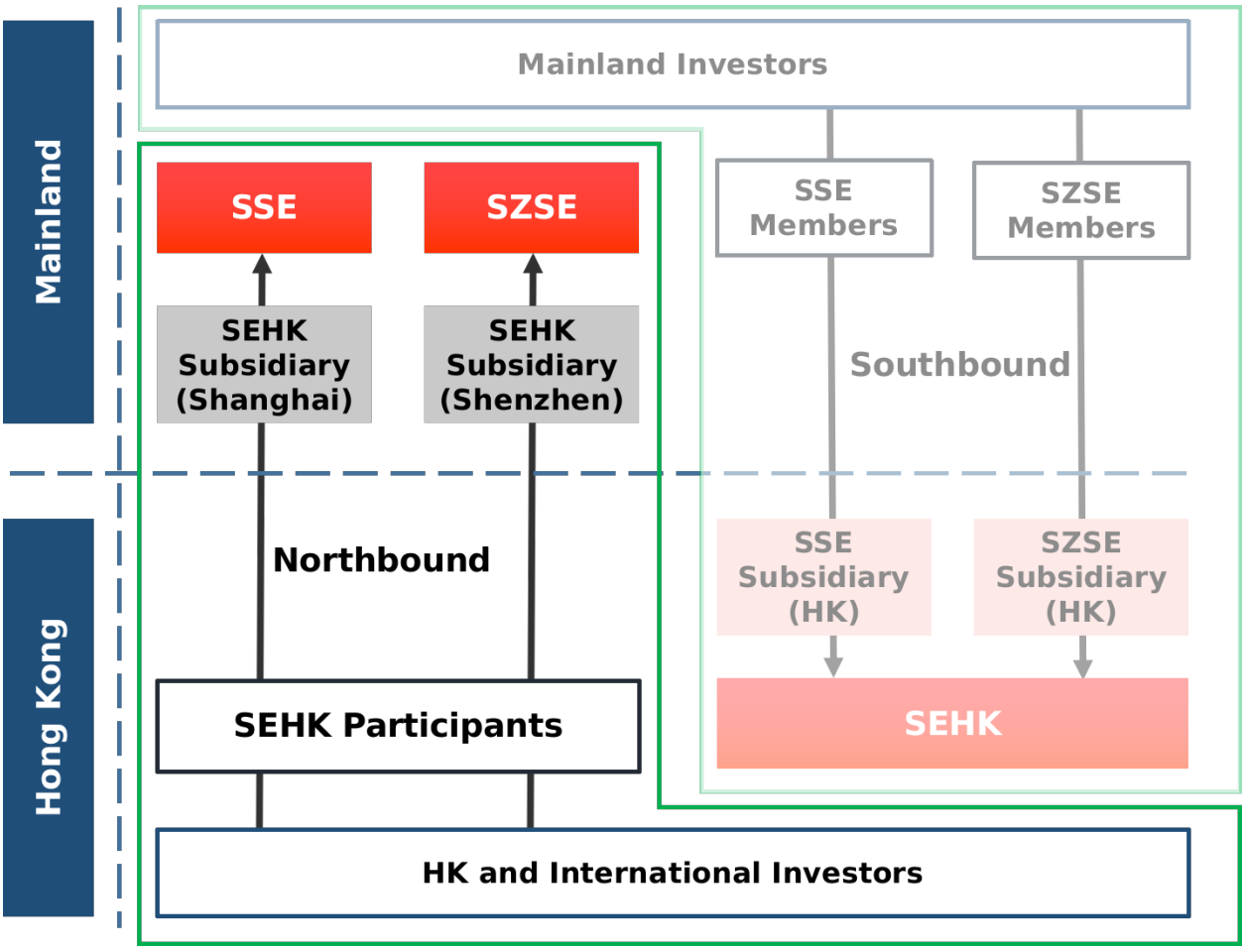


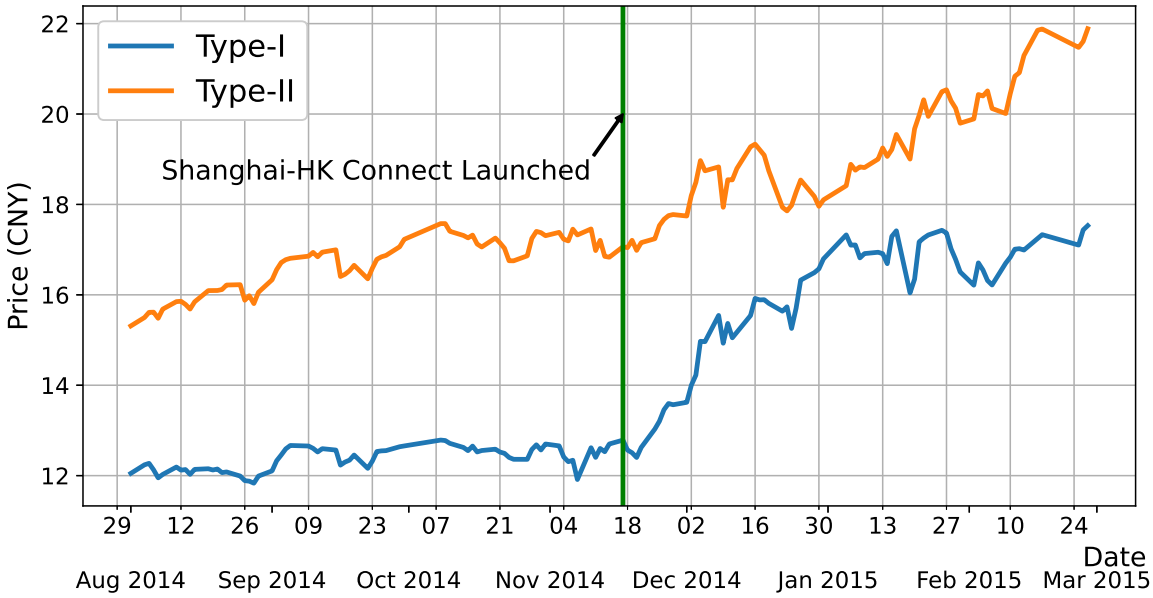
Figure IA.7: Illustration of participants of the Shanghai and Shenzhen Connect programs

overseas investors will be allowed to trade eligible shares listed in Shanghai. However, only Mainland institutional investors and individual investors who have ¥500,000 in their investment and cash accounts are eligible to trade Hong Kong-listed shares. Since its inception, the total trading volume through the Connect has amounted to ¥7.8 trillion, of which ¥4.3 trillion is for northbound trades, and the rest is the trading volume of southbound trades.

IA.J.2 Foreign-Investable Shares at Liberalization Dates

This appendix reports the percentage of foreign-investable shares and market value across markets at the time of stock-market liberalization. Percentages for Shanghai and Shenzhen are calculated using end-of-month values; eligibility is provided by HKSE, and market values by SSE and SZSE. Percentages for other markets are calculated from the raw data in Tables I and II of Chari and Henry (2004).

(a) Aggregate price levels



(b) Annualized volatility

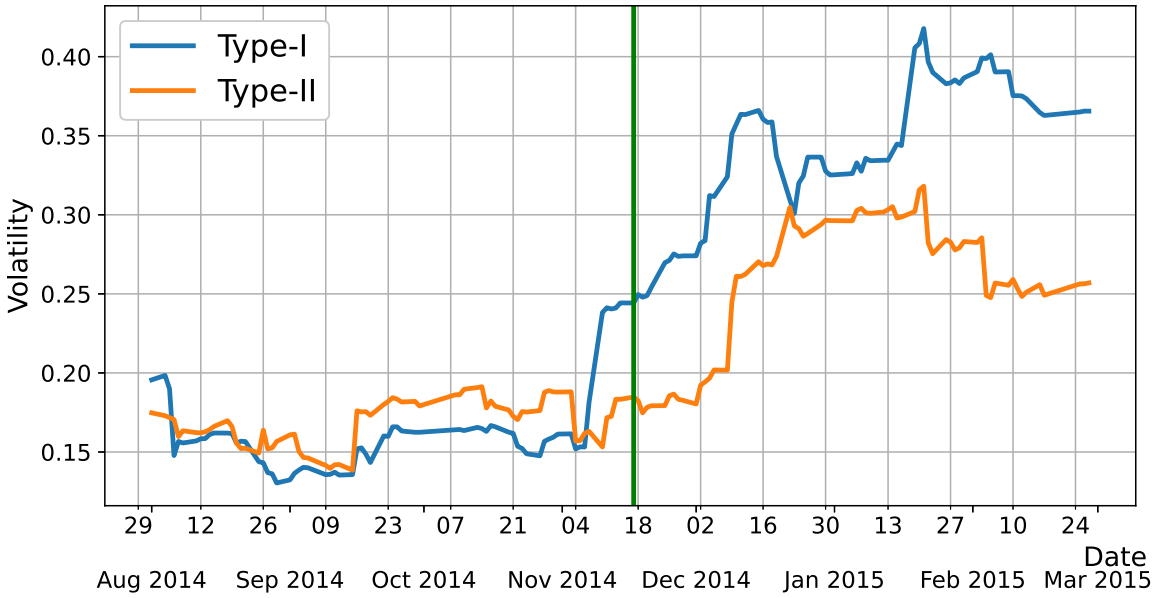


Figure IA.8: Prices and volatilities of A-shares around the launch of the Shanghai-Hong Kong Stock Connect Program. Panel (a) plots the price levels aggregated by negotiable market capitalization and Panel (b) plots the annualized rolling volatility, with a window of 30 trading days. Type-I security is eligible for northbound trade via the Connect while type-II is ineligible.

Table IA.IV

Percentage of foreign-investable shares at market liberalization. Percentages for Shanghai and Shenzhen are calculated using end-of-month values. Eligibility is provided by HKSE, and market values by SSE and SZSE. Percentages for other markets are calculated from the raw data in Tables I and II of Chari and Henry (2004).

Market	Liberalization date	Foreign-investable shares (%)	Foreign-investable market value (%)
Argentina	Sep-89	58.3	-
Brazil	May-91	47.7	-
Chile	Oct-89	34.6	-
Colombia	Dec-91	25.0	-
India	Nov-92	62.9	-
Korea	Jan-92	91.7	-
Mexico	May-89	12.9	-
Pakistan	Feb-91	7.3	-
Taiwan (China)	Jan-91	100.0	100.0
Turkey	Aug-89	64.3	-
Venezuela	Jan-90	38.5	-
Shanghai (China)	Nov-14	59.0	82.8
Shenzhen (China)	Dec-16	53.1	67.0

IA.J.3 Domestic Institutional Holdings and the Investability Split

The calibration adds a moment based on the distribution of domestic institutional holdings across the two asset types. The shares held by domestic institutional investors in type-I (foreign-investable) versus type-II (non-investable) shares are measured from the CSMAR institutional-holdings database as of December 31, 2014. Restricting the institutional categories to professional asset managers (mutual funds, broker-dealers, insurance companies, securities-finance companies, entrusted and managed accounts, finance companies, and banks; excluding industrial capital and QFII accounts), domestic institutions hold approximately 61.5% of their reported holdings in type-I shares and 38.5% in type-II shares, a ratio of roughly 1.60. The calibration targets the log of this ratio, so the estimated domestic allocation $M_I = (\bar{M}, 0)$ is disciplined by the observed holdings distribution through the equilibrium restrictions of the model rather than imposed; the holdings split is not direct evidence for the information allocation.

IA.J.4 Which friction drives the quantitative result?

The counterfactual comparison shows that, in the China calibration, relaxing the short-sale constraint has a much larger effect on price informativeness than expanding foreign investability from partial to full access. The friction-removal exercises reported below shut down the soft-information advantage and the processing-technology gap one at a time. They show that the quantitative result reflects the interaction of asset access with the technology–soft-information endowment, rather than investability in isolation.

IA.J.5 Structural Calibration: Moment Conditions and Explicit Residuals

This appendix reports the explicit residuals used in the minimum-distance computation of the calibration in the main text. The third-stage criterion minimizes ten residuals—three price loadings, three price-noise covariance elements, three ownership shares, and the domestic institutional holdings-ratio residual. The expected-value matrix $\widehat{\boldsymbol{B}}$ is used only in the preliminary step that recovers the fixed dividend-growth matrix $\boldsymbol{G}$; with $\boldsymbol{G}$ fixed, the model-implied $\boldsymbol{B}$ coincides with $\widehat{\boldsymbol{B}}$ by construction, so $\boldsymbol{B}$ residuals carry no identifying information and are not part of the minimized criterion. The residual blocks are stated inline in the calibration section; here we give their full expressions.

IA.J.5.1. Moment identification

The moment restrictions used in the calibration map onto distinct components of the model. Table IA.V summarizes the mapping; the explicit residuals are stated below.

IA.J.5.2. Data Construction for Price-Coefficient Estimation

This subsection details the construction of the variables used in the cross-sectional price regressions of the main text.

Earnings surprises. Unexpected earnings are measured following two methodologies. Under the analyst-forecast approach, all analyst forecasts on earnings per share and the corresponding annual earnings announcement are obtained for the relevant companies; only forecasts reported in the previous fiscal year of the forecast end date are kept; within this set, all but the latest forecast from each analyst are dropped and the median across analysts is taken for each firm and

Table IA.V
Moment-identification mapping.

Moment (residual)	Data target	Identifies
Price loading $\boldsymbol{C}$ (three residuals)	Cross-sectional regression estimate of the price loadings	Information/noise covariance structure; endogenous information acquisition affects equilibrium price sensitivities
Price-noise covariance $\text{vech}[\boldsymbol{D}\boldsymbol{\Sigma}_x\boldsymbol{D}']$ (three residuals)	Residual covariance of the cross-sectional price regression	Price-noise component $\boldsymbol{D}\boldsymbol{\Sigma}_x\boldsymbol{D}'$
Expected-value matrix $\boldsymbol{B}$ (preliminary step, not a third-stage residual)	Used to recover the fixed dividend-growth matrix $\boldsymbol{G}$; with $\boldsymbol{G}$ fixed the model-implied $\boldsymbol{B}$ equals $\widehat{\boldsymbol{B}}$	Growth parameters enter as fixed inputs; $\boldsymbol{B}$ residuals are identically zero and excluded from the criterion
Ownership shares (three residuals, two independent)	Observed holdings of domestic institutions, domestic individuals, and foreign institutions	Relative investor masses $\boldsymbol{\lambda}$ after accounting for equilibrium portfolio demand
Domestic institutional holdings ratio (one residual)	$61.5\%/38.5\%$ split of domestic institutional holdings across type-I and type-II shares	Separation of the investor masses from the information-technology parameters

The cross-asset precision object z is not a moment: it is an endogenous equilibrium object determined by the consistency of the foreign posterior precision with the constrained local price signal, and its equation is reported below as an equilibrium consistency condition rather than an identifying restriction. The matrix $\widehat{\boldsymbol{B}}$ is used exclusively to recover the dividend-growth parameters $\boldsymbol{G}$ through $\boldsymbol{G} = r\widehat{\boldsymbol{B}}(\boldsymbol{I} + \widehat{\boldsymbol{B}})^{-1}$ (fixed inputs to the equilibrium); $\boldsymbol{B}$ residuals are not part of the third-stage criterion, which minimizes the $\boldsymbol{C}$, price-noise-covariance, ownership-share, and holdings-ratio residuals only.

year. Unexpected earnings are the announced earnings minus the median forecast, multiplied by the number of shares outstanding, and divided by total assets ($ue_{i,t}$). Under the time-series approach, annual earnings announcements from 2010 to 2018 are fitted to a VAR(1) model and the residuals are taken as earnings surprises.

Lagged dividends and market prices. Last period's dividend is measured by the previous year's EPS announcement, multiplied by shares outstanding and divided by total assets ($ea_{i,t-1}$). Market prices are normalized identically: the logarithm of market capitalization divided by total assets ($ma_{i,t}$), so the price-loading coefficients measure semi-elasticities of market capitalization with respect to lagged earnings and earnings surprises.

Foreign-investable classification and matching. The foreign-investable shares are the April 10, 2014 Northbound-eligible securities under the Stock Connect rules (Table IA.III): constituents of the SSE 180 and SSE 380 indices plus A-shares of companies listed on both the SSE and the SEHK. Because eligible and ineligible shares are fundamentally different, a propensity score is estimated on total market capitalization, negotiable market capitalization, and industry dummies, and a kNN algorithm matches each type-I asset to its closest type-II neighbor.

Matching robustness of price-coefficient moments. Because the cross-loading C_{12} is central to the cross-asset learning mechanism, we examine its sensitivity to the matching design. Table IA.VI reports the 2014 cross-sectional price-loading moments under three specifications: the baseline propensity-score matching on total market capitalization, negotiable market capitalization, and industry dummies; a no-matching specification in which the cross-earnings term for each type-I asset is the equal-weighted mean of unexpected earnings across all type-II assets; and an alternative matching specification that drops the negotiable-market-capitalization variable from the propensity score. The own-loadings C_{11} and C_{22} are stable across specifications. The cross-loading C_{12} remains positive in all three, but its magnitude is sensitive to the matching design: it is 1.1766 under baseline matching, 0.0342 without matching, and 0.4012 under the alternative variables, with HC1 standard errors of 1.0610, 0.0041, and 1.3770, respectively. The matching design therefore matters for the level of the cross-loading. Given the imprecision of the baseline estimate (HC1 standard error 1.0610 against a coefficient of 1.1766) and the design-dependence of its magnitude (from 1.1766 to 0.0342), we treat C_{12} as a fragile reduced-form target rather than as strong independent evidence for cross-asset learning: its positive sign across designs is suggestive, and the structural conclusions are unchanged when the matching moment vector is replaced (Section IA.J.5).

IA.J.5.3. Sensitivity and robustness of the calibration

This subsection records the calibration sensitivities that have been examined and, explicitly, those that have not, so that the status of each quantitative claim is transparent.

Information technology and soft-information friction. Table IA.VIII shuts down the technology gap ($\beta_I \to \beta$) and the soft-information advantage ($\mathbf{\Omega}_\epsilon \to 0$) one at a time while holding the China calibration fixed. These exercises bound the role of the two endowment asymmetries.

Investability restriction. Removing the investability constraint is the third friction experiment and the full-access counterfactual of the main text. As documented in Section IA.H, the constrained full-access data-choice iteration converges at the current calibration; the unrestricted

Table IA.VI
Matching robustness of price-coefficient moments.

The table reports 2014 price-loading moments under alternative propensity-score matching designs; HC1 standard errors are in parentheses. The first row is the baseline, the second replaces the matched cross-earnings term with the comparison-group mean, and the third omits negotiable market capitalization from the propensity score. C_{22} is identical across rows because its regression contains no matched term.

Specification	C_{11}	C_{12}	C_{22}
(1) Baseline PSM matching	3.7123	1.1766	2.2683
	(1.0374)	(1.0610)	(0.2753)
(2) No matching	3.8319	0.0342	2.2683
	(1.0088)	(0.0041)	(0.2753)
(3) Alternative matching variables	3.7497	0.4012	2.2683
	(1.0304)	(1.3770)	(0.2753)

Table IA.VII
Calibration sensitivity.

Each row perturbs the indicated primitive around its estimate, holds the other primitives fixed, and re-solves the constrained partial-access equilibrium. M_K is foreign data allocation, $\Pr(zero, K)$ is the foreign Type-I zero-holding probability, and VR^p is price-based variance reduction. Panel B varies φ while holding $\mathbf{\Pi}_e$ fixed. All rows converge.

Perturbation	$M_{K,(\mathrm{I})}$	$M_{K,(\mathrm{II})}$	$\Pr(zero, K)$	VR^p_{I}	VR^p_{II}
Baseline (estimated)	2.000	0.000	92.8%	9.32%	4.15%
$\Omega_{x,(\mathrm{I})}, \Omega_{x,(\mathrm{II})} \times 0.50$	2.000	0.000	92.9%	4.88%	2.12%
$\Omega_{x,(\mathrm{I})}, \Omega_{x,(\mathrm{II})} \times 0.75$	2.000	0.000	92.8%	7.15%	3.15%
$\Omega_{x,(\mathrm{I})}, \Omega_{x,(\mathrm{II})} \times 1.25$	2.000	0.000	92.7%	11.39%	5.13%
$\Omega_{x,(\mathrm{I})}, \Omega_{x,(\mathrm{II})} \times 1.50$	2.000	0.000	92.6%	13.37%	6.09%
$\Pi_e \times 0.50$ (Ω_e scaled)	2.000	0.000	85.7%	8.33%	4.14%
$\Pi_e \times 0.75$ (Ω_e scaled)	2.000	0.000	90.0%	8.87%	4.15%
$\Pi_e \times 1.25$ (Ω_e scaled)	2.000	0.000	94.7%	9.72%	4.16%
$\Pi_e \times 1.50$ (Ω_e scaled)	2.000	0.000	96.0%	10.08%	4.16%
Panel B. Relative processing efficiency $\varphi = \beta_I/\beta$					
$\varphi = 1/4$ (i.e. $\beta = 4$)	2.000	0.000	92.7%	8.53%	4.14%
$\varphi = 1/3$ (baseline, $\beta = 3$)	2.000	0.000	92.8%	9.32%	4.15%
$\varphi = 1/2$ (i.e. $\beta = 2$)	2.000	0.000	92.5%	11.24%	4.17%
$\varphi = 2/3$ (i.e. $\beta = 3/2$)	2.000	0.000	91.9%	13.24%	4.19%
$\varphi = 1$ (no technology gap, $\beta = 1$)	2.000	0.000	90.3%	16.20%	4.23%

full-access equilibrium (short selling allowed) is also fully converged.

Ownership-share weighting. The minimum-distance criterion uses the sampling covariance

of the price-coefficient and price-noise moments and a ten-percentage-point economic scale for ownership shares. The main text notes that the reported objective is therefore not a conventional GMM J statistic.

Supply-noise and data-precision sensitivity. Table IA.VII varies the supply-noise precisions $\Omega_{x,(\mathrm{I})}, \Omega_{x,(\mathrm{II})}$ and the effective data precision $\Pi_e \equiv \beta^2 \Omega_e$ (via $\Omega_{e,(\mathrm{I})}, \Omega_{e,(\mathrm{II})}$, holding the type-I effective precision at its estimate and the type-II precision at its corner-supporting evaluation point) by $\pm 25\%$ and $\pm 50\%$ around the estimated values, and reports the foreign data allocation, the foreign type-I zero-holding probability, and the price-based variance reduction at the fully converged constrained partial-access equilibrium. The corner allocation $M_K = (\bar{M}, 0)$—foreign institutions allocate all processing capacity to type-I data—is unchanged at every perturbation point, the zero-holding probability moves only moderately (between 85.7% and 96.0%), and the ranking $VR^p_{\mathrm{I}} > VR^p_{\mathrm{II}}$ is preserved throughout. The mechanism documented in the main text is therefore not knife-edge in these two primitive dimensions.

Welfare decomposition across policy margins. A decomposition of the welfare change into "remove the access restriction only," "remove the short-sale constraint only," and the information-allocation increment is not reported. The constrained full-access equilibrium converges at the current calibration, so the "remove access only" step is computable in principle. We nevertheless leave the full policy-margin decomposition to a future revision: it would require an explicit convention for assigning the certainty-equivalent gains across the three margins, and a decomposition built on the current welfare numbers (Table IV of the main text) would therefore carry a convention-dependent allocation rather than an unambiguous accounting identity.

Diagnostics on the estimated precision ratio and the processing technology. The working calibration files also record diagnostics on the precision ratio and the processing technology, which are summarized here because they directly address referee concerns. First, the domestic-institutional holdings ratio disciplines the domestic corner through the equilibrium restrictions of the model: the moment (61.5% of domestic institutional holdings in type-I shares) is matched at $M_I = (\bar{M}, 0)$, but the mapping from the observed holdings split to a $100/0$ information allocation runs entirely through the equilibrium, so the holdings distribution is not direct evidence for the corner. The estimated type-I effective precision is much higher than the type-II evaluation point, but the type-II parameter is set- rather than point-identified: only the corner-supporting region of type-II precision is identified from the data. Second, the sensitivity table shows that the allocation corners $M_I = (\bar{M}, 0)$ and $M_K = (\bar{M}, 0)$ are unchanged when the supply-noise precisions and the effective data precision are scaled by $\pm 25\%$ and $\pm 50\%$ around the estimates, and the partial profiles below show that they are unchanged over wide ranges of λ_K and ρ: the specialization

result does not hinge on the exact estimated ratio. Third, Table IA.VII reports a profile over the relative processing efficiency $\varphi = \beta_I/\beta \in \{1/4, 1/3, 1/2, 2/3, 1\}$; the allocation corners are unchanged at every value, and the type-I price-based variance reduction rises from 8.5% to 16.2% as φ increases (as domestic processing becomes relatively more efficient, the type-I information advantage is more fully transmitted into prices). The mechanism therefore does not depend on the specific estimated ratio or on the fixed value $\varphi = 1/3$. Finally, the capacity normalization $\bar{M} = 2$ is immaterial: the equilibrium is numerically invariant to rescaling $\bar{M} \mapsto a\bar{M}$ together with $\mathbf{\Pi}_e \mapsto \mathbf{\Pi}_e/a$, so the data choices can be read as shares of a unit processing budget.

Table IA.VIII

Friction-removal experiments. Each experiment is evaluated relative to its closed-market counterfactual. The no-soft-information exercise uses $\Omega_\epsilon = 10^{-4}$, the smallest value at which the constrained solver converges.

Experiment	$M_{K,(\mathrm{II})}$	Δ Type-I SNR	Δ average posterior precision
Baseline China (partial)	0.000	-0.132	$+115.740$
No soft information: $\Omega_\epsilon \to 0$	0.000	-6.939	$+150.015$
No technology gap: $\beta_I \to \beta$	0.000	-0.132	$+85.292$
Full opening with constraint	0.000	-0.132	$+115.740$

Under the unrestricted benchmark the model price coefficients admit closed-form expressions, which we state below to make the residual definitions explicit. In the estimated constrained partial-access economy, the same residuals are evaluated at the model price coefficients obtained numerically from the constrained local fixed point (Internet Appendix Section IA.F), with the no-short-sale constraints imposed asset by asset and the data choices solved as an inner fixed point; the residual definitions—model coefficient minus data moment—are identical in the two

implementations, and only the numerical evaluation of $\boldsymbol{C}$ and $\boldsymbol{D}$ differs.

$$
\begin{aligned}
g_C(\boldsymbol{\theta}, r, \boldsymbol{X}) = (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \Big[& \bar{\hat{\boldsymbol{\Omega}}}_y - \boldsymbol{\Omega}_y + \frac{\lambda_K}{\lambda_2}(\Omega_{y,12} + \Omega_{p,12})\boldsymbol{E}_{12} + \frac{\lambda_K \hat{\mathcal{C}}_{Ky}^2}{\lambda_2 \hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{11} \\
& - \frac{\lambda_K \hat{\mathcal{C}}_{Ky}}{\lambda_2 \hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12}(\boldsymbol{\Omega}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e) \Big] - \boldsymbol{C}, \qquad \text{(IA.81)}
\end{aligned}
$$

$$
\begin{aligned}
\boldsymbol{D}(\boldsymbol{\theta}, r, \boldsymbol{X}) = & -\rho r (r\boldsymbol{I} - \boldsymbol{G})^{-1\prime}\bar{\hat{\boldsymbol{\Sigma}}}_y \Big[\bar{\hat{\boldsymbol{\Omega}}}_y - \boldsymbol{\Omega}_y + \frac{\lambda_K}{\lambda_2}(\Omega_{y,12} + \Omega_{p,12})\boldsymbol{E}_{12} + \frac{\lambda_K \hat{\mathcal{C}}_{Ky}^2}{\lambda_2 \hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{11} \\
& - \frac{\lambda_K \hat{\mathcal{C}}_{Ky}}{\lambda_2 \hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12}(\boldsymbol{\Omega}_p + \boldsymbol{M}_K \boldsymbol{\Pi}_e) \Big] \\
& \times \left[\left(\lambda_I \varphi^2 \boldsymbol{M}_I + \lambda_K \left(\boldsymbol{E}_{11} - \frac{\hat{\mathcal{C}}_{Ky}}{\hat{\Omega}_{Ky,(\mathrm{II})}}\boldsymbol{E}_{12} \right) \boldsymbol{M}_K \right) \boldsymbol{\Pi}_e + \lambda_1 \boldsymbol{\Omega}_\epsilon \right]^{-1} (r\boldsymbol{I} - \boldsymbol{G})^{-1}, \qquad \text{(IA.82)}
\end{aligned}
$$

$$
g_D(\boldsymbol{\theta}, r, \boldsymbol{X}) = \mathrm{vech}\big[\boldsymbol{D}(\boldsymbol{\theta}, r, \boldsymbol{X}) \boldsymbol{\Sigma}_x(\boldsymbol{\theta}) \boldsymbol{D}(\boldsymbol{\theta}, r, \boldsymbol{X})' - \widehat{\boldsymbol{\Sigma}}_p \big], \qquad \text{(IA.83)}
$$

$$
\begin{aligned}
g_z(\boldsymbol{\theta}, r, \boldsymbol{X}) = & \mathcal{C}_y - \left(\frac{r - g_1}{\rho r} \right)^2 \lambda_K z M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} \Big[(\lambda_I \varphi^2 M_{I,(\mathrm{I})} \\
& + \lambda_K M_{K,(\mathrm{I})}) \Pi_{e,(\mathrm{I})} + \lambda_1 \Omega_{\epsilon,(\mathrm{I})} \Big] \Omega_{x,(\mathrm{I})} \\
& - z \Bigg\{ \Omega_{y,(\mathrm{II})} + M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} + \left[\frac{r - g_1}{\rho r} \lambda_K z M_{K,(\mathrm{II})} \Pi_{e,(\mathrm{II})} \right]^2 \Omega_{x,(\mathrm{I})} \\
& + \left[\frac{r - g_2}{\rho r} \left[\lambda_I \varphi^2 M_{I,(\mathrm{II})} \Pi_{e,(\mathrm{II})} + \lambda_1 \Omega_{\epsilon,(\mathrm{II})} \right] \right]^2 \Omega_{x,(\mathrm{II})} \Bigg\}, \qquad \text{(IA.84)}
\end{aligned}
$$

The last condition (IA.84) is the equilibrium consistency of the foreign posterior cross-precision z: given the parameters and the constrained local price signal, it determines the endogenous cross-asset learning object rather than matching a separately observed empirical moment $\widehat{z}$. The identifying content of the moment system therefore comes from the price-loading restrictions g_C and the price-noise covariance restriction g_D, together with the ownership-share and holdings-ratio conditions of the main text; the three ownership-share conditions contain two independent restrictions (shares sum to one), and the domestic-institutional holdings ratio (Section IA.J.3 of the data appendix) adds a further independent restriction, so the equality system provides nine independent restrictions. As documented in the main text and in the numerical local-identification analysis below, the Type-II effective-precision direction is locally null at the estimated corner, so eight structural directions are locally point-identified while $\Pi_{e,(\mathrm{II})}$ is restricted by the corner-optimality inequality rather than point identified. The holdings-ratio

moment is the restriction that separates the investor masses from the information-technology parameters: it pins down how institutional processing capacity is allocated across the two asset types, and it is matched at the estimated domestic corner $M_I = (\bar{M}, 0)$.

To assess how tightly the estimated parameters are pinned by the moments, we compute partial profiles for risk aversion ρ and the foreign institutional mass λ_K, holding all other parameters at their estimated values and re-solving the constrained equilibrium at every point. Two findings stand out. First, the equilibrium converges at every profile point, and the estimated information-allocation corners $M_I = (\bar{M}, \mathbf{0})$ and $M_K = (\bar{M}, \mathbf{0})$ are unchanged throughout, so the benchmark specialization result does not hinge on the exact values of either parameter. Second, the objective is substantially more sensitive to λ_K than to ρ: deviating from the estimated $\lambda_K = 0.0195$ in either direction raises the objective—at $\lambda_K = 0.01$ it is 1.380 and at 0.05 it is 2.253, versus 1.2587 at the estimate (with λ_I, λ_J rescaled proportionally)—while moving ρ within $[0.8, 1.6]$ changes the objective by at most 1.2% (1.273 at $\rho = 0.8$ declining to 1.2587 at $\rho = 1.6$). The ownership-share and holdings-ratio conditions therefore carry the identifying content for the investor masses, whereas ρ is pinned jointly with the price-loading moments. The ρ-profile is monotone declining over $[0.8, 1.6]$, which reflects the partial nature of the exercise (other parameters are held fixed rather than re-estimated) rather than evidence against joint identification.

We also examine the numerical local identification of the moment system directly at the estimated parameter vector. Let $\boldsymbol{r}(\boldsymbol{\theta})$ denote the ten weighted moment residuals—three price loadings $\boldsymbol{C}$, three price-noise covariance elements, three ownership shares, and the holdings-ratio residual—which constitute exactly the residual vector minimized in the third-stage criterion, and let $\boldsymbol{J} = \partial \boldsymbol{r} / \partial \boldsymbol{\theta}'$ be the 10×9 Jacobian evaluated by central finite differences at the estimate. Because raw Jacobian singular values depend on parameter units (the estimated parameters range from 0.02 to more than $7{,}000$), we report the singular values of the parameter-scaled Jacobian $\boldsymbol{J}\,\mathrm{diag}(\boldsymbol{\theta})$: $\{881.8,\ 27.5,\ 3.16,\ 0.774,\ 0.586,\ 0.293,\ 7.0 \times 10^{-3},\ 1.1 \times 10^{-3},\ 8.5 \times 10^{-12}\}$. The weakest right-singular vector is, to numerical precision, the coordinate direction of the Type-II effective precision $\Pi_{e,II}$ (loading 1.0 on $\Pi_{e,II}$ and at most 10^{-8} on every other parameter). Consistent with this direction, the objective is essentially flat along $\Pi_{e,II}$: perturbing $\Pi_{e,II}$ by $\pm 20\%$ leaves the objective at 1.2587, unchanged to three decimals. This flatness is a direct consequence of the corner solution $M_I = (\bar{M}, \mathbf{0})$: at the estimated allocation no investor type processes Type-II data in equilibrium, so $\Pi_{e,II}$ enters the objective only through the corner-optimality boundary condition—the Type-II data-allocation optimality inequality must hold at the boundary—rather than through any equilibrium quantity reported in the paper. Within the region in which that inequality holds, every reported equilibrium object is invariant to the exact value of $\Pi_{e,II}$. The in-

vestor masses λ_I and λ_J are, by contrast, strongly identified: perturbing either by $\pm 5\%$ raises the objective to between 1.42 and 2.74, consistent with the ownership-share and holdings-ratio restrictions carrying the identifying content for the masses. The next-weakest scaled directions load on the difference $\Omega_{x,(\mathrm{I})} - \Omega_{x,(\mathrm{II})}$ (loadings ± 0.707) and on a λ_J/λ_I combination ($-0.959/-0.282$); excluding the $\Pi_{e,(\mathrm{II})}$ direction leaves a scaled condition number of roughly 8.1×10^5. The Ω_x-difference direction being weakly identified is consistent with the two estimated supply-noise precisions being numerically close, and both parameters are coded independently (separate coordinates, no common bound or transformation), so their proximity is a property of the optimum rather than a coding artifact.Because the sensitivity checks elsewhere in this section scale the two supply-noise precisions together, we examine the $\Omega_{x,(\mathrm{I})}/\Omega_{x,(\mathrm{II})}$ ratio directly. Holding $\Omega_{x,(\mathrm{II})}$ at its estimate and varying $\Omega_{x,(\mathrm{I})}$ over $\{0.003, 0.004, 0.005, 0.010, 0.015, 0.020\}$ with the other primitives fixed raises the objective from 1.2587 at the estimate to 1.620 at 0.005 and 1.276 at 0.010, and to 2.06–2.90 at the extremes; the allocation corner $M_I = M_K = (\bar{M}, \mathbf{0})$ is unchanged at every profile point. Constraining $\Omega_{x,(\mathrm{I})} = \Omega_{x,(\mathrm{II})} = 0.007692$ and re-estimating the remaining seven parameters yields an objective of 1.25802 (versus 1.25865 at the unconstrained estimate) with the other estimates essentially unchanged, confirming that the weakly identified supply-noise difference is not what produces the benchmark conclusions. Identification is strong along the share-moment directions; we interpret the system as having eight locally point-identified structural directions, with the corner-irrelevant Type-II precision direction restricted by the corner-optimality inequality rather than point identified.

The price-noise block is the weakest-fitting block of the moment system (marginal residuals $+3.00$ and -3.51 for the two diagonal elements, expressed relative to the tight sampling standard errors of the underlying moments). The joint objective is nonetheless only 1.2587 because the criterion weights the price-noise residuals by unit weight rather than by their sampling precision: the residuals are small in level (0.066 and -0.103 in precision units), so their unit-weighted contributions are 0.004 and 0.011, and the joint criterion is dominated by the other moment blocks. The large marginal statistics reflect the very tight sampling precision of the price-noise moments (the empirical covariances are estimated on several hundred matched pairs), not a large absolute miss in the model's noise covariance. We do not report a block-specific Mahalanobis statistic because the estimation weights each residual by unit weight rather than by the sampling covariance of the blocks (Section IA.J.5).Robustness to the measurement assumption: halving the weight on the price-noise block and re-estimating all nine parameters leaves every estimate and the allocation corners $M_I = M_K = (\bar{M}, \mathbf{0})$ unchanged (the criterion falls to 1.251 only because the block is downweighted), so the imperfect fit of the two diagonal noise moments does not

drive the estimated information allocation. The exercise demonstrates robustness to the block's weight; it does not validate the maintained measurement mapping from the regression residual covariance to the aggregate supply-noise covariance, which also absorbs omitted fundamentals, measurement error, and misspecification.

Table IA.IX reports three re-estimation exercises that probe whether the benchmark survives less favorable information-technology and weighting assumptions. Panel A re-estimates all parameters under alternative values of the maintained foreign processing advantage $\beta \in \{1, 1.5, 2, 3\}$; Panel B re-estimates all remaining parameters with the effective-precision ratio fixed at $\Pi_{e,(\mathrm{I})}/\Pi_{e,(\mathrm{II})} \in \{50, 20, 10, 5, 3, 1\}$; Panel C varies the economic scale of the ownership-share moments (scale $\in \{0.20, 0.10, 0.04\}$) and re-estimates. Conditional re-estimation at every $\beta \in \{1, 1.5, 2, 3\}$ returns the same best-converged ten-moment objective (1.2405) and the same allocation corners, because the calibration estimates the effective data precision $\mathbf{\Pi}_e = \beta^2 \mathbf{\Omega}_e$ directly. Multi-start re-estimation at every fixed β and at every ratio $R \in \{3, \ldots, 50\}$ locates a lower-objective basin (1.2405–1.2469) in which risk aversion collapses to ρ between 10^{-4} and 10^{-3} (and to 0.29–0.56 at $R = 5, 3$), an economically extreme region relative to the benchmark value $\rho = 1.46$; the economically relevant comparison therefore retains the benchmark's locally selected solution (objective 1.2587), with the best-converged multi-start objectives and their parameter relocations reported in Table IA.IX and its notes. Because the criterion is not a conventional J-statistic, the reported objectives are comparisons of fit rather than formal tests. The benchmark allocation corners $M_I = M_K = (\bar{M}, \mathbf{0})$ and the ranking $VR^p_{\mathrm{I}} > VR^p_{\mathrm{II}}$ are preserved in every economically relevant re-estimation except $R = 1$ (equal Type-I and Type-II effective precision), where the domestic corner flips to $M_I = (0, \bar{M})$ (with $\rho = 0.71$, below the benchmark value 1.46). The best-converged objective at $R = 1$ (1.2919) is about 2.6% above the benchmark local solution, so the fit itself does not strongly reject equal precision; excluding this region requires an external restriction—on the admissible range of risk aversion ρ or on the relative precision itself—rather than a statistical rejection. Without such a restriction, the benchmark conclusions are documented as locally robust and over $R \in [3, 50]$, but not globally robust to the equal-precision specification. Over the empirically relevant range $R \in [3, 50]$, the solution preserving the benchmark primitives has objective 1.2587 at every ratio; the earlier R-to-R alternation between 1.258 and 1.300 traced to path dependence in the inner constrained-equilibrium fixed-point solve across profile evaluations rather than to any economic response, and the equilibrium objects at each such local solution (Table IA.X) are identical to the benchmark. Table IA.X takes the analysis one step further and re-solves all four counterfactual regimes at each economically relevant re-estimated parameter vector, so that the two quantitative con-

clusions of the paper are tested directly rather than through the auxiliary allocation-corner and variance-reduction-ranking statistics.

Table IA.IX
Re-estimation robustness of the benchmark.

Panels A–C re-estimate under alternative technology and weighting assumptions using the benchmark ten-moment criterion and frozen empirical moments. $R \equiv \Pi_{e,(\mathrm{I})}/\Pi_{e,(\mathrm{II})}$; Scale is the ownership-share scale. Objectives are the best converged values across three starts. The ten-moment criterion is invariant to β.

	Objective	M_I	M_K	VR^p_{I}	VR^p_{II}
Panel A: Processing advantage β (re-estimated)					
$\beta = 1$	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.85%	3.01%
$\beta = 1.5$	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.77%	3.01%
$\beta = 2$	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.60%	3.01%
$\beta = 3$ (benchmark)	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.55%	3.01%
Panel B: Effective-precision ratio R (re-estimated)					
$R = 50$	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.54%	3.01%
$R = 20$	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.55%	3.01%
$R = 10$	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.55%	3.01%
$R = 5$	1.2408	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.25%	3.04%
$R = 3$	1.2469	$(\bar{M}, 0)$	$(\bar{M}, 0)$	9.58%	3.53%
$R = 1$	1.2919	$(0, \bar{M})$	$(\bar{M}, 0)$	9.06%	7.43%
Panel C: Ownership-share moment scale (re-estimated)					
0.20	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.55%	3.01%
0.10 (benchmark)	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.55%	3.01%
0.04	1.2405	$(\bar{M}, 0)$	$(\bar{M}, 0)$	11.54%	3.01%

Table IA.X
Calibration robustness and quantitative conclusions.

The table re-solves closed, partial-access, constrained-full-access, and unrestricted-full-access counterfactuals at the re-estimated parameters of Table IA.IX. Active foreign mass equals λ_K times Type-I participation. [§] denotes a partial-to-unrestricted comparison because constrained full access does not converge at $R = 50$; [‡] denotes an equal-precision case not evaluated in unrestricted full access.

	$\Delta VR_{\mathrm{I}}^{p}$ (closed→partial)	$\Delta VR_{\mathrm{II}}^{p}$ (closed→partial)	Foreign participation	Active foreign mass	$\Delta VR_{\mathrm{I}}^{p}$ (partial→full^{c})	$\Delta VR_{\mathrm{I}}^{p}$ (full^{c}→full^{u})
Benchmark	$+0.47$ pp	-0.03 pp	7.2%	0.14%	≈ 0	$+56.6$ pp
$\beta = 1.5$ (re-estimated)	$+1.21$ pp	-0.04 pp	12.0%	0.41%	≈ 0	$+61.7$ pp
$\beta = 2$ (re-estimated)	$+0.68$ pp	-0.02 pp	5.3%	0.11%	≈ 0	$+66.0$ pp
$R = 5$ (re-estimated)	$+0.47$ pp	-0.03 pp	7.2%	0.14%	-0.1 pp	$+54.3$ pp
$R = 50$ (re-estimated)	$+0.47$ pp	-0.03 pp	7.2%	0.14%	(not converged)	$+56.6$ pp[§]
Scale 0.20 (re-estimated)	$+0.03$ pp	-0.02 pp	0.3%	0.003%	≈ 0	$+73.1$ pp
Scale 0.04 (re-estimated)	$+0.47$ pp	-0.03 pp	7.2%	0.14%	≈ 0	$+56.6$ pp
$R = 1$ (equal precision)[‡]	$+0.00$ pp	-0.04 pp	$\approx 0.0\%$	$\approx 0.0\%$	≈ 0	n.a.

Table IA.X shows that both quantitative conclusions of the paper survive all reported fully converged economically relevant re-estimated specifications. The $R = 1$ row (equal effective precision) is reported with an exclusion note: at that point the best-converged solution flips the domestic corner to $M_I = (0, \bar{M})$ (with $\rho = 0.71$, below the benchmark 1.46), and the unrestricted full-access equilibrium does not converge, so neither conclusion is evaluated for that row. As discussed above, the $R = 1$ fit is only about 2.6% worse, so excluding this region rests on the flipped information allocation and the relocation of the other primitives rather than on a statistical rejection. The $\beta = 1$ point is likewise excluded from the table because it fits substantially worse under the chosen eleven-moment criterion (objective 1.947, driven by the forecast-error moment). First, the closed-to-partial change in type-I price-based variance reduction is positive in all seven rows ($+0.03$ to $+1.21$ percentage points) while the active foreign mass that produces it is at most 0.41%—and as little as 0.003% under the extreme ownership-weighting point—so the price-informativeness movement is disproportionate to the participating foreign sector. As the main text emphasizes, the absolute magnitude is moderate: the mechanism is that a quantitatively tiny foreign sector can exert a non-proportional informational influence, not that it produces large movements in every market statistic. Second, in every converged row the constrained full-access regime delivers essentially the same type-I price informativeness as the partial-access benchmark (differences of at most 0.1 percentage point), whereas removing the short-sale constraint raises type-I price-based variance reduction by roughly 54 to 73 percentage points. The conclusion that asset access and trading flexibility are complementary—with portfolio flexibility the dominant margin at the Chinese calibration—is therefore not an artifact of the benchmark parameter point.The most defensible statement of the two quantitative conclusions is accordingly narrower: within a locally selected, severe-constraint calibration and over the reported $R \geq 3$ profiles, relaxing short-sale constraints has a much larger modeled effect on price-information aggregation than expanding investability, while the evidence does not yet globally select this mechanism over the equal-precision alternative. The single non-converged cell ($R = 50$, constrained full access) is reported as such; because that row lacks a valid constrained full-access equilibrium, the full-access short-sale contrast for this row is reported as partial-constrained-to-unrestricted-full ($+56.6$ percentage points), which does not require the non-converged equilibrium; the benchmark row, where the constrained full-access equilibrium does converge, reports the full-constrained-to-unrestricted contrast of $+56.6$ percentage points (the partial- and full-constrained type-I price-informativeness values coincide at 9.32%). Matching-design sensitivity is documented separately in Table IA.VI: the sign of the cross-loading C_{12} is stable across matching designs, although its level is sensitive to the design. A full structural re-estimation under the

alternative matching moment vectors has now been carried out: re-estimating all nine parameters with the no-matching and alternative-matching price-loading targets of Table IA.VI (holding the price-noise, ownership, and holdings-ratio moments at their baseline values) leaves the allocation corners $M_I = M_K = (\bar{M}, \mathbf{0})$ and the type-I specialization ranking unchanged, with objective values of 0.0002 and 0.1185 respectively (versus 1.2587 at the baseline moments), all under the same weighting matrix (Section IA.J.5); type-I price-based variance reduction moves to 12.6% and 12.0% under the two designs (type-II to 2.9% and 3.0%), while the allocation corners and the type-I specialization ranking are unchanged. The cross-loading-level sensitivity documented in Table IA.VI therefore does not propagate to the structural conclusions.Because the large short-sale effects are computed under calibrations in which the constraint binds frequently (benchmark foreign participation of 7.2%, ranging from 0.3% to 12.0% in Table IA.X), we searched for parameterizations with materially lower binding probabilities by raising the foreign institutional mass (λ_K up to 0.20), lowering risk aversion, widening the supply-noise variance, and removing the processing-technology gap ($\beta = 1$). Participation never exceeded about 10% while preserving convergence of the constrained equilibrium, so the short-sale result remains documented under severe-constraint calibrations; within that family the margin remains dominant—at the least-binding points found ($\lambda_K = 0.20$: participation 8.7%; $\lambda_K = 0.10$ with $\beta = 1$: participation 10.0%), constrained full access still reproduces the partial-access benchmark ($15.19\%/3.83\%$ and $19.23\%/4.09\%$), while unrestricted full access reaches $88.93\%/29.23\%$ and $93.46\%/34.24\%$, Type-I gaps of roughly 74 percentage points.

IA.J.6 Reduced-Form Evidence: Data and Robustness

This appendix provides implementation details for the reduced-form analysis in Section F. The main text uses this evidence descriptively, as a complement to the structural analysis. It reports the data construction, alternative specifications, and robustness exercises.

IA.J.6.1. Institutional Timeline and Sample Construction

The empirical setting is the Shanghai–Hong Kong Stock Connect, which was announced on April 10, 2014 and launched on November 17, 2014. The eligible group is defined using announcement-date eligibility: it comprises the full set of stocks in the April 10, 2014 reference universe satisfying the Northbound eligibility criteria, and eligible status is retained for the entire sample window regardless of subsequent changes in Stock Connect eligibility. The comparison group consists of stocks that never became eligible, drawn from both the Shanghai and Shenzhen exchanges: of the 2,243 comparison stocks, 443 are Shanghai-listed and 1,800 are Shenzhen-listed

(1,254 Shenzhen main-board and SME stocks and 546 ChiNext stocks). This definition fixes the eligible group using eligibility information known at the announcement and avoids conditioning eligible-group membership on post-treatment eligibility outcomes; comparison-group membership is by construction defined using eligibility over the full sample, so the estimates should be read with that qualification. The estimation sample covers trading days from January 1, 2013 to December 4, 2016. The April 10, 2014 reference universe contains 568 stocks. Of these, 118 stocks that later lost eligibility are retained in the eligible group under the announcement-date eligibility definition, and 119 stocks that became eligible only after launch are excluded from the baseline estimation sample. Six stocks from this reference universe are missing from the estimation panel because of delisting or missing daily data. The estimation sample therefore comprises 562 eligible stocks and 2,243 comparison stocks, with 2,067,269 firm-day observations. The announcement period runs from April 10, 2014 through November 16, 2014, and the post-launch period begins on November 17, 2014. The mean idiosyncratic volatility of eligible stocks during the pre-announcement period is 0.317.

IA.J.6.2. Variable Construction

All variables are constructed from daily trading data for A-shares in normal trading status (daily returns, market values, and trading values) over the sample window. The primary outcome is idiosyncratic volatility: the annualized standard deviation (scaled by $\sqrt{252}$) of the residuals from a market-model regression of daily returns on the equal-weighted market return (the cross-sectional mean of daily returns across all A-shares in normal trading status over the sample window), computed over rolling 30-trading-day windows requiring at least 30 observations. Total volatility is the 30-trading-day annualized standard deviation of daily returns. The Amihud illiquidity measure is the daily absolute return divided by trading value (in yuan). The absolute market-model residual is the 30-trading-day mean of the absolute market-model residual. Return synchronicity is the logit of the R^2 from a rolling 120-trading-day market-model regression requiring at least 60 observations. All rolling outcome measures are constructed on each firm's complete daily series before the estimation-sample restrictions are imposed, so every rolling window initializes on the same per-firm history; the non-missing observation counts therefore differ across outcomes only through the window lengths (30 trading days for idiosyncratic volatility, total volatility, and the absolute market-model residual, and 120 trading days for synchronicity). All outcome and control variables are winsorized at the 1st and 99th percentiles. The controls are log market capitalization, turnover, and daily return. Amihud illiquidity is reported in raw units (daily absolute return divided by trading value in yuan). Standard errors are two-way clus-

tered by firm and date in all reported specifications, including the baseline, alternative-outcome, event-study, and robustness tables.

IA.J.6.3. Reduced-Form Specifications

The baseline specification is

$$Y_{it} = \alpha_i + \gamma_{j(i),t} + \beta_A(\text{Eligible}_i \times \text{Announcement}_t) + \beta_L(\text{Eligible}_i \times \text{Post}_t) + \varepsilon_{it},$$

where $\gamma_{j(i),t}$ denotes the industry-by-day fixed effect in the preferred design. We report implementation-date-only specifications, announcement-period specifications, and alternative event-study formulations that separately describe the announcement-to-launch and post-implementation dynamics. As a robustness exercise, we augment this specification with contemporaneous controls $\boldsymbol{X}_{it}$ (log market capitalization, turnover, and the daily return); the preferred specification absorbs firm and industry-by-day fixed effects and omits them, because these variables can themselves respond to the liberalization and are mechanically related to return-based outcomes. Three estimands are reported throughout: β_A, the announcement-to-launch differential relative to the pre-announcement baseline; β_L, the post-launch level differential relative to the same baseline; and $\beta_L - \beta_A$, the incremental post-launch contrast. Because the April 10, 2014 announcement was known to market participants before trading began, $\beta_L - \beta_A$ isolates the additional differential change after implementation relative to the announcement period, and we report it as the incremental post-launch contrast.

IA.J.6.4. Robustness and Additional Outcomes

This appendix reports alternative return-based market outcomes, placebo dates, and alternative sample windows. These exercises are designed to assess the robustness of the reduced-form association and are not interpreted as structural identification exercises. The full-sample specifications reported in this appendix are additional robustness exercises rather than alternative headline specifications; the preferred design is the Shanghai-only comparison of Table V in the main text.

IA.J.6.5. Baseline Regression Tables

Adding the contemporaneous firm-level controls materially changes the post-launch coefficients in column (4) of Table IA.XI: the post-launch differential reverses sign and the incremental contrast becomes strongly significant. Because these variables can themselves respond to the liber-

Table IA.XI
Baseline reduced-form specifications.

The outcome is idiosyncratic volatility. Column (1) is the preferred Shanghai-only specification; columns (2)–(4) are full-sample robustness specifications, with exchange-by-day effects in column (2) and contemporaneous controls in column (4). Standard errors are two-way clustered by firm and date. Post-launch − Announcement is $\beta_L - \beta_A$.

	(1)	(2)	(3)	(4)
Eligible × Announcement-to-launch period	-0.0388^{***}	-0.0365^{***}	-0.0225^{***}	-0.0103^{***}
	(0.0066)	(0.0064)	(0.0044)	(0.0036)
Eligible × Post-launch period	-0.0385^{***}	-0.0363^{***}	-0.0245^{***}	0.0051
	(0.0065)	(0.0063)	(0.0045)	(0.0041)
Post-launch − Announcement	0.0002	0.0002	-0.0019	0.0155^{***}
	(0.0061)	(0.0060)	(0.0041)	(0.0036)
Sample	Shanghai	Full	Full	Full
Firm fixed effects	Yes	Yes	Yes	Yes
Industry × day fixed effects	Yes	Yes	Yes	Yes
Exchange × day fixed effects	No	Yes	No	No
Controls	No	No	No	Yes
Clustering	Firm & date	Firm & date	Firm & date	Firm & date
Observations	761,168	2,067,269	2,067,269	2,067,269
Within R^2	0.0044	0.0013	0.0012	0.2193

alization and are mechanically related to return-based outcomes, we regard the controlled specification as a bad-control sensitivity exercise rather than an alternative preferred estimate.

IA.J.6.6. Alternative Outcome Measures

Table IA.XII
Alternative reduced-form outcomes.

Each column uses the preferred Shanghai-only specification with firm and industry-by-day fixed effects and no contemporaneous controls. Standard errors are two-way clustered by firm and date. Amihud illiquidity is in raw units; Post-launch − Announcement is $\beta_L - \beta_A$.

Dependent variable	Total volatility	Synchronicity	Amihud illiquidity	Abs. market-model residual
Eligible × Announcement-to-launch period	-0.0432^{***}	0.0333	$1.98 \times 10^{-7***}$	-0.0008^{***}
	(0.0063)	(0.0448)	(2.45×10^{-8})	(0.0002)
Eligible × Post-launch period	-0.0600^{***}	-0.0396	$3.16 \times 10^{-7***}$	-0.0004^{***}
	(0.0062)	(0.0408)	(2.49×10^{-8})	(0.0001)
Post-launch − Announcement	-0.0168^{***}	-0.0729^{*}	$1.18 \times 10^{-7***}$	0.0004^{**}
	(0.0058)	(0.0434)	(1.76×10^{-8})	(0.0001)
Firm fixed effects	Yes	Yes	Yes	Yes
Industry × day fixed effects	Yes	Yes	Yes	Yes
Controls	No	No	No	No
Clustering	Firm & date	Firm & date	Firm & date	Firm & date
Observations	761,168	731,138	761,168	761,168
Within R^2	0.0116	0.0005	0.0261	0.0010

IA.J.6.7. Event Studies, Placebos, and Alternative Specifications

The event-study specification is $Y_{it} = \alpha_i + \gamma_{j(i),t} + \sum_{k \neq -1} \beta_k (\text{Eligible}_i \times \mathbf{1}[t = k]) + \varepsilon_{it}$, with firm and industry-by-day fixed effects, no contemporaneous controls, and standard errors two-way clustered by firm and date. Event time k is measured in 21-trading-day bins relative to the April 10, 2014 announcement, with the bin immediately preceding the announcement (March 11–April 9, 2014, $k = -1$) as the omitted reference; the launch corresponds to bin $+7$. Anchoring bins on exact trading dates avoids calendar-month contamination: April 2014 mixes pre-announcement and post-announcement trading days, and November 2014 mixes pre-launch and post-launch trading days. The pre-announcement bins $-12, \ldots, -2$ are all before the announcement and form the parallel-trends diagnostic. On the full sample the joint F-test that all pre-announcement leads are zero rejects at the 1% level ($F = 7.69$, $p < 0.001$), with the rejection driven by elevated coefficients from bin -12 through bin -5. On the preferred Shanghai-only subsample the same test does not reject ($F = 1.497$, $p = 0.127$); the pre-event rejection attenuates substantially when the comparison group is restricted to Shanghai-listed stocks, suggesting that the full-sample pre-event differences are partly associated with the broader cross-exchange comparison. The bin-level dynamics are nonmonotonic: on the Shanghai-only subsample the announcement bin is significantly negative (-0.0171^{**}) and the announcement-centered launch bin is negative and statistically indistinguishable from zero (-0.0170, s.e. 0.0132), whereas the full-sample announcement-centered launch bin is positive and significant (0.0242^{***}); the launch-centered specification, which uses the bin immediately preceding the launch as the omitted reference, identifies a positive local movement at implementation on the Shanghai-only subsample (0.0343^{***}). Because the size-adjusted design is column (2) of the main-text table, we also estimate the Shanghai-only event study with pre-treatment size-decile-by-day fixed effects on the column-(2) common sample. The joint test that the pre-announcement leads are zero gives $F = 0.685$ ($p = 0.754$), substantially flatter than the unadjusted path, and the announcement bin (-0.0024, s.e. 0.0084) and launch bin (-0.0246, s.e. 0.0161) are both statistically insignificant. The size-adjusted event-time path therefore provides no evidence of systematic differential pre-event dynamics, consistent with size-related differential shocks accounting for part of the unadjusted pre-event pattern and the larger baseline announcement-period coefficient. The event-time dynamics are therefore sensitive to both the comparison sample and the reference period. Figure IA.9 plots the Shanghai-only announcement-centered coefficients with 95% confidence intervals. The launch-centered event study repeats the Shanghai-only specification relative to November 17, 2014; its coefficients therefore answer a different local comparison from the announcement-centered coefficients. Because idiosyncratic volatility is constructed over rolling

30-trading-day windows, coefficients in the bins immediately around the event boundaries incorporate returns from before the boundary and should be interpreted as local dynamics rather than sharp one-day event responses.

Placebo tests use pseudo events one year before the announcement (April 10 and November 17, 2013) within the pre-announcement subsample (before April 10, 2014). Pseudo events inside the post-launch window are not valid placebos because treated stocks are already treated there, so they are not reported.

The sample-window rows drop the first 30 or 120 trading days after each treatment boundary, so that no rolling window mixes pre-treatment and treated returns, and truncate the sample on August 15, 2016, the trading day before the aggregate Stock Connect quota was abolished and other contemporaneous liberalization changes took effect. Because these rows are estimated on the Shanghai-only subsample, they do not involve Shenzhen comparison stocks; the Shenzhen–Hong Kong Stock Connect announcement on August 16, 2016 affects only the full-sample designs, reported separately. All rows are estimated on the Shanghai-only subsample under the preferred specification (firm and industry-by-day fixed effects, no contemporaneous controls, and standard errors two-way clustered by firm and date). Applied to synchronicity itself, the preferred Shanghai-only specification leaves both period differentials statistically insignificant while the incremental contrast is negative and only marginally significant (-0.0729, $s.e. = 0.0434$); synchronicity therefore provides limited corroborating evidence.

Table IA.XIII
Placebo tests and robustness specifications.

Placebo rows use pseudo events one year before the announcement. Boundary-clean and washout rows exclude 30 and 120 trading days after treatment boundaries, respectively; the quota row ends on August 15, 2016. All estimates use the preferred Shanghai-only specification with two-way clustered standard errors.

Specification	Eligible × Ann.	Eligible × Post	$\beta_L - \beta_A$
Panel A. Placebo tests (pre-announcement pseudo events, 2013; Shanghai-only subsample)			
Pseudo announcement, 2013 pre-period	-0.0003	—	—
	(0.0077)	—	—
Pseudo post-launch, 2013 pre-period	—	-0.0053	—
	—	(0.0096)	—
Panel B. Robustness specifications (Shanghai-only subsample, preferred design)			
Boundary-clean: first 30 trading days	-0.0422^{***}	-0.0410^{***}	0.0012
	(0.0071)	(0.0065)	(0.0064)
120-day washout-window sensitivity	-0.0612^{***}	-0.0515^{***}	0.0097
	(0.0100)	(0.0067)	(0.0093)
Sample through August 15, 2016 (pre-quota removal)	-0.0391^{***}	-0.0356^{***}	0.0035
	(0.0066)	(0.0067)	(0.0063)

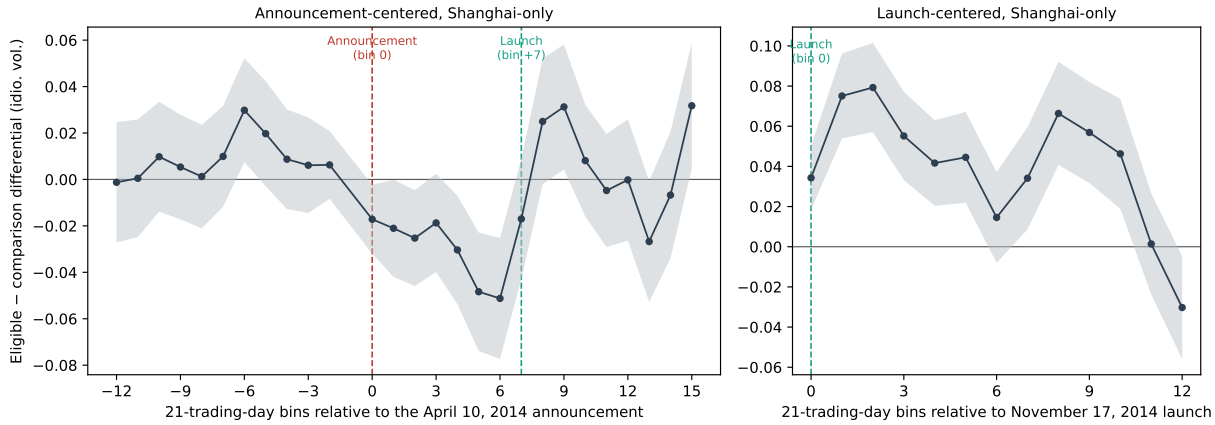


Figure IA.9: **Same-Exchange (Shanghai-Only) Event Study.** The figure plots the differential in idiosyncratic volatility between announcement-eligible and comparison stocks on the Shanghai-only subsample in 21-trading-day bins, estimated with firm and industry-by-day fixed effects, no contemporaneous controls, and standard errors two-way clustered by firm and date. The left panel is announcement-centered, with event time measured in 21-trading-day bins relative to April 10, 2014 (bin -1, the 21 trading days immediately preceding the announcement, omitted) and the dashed lines marking the April 10, 2014 announcement (bin $+0$) and the November 17, 2014 launch (bin $+7$); the right panel is launch-centered. Shaded bands indicate 95% confidence intervals.

IA.J.6.8. Same-Exchange and Size Robustness

Because the eligible stocks are all Shanghai-listed while roughly 80% of the comparison pool is listed on the Shenzhen exchange, the baseline differentials could in principle reflect time-varying Shanghai–Shenzhen dynamics rather than Stock Connect eligibility. Table IA.XIV addresses this concern in five specifications. Column (1) restricts the sample to Shanghai-listed stocks, comparing the eligible stocks with the never-eligible Shanghai comparison stocks under the preferred specification. Column (2) augments the Shanghai-only preferred fixed effects with pre-treatment size-decile-by-day effects. Column (3) keeps the Shanghai-only sample but replaces the comparison group with the not-yet-eligible stocks: a stock that becomes eligible after the announcement is retained as a comparison only until its first admission date, so comparison-group membership does not condition on future eligibility outcomes. Column (4) keeps the full sample and augments the preferred specification with exchange-by-day fixed effects, absorbing Shanghai-wide daily movement so that identification comes from announcement-eligible versus comparison stocks within Shanghai. Column (5) keeps the full sample and augments the preferred fixed effects with both exchange-by-day and pre-treatment size-decile-by-day effects; firms without a 2013 market capitalization are dropped. Because column (2) is estimated on the subsample of Shanghai

firms with a 2013 market-capitalization observation (723,108 firm-days, versus 761,168 in column (1)), we also re-estimate the baseline specification on the column-(2) common sample without the size fixed effects: the common-sample announcement-period coefficient is -0.0375^{***} (s.e. 0.0066), essentially unchanged from the full Shanghai sample (-0.0388^{***}); the common-sample post-launch coefficient is -0.0385^{***} (s.e. 0.0065) and the incremental post-launch contrast is -0.0011 (s.e. 0.0062), likewise essentially unchanged from the full-sample baseline (-0.0385^{***} and 0.0002). The attenuation to -0.0148^{*} therefore comes from the pre-treatment size-decile-by-day fixed effects themselves rather than from the smaller sample. In the Shanghai-only broad-period designs the announcement-to-launch and post-launch differentials are strongly negative ($\beta_A = -0.0388^{***}$ and $\beta_L = -0.0385^{***}$ in column (1), and $\beta_A = -0.0148^{*}$ and $\beta_L = -0.0277^{***}$ with size-decile effects in column (2)), and the incremental post-launch contrast is insignificant in column (1) and only marginally significant in column (2). Retaining the later-eligible Shanghai stocks only until their first admission date (column (3)) delivers broad-period differentials essentially identical to the baseline ($\beta_A = -0.0368^{***}$ and $\beta_L = -0.0368^{***}$), confirming that the announcement-ineligible comparison group, including stocks that become eligible only later, does not drive the pattern. Augmenting the full sample with exchange-by-day effects leaves the broad-period pattern essentially unchanged ($\beta_A = -0.0365^{***}$ and $\beta_L = -0.0363^{***}$), while adding pre-treatment size-decile-by-day effects weakens the announcement-to-launch coefficient ($\beta_A = -0.0119^{*}$ and $\beta_L = -0.0144^{**}$). The Shanghai-only announcement-centered event study materially weakens the full-sample pre-trend rejection: the joint test that the pre-announcement leads are zero gives $F = 1.497$ ($p = 0.127$) in 21-trading-day bins, compared with $F = 7.693$ ($p < 0.001$) on the full sample, and the Shanghai-only announcement-centered launch-bin coefficient is negative rather than positive. The same-exchange comparisons therefore reinforce the broad-period pattern, while the pre-event rejection attenuates substantially when the comparison group is restricted to Shanghai-listed stocks, suggesting that the full-sample pre-event differences are partly associated with the broader cross-exchange comparison.

Table IA.XIV
Same-exchange and size robustness.

Columns (1)–(3) use Shanghai-only comparisons; column (2) adds pre-treatment size-decile-by-day effects and column (3) uses not-yet-eligible controls. Columns (4)–(5) use the full sample with exchange-by-day effects, with size effects additionally included in column (5). The bottom rows summarize announcement-centered event studies.

	(1)	(2)	(3)	(4)	(5)
Eligible × Announcement-to-launch period	-0.0388^{***}	-0.0148^{*}	-0.0368^{***}	-0.0365^{***}	-0.0119^{*}
	(0.0066)	(0.0080)	(0.0062)	(0.0064)	(0.0069)
Eligible × Post-launch period	-0.0385^{***}	-0.0277^{***}	-0.0368^{***}	-0.0363^{***}	-0.0144^{**}
	(0.0065)	(0.0078)	(0.0062)	(0.0063)	(0.0067)
Post-launch − Announcement	0.0002	-0.0129^{*}	0.0000	0.0002	-0.0025
	(0.0061)	(0.0073)	(0.0058)	(0.0060)	(0.0063)
Sample	Shanghai	Shanghai	Shanghai	Full	Full
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry × day fixed effects	Yes	Yes	Yes	Yes	Yes
Exchange × day fixed effects	No	No	No	Yes	Yes
Pre-treatment size decile × day fixed effects	No	Yes	No	No	Yes
Not-yet-eligible comparison (censored at first admission)	No	No	Yes	No	No
Controls	No	No	No	No	No
Clustering	Firm & date	Firm & date	Firm & date	Firm & date	Firm & date
Observations	761,168	723,108	800,347	2,067,269	1,929,993
Within R^2	0.0044	0.0015	0.0041	0.0013	0.0002
Shanghai-only announcement-centered event study					
Joint F-test: all pre-announcement leads equal zero	$F = 1.497$, $p = 0.127$	$F = 0.685$, $p = 0.754$			
Announcement bin	-0.0171^{**} (0.0076)	-0.0024 (0.0084)			
Announcement-centered: launch (bin +7)	-0.0170 (0.0132)	-0.0246 (0.0161)			

IA.K Supplementary Technical Corollaries

The following corollaries provide technical characterizations supporting the main propositions. They are not separate main-text contributions.

Corollary 1 (Domestic Type-I specialization across access regimes). *In equilibrium,*

1) in the closed-access equilibrium, if the equilibrium equivalent data amount with zero data processing for type-I risky asset is still larger than or equal to the equilibrium equivalent data amount at data processing capacity for type-II formally $m_{e,(\mathrm{I});\mathrm{cl}}(0) \geq m_{e,(\mathrm{II});\mathrm{cl}}(\bar{M})$*, then domestic institutional investors will not acquire and process data for type-I;*

2) if in addition the effective precision of a directly processed type-I data point is no larger than the type-I-equivalent effective precision obtained from a type-II data point through cross-asset learning, i.e. $\Pi_{e,(\mathrm{I})} \leq z^2 \Pi_{e,(\mathrm{II})}$*, then domestic institutional investors will not acquire and process data for type-I when the market is partially liberalized.*

Corollary 2 (Type-I own-asset SNR under partial access). *Suppose* $\kappa(0) \leq 1$*, so foreign institutional investors allocate all processing capacity to type-I data. Corollary 8 provides a primitive sufficient condition for this corner.*

1) A sufficient condition for partial market access to have a positive effect on the Type-I self-signal-to-noise ratio is $\lambda_K > \lambda_I \varphi^2$*.*

2) If the foreign institutional mass satisfies $\lambda_K < ((M_{(\mathrm{I});\mathrm{cl}} - M_{(\mathrm{I})})\varphi^2/\bar{M})\lambda_I$ *and entry induces domestic institutional investors to learn less about type-I risky assets, then the Type-I self-signal-to-noise ratio decreases once market access is granted.*

3) The cross-signal-to-noise is zero.

Corollary 3 (Price and private information under full access). *For type-I,*

$$\Delta_{\mathrm{fu}} \bar{\hat{\Omega}}_{y,(\mathrm{I})} = \Delta_{\mathrm{fu}} \hat{\Omega}_{p,(\mathrm{I})} + \mathcal{P}_{(\mathrm{I});\mathrm{fu}} - \mathcal{P}_{(\mathrm{I})} + \mathcal{Q}_{\mathrm{I}}, \tag{IA.85}$$

where $\Delta_{\mathrm{fu}} \hat{\Omega}_{p,(\mathrm{I})} = \left(\zeta^2_{(\mathrm{I});\mathrm{fu}} - \zeta^2_{(\mathrm{I})}\right) \Omega_{x,(\mathrm{I})}$*. For type-II,*

$$\Delta_{\mathrm{fu}} \bar{\hat{\Omega}}_{y,(\mathrm{II})} = \Delta_{\mathrm{fu}} \hat{\Omega}_{p,(\mathrm{II})} + \mathcal{P}_{(\mathrm{II});\mathrm{fu}} - \mathcal{P}_{(\mathrm{II})}, \tag{IA.86}$$

with

$$\Delta_{\mathrm{fu}} \hat{\Omega}_{p,(\mathrm{II})} = \left(\zeta^2_{(\mathrm{II});\mathrm{fu}} - \zeta^2_{(\mathrm{II})}\right) \Omega_{x,(\mathrm{II})} - \zeta^2_{(\mathrm{I}\leftarrow\mathrm{II})} \Omega_{x,(\mathrm{I})}. \tag{IA.87}$$

Thus full access removes the type-I Schur-complement discount $\mathcal{Q}_{\mathrm{I}}$*, while for type-II it removes the partial-access cross-price information term. Consequently, for type-II a change in the self-SNR need*

not imply a change in price informativeness in the same direction, and for either asset a change in price informativeness need not imply a change in average posterior precision in the same direction because eligible-investor private-information density also changes.

COROLLARY 4 (Type-II own-asset SNR under full access). *Suppose full access induces foreign institutional investors to process type-II data. The change in the type-II self-signal-to-noise ratio is negative if and only if the loss of domestic type-II effective information exceeds the new foreign type-II effective information:*

$$\lambda_I \left(M_{I,(\mathrm{II})} - M_{I,(\mathrm{II});\mathrm{fu}}\right) \varphi^2 \Pi_{e,(\mathrm{II})} > \lambda_K M_{K,(\mathrm{II});\mathrm{fu}} \Pi_{e,(\mathrm{II})}.$$

COROLLARY 5 (Type-I own-asset SNR under full access). *For type-I risky assets, the signal-to-noise ratio*

1) decreases if domestic institutional investors learn less about them;

2) increases if domestic institutional investors learn more about them and $\lambda_K < ((M_{I,(\mathrm{I});\mathrm{fu}} - M_{I,(\mathrm{I})})\varphi^2/(\bar{M} - M_{K,(\mathrm{I});\mathrm{fu}}))\lambda_I$.

COROLLARY 6 (No cross-asset learning without fundamental correlation). *If* $\mathcal{C}_y = 0$, *then the equilibrium cross-precision satisfies* $z = 0$ *and*

$$\kappa(M) = 0 \qquad \text{for all } M \in [0, \bar{M}].$$

Hence the foreign institutional investor processes no Type-II data, $M^*_{k,(\mathrm{II})} = 0$. *Because the processing-capacity constraint binds,* $M^*_{k,(\mathrm{I})} = \bar{M}$.

COROLLARY 7 (Domestic information allocation). *Conditional on the equilibrium price signal, domestic investor* i *allocates more data-processing capacity to type-I than to type-II if and only if* $m_{e,(\mathrm{I})}(0) < m_{e,(\mathrm{II})}(0)$. *If* $m_{e,(\mathrm{I})}(0) = m_{e,(\mathrm{II})}(0)$, *the investor splits capacity equally,* $M^*_i = \bar{M}/2$; *if* $m_{e,(\mathrm{I})}(0) > m_{e,(\mathrm{II})}(0)$, *the investor allocates more capacity to type-II than to type-I.*

COROLLARY 8 (Primitive condition for all-Type-I foreign processing). *A sufficient condition for foreign institutional investors to allocate all processing capacity to type-I data is*

$$\frac{\Pi_{e,(\mathrm{I})}}{\Pi_{e,(\mathrm{II})}} \geq \left(\frac{\mathcal{C}_y}{\Omega_{y,(\mathrm{II})}}\right)^2. \tag{IA.88}$$

Under this condition, $M^*_{K,(\mathrm{I})} = \bar{M}$ *and* $M^*_{K,(\mathrm{II})} = 0$, *independently of the precision of domestic soft information.*

COROLLARY 9 (Information reallocation and own-asset SNRs). *Suppose $\kappa(\bar{M}) \geq 1$, so foreign institutional investors allocate all processing capacity to type-II data. Partial market access moves the two self-signal-to-noise ratios in opposite directions whenever domestic institutional investors reallocate data processing across the two asset types.*

IA.K.1 Technical Corollaries Moved from the Manuscript

The following result provides a technical characterization supporting the main price-informativeness proposition. It is not a separate main-text contribution.

COROLLARY 10 (Type-I learning and posterior precision). *Suppose foreign institutions are at the all-Type-I processing corner. Define*

$$\mathcal{H}_{\mathrm{I}} \equiv \left[\lambda_I \varphi^2 \left(M_{(\mathrm{I})} - M_{(\mathrm{I});\mathrm{cl}}\right) + \lambda_K \bar{M}\right] \Pi_{e,(\mathrm{I})},$$

$$A_{\mathrm{I}} \equiv \frac{r - g_1}{\rho r} \Omega_{x,(\mathrm{I})} \left(\left|\zeta_{(\mathrm{I})}\right| + \left|\zeta_{(\mathrm{I});\mathrm{cl}}\right|\right) + \frac{1}{\lambda_2} > 0,$$

and

$$C_{\mathrm{I}} \equiv \frac{\lambda_I}{\lambda_1} M_{(\mathrm{I});\mathrm{cl}} \varphi^2 \Pi_{e,(\mathrm{I})} + \Omega_{\epsilon,(\mathrm{I})} > 0, \qquad \mathcal{Q}_{\mathrm{I}} \equiv \frac{\lambda_K}{\lambda_2} z^2 \hat{\Omega}_{Ky,(\mathrm{II})}.$$

Then

$$\Delta\bar{\hat{\Omega}}_{y,(\mathrm{I})} = A_{\mathrm{I}} \mathcal{H}_{\mathrm{I}} - \frac{\lambda_K}{\lambda_2} C_{\mathrm{I}} - \mathcal{Q}_{\mathrm{I}}. \tag{IA.89}$$

Hence a weak decline in Type-I price informativeness implies a strict decline in average Type-I posterior precision.

For Type II, define

$$\mathcal{H}_{\mathrm{II}} \equiv \lambda_I \varphi^2 \left(M_{(\mathrm{II})} - M_{(\mathrm{II});\mathrm{cl}}\right) \Pi_{e,(\mathrm{II})},$$

and

$$A_{\mathrm{II}} \equiv \frac{r - g_2}{\rho r} \left(\left|\zeta_{(\mathrm{II})}\right| + \left|\zeta_{(\mathrm{II});\mathrm{cl}}\right|\right) \Omega_{x,(\mathrm{II})} + \frac{1}{\lambda_1} > 0.$$

COROLLARY 11 (Type-II learning, cross-price information, and posterior precision). *Relative to the closed-market counterfactual,*

$$\Delta\bar{\hat{\Omega}}_{y,(\mathrm{II})} = A_{\mathrm{II}} \mathcal{H}_{\mathrm{II}} + \zeta^2_{(\mathrm{I}\leftarrow\mathrm{II})} \Omega_{x,(\mathrm{I})}. \tag{IA.90}$$

Hence:

1) If $\mathcal{H}_{\text{II}} \geq 0$, average type-II posterior precision weakly increases, with a strict increase whenever $\mathcal{H}_{\text{II}} > 0$ or $\zeta_{(\text{I}\leftarrow\text{II})} \neq 0$.

2) If $\mathcal{H}_{\text{II}} < 0$, average type-II posterior precision decreases if and only if

$$\zeta^2_{(\text{I}\leftarrow\text{II})}\Omega_{x,(\text{I})} < -A_{\text{II}}\mathcal{H}_{\text{II}}. \tag{IA.91}$$

Thus the cross-price channel can offset, and potentially more than offset, a decline in domestic type-II information production. Absent this channel, average type-II posterior precision moves in the same direction as domestic type-II effective information production.

IA.K.2 OLS in calibration

Grouped by their eligibility for northbound trade, shares are aggregated by negotiable market capitalization. The variance-covariance structure of the two aggregate shares is $\left(\begin{smallmatrix} 4.1721 & 2.9818 \\ 2.9818 & 2.5778 \end{smallmatrix}\right)$.

Following the measurement methodology in the literature, unexpected earnings proxy for the dividend innovation $\widetilde{y}$. Earnings surprises are constructed from analyst forecasts and, alternatively, from a time-series model of annual earnings announcements; last-period dividends are measured by the previous year's EPS announcement, and market prices are normalized identically by total assets. The foreign-investable shares are the April 10, 2014 Northbound-eligible securities under the Stock Connect rules: constituents of the SSE 180 and SSE 380 indices plus A-shares of companies listed on both the SSE and the SEHK (Table IA.III). Because eligible and ineligible shares differ systematically, the analysis uses propensity-score matching on total market capitalization, negotiable market capitalization, and industry dummies; a kNN algorithm matches each type-I asset to its closest type-II neighbor. Section IA.J.5 provides additional details on variable construction.